

Second Edition

Solutions Manual of
Introduction to

TOPOLOGY

For

BS 4-Years, M.Sc. Mathematics

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Chapter 1

TOPOLOGICAL SPACES

EXERCISE 1

Short Questions

Q.3

(i)

Sol:

Solve / answer the following short questions:
Define a topological space.

Let X be a nonempty set. A collection τ of subsets of X is said to be a topology on X if the following conditions are satisfied:

- (i) The union of any number of members of τ belongs to τ .
- (ii) The intersection of finite number of members of τ belongs to τ .
- (iii) The empty set ϕ and the set X itself belong to τ .

The pair (X, τ) is called a topological space.

(ii)

List all possible topologies on the set $X = \{0, 1\}$.

Sol:

$$X = \{0, 1\}$$

All topologies on X are

$$\tau_1 = \{\phi, X\}, \quad \tau_2 = \{\phi, \{0\}, X\},$$

$$\tau_3 = \{\phi, \{1\}, X\}, \quad \tau_4 = \{\phi, \{0\}, \{1\}, X\}$$

(iii)

If $X = \{0, 1\}$, find all possible Sierpinski spaces.

Sol:

$$X = \{0, 1\}$$

Consider the following topologies on X :

$$\tau_1 = \{\phi, \{0\}, X\}, \quad \tau_2 = \{\phi, \{1\}, X\}$$

$(X, \tau_1), (X, \tau_2)$ are all possible Sierpinski spaces.

(iv)

If $X = \{a, b, c, d, e, f\}$, find three Sierpinski spaces.

Sol:

$$X = \{a, b, c, d, e, f\}$$

Consider the following three topologies on X :

$$\tau_1 = \{\phi, \{a\}, X\}, \quad \tau_2 = \{\phi, \{a, f\}, X\}$$

$$\tau_3 = \{\phi, \{b, c, f\}, X\}$$

$(X, \tau_1), (X, \tau_2), (X, \tau_3)$ are three Sierpinski spaces.

- (v) Let $\tau = \{\emptyset, A_q, R\}$, where $A_q = (q, \infty)$, $q \in Q$ (Q is the set of rational numbers). Whether $\cup \{A_q : q \in Q, q > \sqrt{2}\} \in \tau$.

Sol: It is clear by the definition of A_q that

$$\cup \{A_q : q \in Q, q > \sqrt{2}\} = (\sqrt{2}, \infty)$$

Since $\sqrt{2}$ is irrational, so $(\sqrt{2}, \infty) \notin \tau$. This shows that

$$\cup \{A_q : q \in Q, q > \sqrt{2}\} \notin \tau$$

- (vi) Let $A_q = (-\infty, q)$, $q \in Q$, where Q is the set of rational numbers. Show that $\tau = \{\emptyset, A_q, R\}$ is not a topology on the set of real numbers R .

Sol: We observe that

$$A = \cup \{A_q : q \in Q, q < \sqrt{2}\} = (-\infty, \sqrt{2})$$

is the union of members of τ , but $A \notin \tau$, because $\sqrt{2}$ is irrational. This shows that union of members of τ is not in τ , so τ is not a topology on R .

- (vii) List all topologies on $X = \{a, b\}$ which consist of exactly four members.

Sol: $X = \{a, b\}$

All the possible subsets of X are

$$\emptyset, \{a\}, \{b\}, X$$

Since $\{a\}, \{b\}$ form the partition of X , so

$$\tau = \{\emptyset, \{a\}, \{b\}, X\}$$

is a topology on X .

Since the four subsets of X are not totally ordered by inclusion, so τ is the only topology on X consisting of four members.

- (viii) Let X be an indiscrete topological space and A be a singleton subset of X . Find A^d .

Sol: Since A is a singleton set, so let $A = \{a\}$. Since X is an indiscrete topological space, so X itself is the only nonempty open set. Obviously, a is not the limit point of A because open set X does not contain any point of A different from a . Any point $x \neq a$ of X is a limit point of A because the open set X containing x contains a point a of A different from x . Hence

$$A^d = X - A = A^c$$

- (ix) If A is any subset of a cofinite topological space X , then find $\overline{A}$.

Sol: Finite subsets of X and X itself are the only closed sets in X . Since $\overline{A}$ is the smallest closed superset of A , so

$$\overline{A} = \begin{cases} A & \text{if } A \text{ is finite} \\ X & \text{if } A \text{ is infinite} \end{cases}$$

- (x) Let τ be a collection of subsets of X totally ordered by set inclusion. Show that the intersection of any two members of τ is in τ .

Sol: Let $A, B \in \tau$. Since τ is totally ordered by set inclusion, so

$$\text{either } A \subset B \quad \text{or} \quad B \subset A$$

That is

$$\text{either } A \cap B = A \quad \text{or} \quad A \cap B = B$$

In either case $A \cap B \in \tau$.

- (xi) Prove that if x is a limit point of the set A , then x is also a limit point of $A \setminus \{x\}$.

Sol: Since x is a limit point of A , so for any open set U containing x , we have

$$U \cap (A \setminus \{x\}) \neq \emptyset$$

This shows that any open set U containing x contains a point of $A \setminus \{x\}$. This point is obviously different from x , so x is a limit point of $A \setminus \{x\}$.

- (xii) Prove that if $A \subset B$, then $\overline{A} \subset \overline{B}$.

Sol: Since the closure of a set is its superset, so

$$B \subseteq \overline{B} \quad \dots(1)$$

Since $A \subseteq B$, so combining it with (1), we have

$$A \subseteq \overline{B} \quad \dots(2)$$

Since the closure of a set is always a closed set, so (2) shows that $\overline{B}$ is the closed superset of A . But $\overline{A}$ is the smallest closed superset of A , so $\overline{A}$ must be a subset of $\overline{B}$, i.e. $\overline{A} \subseteq \overline{B}$.

- (xiii) Find indiscrete and discrete topologies in $X = \{1, 2, 3\}$.

Sol: Indiscrete topology is

$$\tau = \{\emptyset, X\}$$

All possible subsets of $X = \{1, 2, 3\}$ are

$$\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, X$$

Discrete topology is

$$\tau = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, X\}$$

- (xiv) Define metrizable space. Give an example to show that a topological space need not be a metrizable space.

Sol: A topological space (X, τ) is said to be the metrizable space if there exists at least one metric d on X such that the collection of all open sets generated by d is exactly the topology τ .

Example:

Let $X = \{a, b\}$, then $\tau = \{\emptyset, \{a\}, X\}$ is a topology on X , and the space (X, τ) is the Sierpinski space. We show that this space is not a metrizable space. For this suppose on contrary that X is a metrizable space, then there exists a metric d on X such that each open set generated by d must be a member of τ and vice versa, i.e. each member of τ must be an open set generated by d .

Therefore, $\{a\}$, being the member of τ , is an open set generated by d , so there must exist an open sphere $S_r(a)$ such that

$$S_r(a) \subseteq \{a\}$$

This is only possible if the open sphere $S_r(a)$ contains only a , i.e.

$$d(a, b) \geq r$$

$$\Rightarrow a \notin S_r(b)$$

$$\Rightarrow S_r(b) = \{b\}$$

This shows that $\{b\}$ is open in the metric space (X, d) . But this is not in τ , so the open set generated does not belong to τ . Hence, X is not a metrizable space.

(xv) **Define A-inclusion topology on X .**

Sol: Let X be a nonempty set and $A \subset X$ be an arbitrary subset of X . Let τ consist of $\emptyset$ and every superset of A , then τ is said to be an A-inclusion topology on X .

(xvi) **Define A-exclusion topology on X .**

Sol: Let X be a nonempty set and $A \subset X$ be an arbitrary subset of X . Let τ consist of X and every subset of X which does not intersect A , then τ is said to be an A-exclusion topology on X .

(xvii) **Define the subspace of a topological space.**

University of Gujrat, 2013 (M.Sc. Math)

Sol: Let (X, τ_X) be a topological space and $Y \subset X$ a nonempty subset, then

$$\tau_Y = \{V \cap Y : V \in \tau_X\}$$

is a topology on Y , called the topology induced by τ_X on Y or relative topology on Y . The pair (Y, τ_Y) is called the subspace of X .

(xviii) If $X = \{a, b, c, d, e\}$ with $\tau_X = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}, X\}$ and $Y = \{b, c, d, e\}$, find the relative topology on Y .

Sol: Taking the intersection of each member of τ_X with Y we have

$$\emptyset \cap Y = \emptyset, \{a\} \cap Y = \emptyset, \{b\} \cap Y = \{b\},$$

$$\{c\} \cap Y = \{c\}, \{a, b\} \cap Y = \{b\}, \{a, c\} \cap Y = \{c\},$$

$$\{b, c\} \cap Y = \{b, c\}, \{a, b, c\} \cap Y = \{b, c\}, X \cap Y = Y$$

This shows that the members of τ_Y are $\phi, \{b\}, \{c\}, \{b, c\}, Y$, so

$$\tau_Y = \{\phi, \{b\}, \{c\}, \{b, c\}, Y\}$$

This is the required relative topology on Y .

(xix) **Show that the subspace of a discrete topological space is also discrete space.**

Sol: Let X be a discrete topological space, then the power set of X , $P(X)$, is the topology on X . Let Y be a non-empty subset of X and τ_Y be the relative topology on Y , then

$$\tau_Y = \{V \cap Y : V \in P(X)\}$$

In order to show that $\tau_Y = P(Y)$, let $A \subset Y$ such that $A \notin \tau_Y$. Then $A \neq V \cap Y$ for all $V \in P(X)$, which is not possible, because each subset of Y is the intersection of some member of $P(X)$ with Y . Thus, each subset of Y is a member of τ_Y , i.e., $\tau_Y = P(Y)$. This shows that τ_Y is discrete topology on Y and so Y is a discrete topological space. This completes the proof.

(xx) **Show that the subspace of an indiscrete topological space is also indiscrete space.**

Sol: Let X be a nonempty set and τ_X be an indiscrete topology on X , then

$$\tau_X = \{\phi, X\}$$

Let Y be a nonempty subset of X , then

$$\phi \cap Y = \phi, X \cap Y = Y$$

are the only members of relative topology τ_Y , i.e.

$$\tau_Y = \{\phi, Y\}$$

This shows that τ_Y is also an indiscrete topology on Y .

Hence, Y is an indiscrete space.

Q.4 Solve / answer the following short questions:

(i) **Define open and closed sets in a topological space.**

Sol: Let (X, τ) be a topological space, then a member of τ is said to be an open set in X .

The subset of X whose complement is a member of τ is said to be a closed set in X .

(ii) **Show that every subset of a discrete topological space is closed.**

Sol: Let X be a discrete topological space, then the topology on X is its power set, so every subset of X is a member of topology and, therefore, the complement of every subset of X is also a member of topology. This shows that every subset of X is a closed subset of X . This completes the proof.

(iii) **If Y is an open subspace of a topological space X , then show that each open subset of Y is also open in X .**

Sol: Let U be an open subset of Y , then open subset V of X is defined as

$$U = V \cap Y$$

Since V and Y are open subsets of X and the intersection of two open sets is also open, so U , being the intersection of two open sets is open, in X .

(iv) **Show that every subset of a topological space is open if and only if its each singleton subset is open.**

Sol: Let (X, τ) be a topological space. Let every subset of X be open, then its singleton subsets are also open in X .

Conversely, let the singleton subsets of X be open, then each subset of X , being the union of its singleton subsets, is open.

(v) **Is the union of two topologies on X a topology on X ? Justify your answer.**

PU, 2013 (BS Math)

Sol: Let $X = \{a, b, c\}$ with topologies

$$\tau_1 = \{\emptyset, \{a\}, \{b, c\}, X\} \text{ and } \tau_2 = \{\emptyset, \{a, b\}, X\}$$

$$\text{Now } \tau_1 \cup \tau_2 = \{\emptyset, \{a\}, \{b, c\}, \{a, b\}, X\}$$

Since $\{b, c\}, \{a, b\} \in \tau_1 \cup \tau_2$ and $\{b, c\} \cap \{a, b\} = \{b\} \notin \tau_1 \cup \tau_2$, so $\tau_1 \cup \tau_2$ is not a topology on X .

This shows that the union of two topologies on X need not be a topology on X .

(vi) **Define the neighbourhood of a point in a topological space.**

Sol: Let (X, τ) be a topological space. A subset N of X containing $x \in X$ is said to be the neighbourhood of x if there exists an open set U containing x such that N contains U , i.e.

$$x \in U \subseteq N$$

(vii) **Let $X = \{a, b, c, d\}$ with topology $\tau = \{\emptyset, \{a\}, \{b\}, \{a, b\}, X\}$, then find all the possible neighbourhoods of a .**

University of Gujrat, 2013 (M.Sc. Math)

Sol: $X = \{a, b, c, d\}$, $\tau = \{\emptyset, \{a\}, \{b\}, \{a, b\}, X\}$

The subsets of X containing a are

$$\{a\}, \{a, b\}, \{a, c\}, \{a, d\}, \{a, b, c\}, \{a, b, d\}, \{a, c, d\}, X$$

Since $\{a\}$ is an open set such that

$$a \in \{a\} \subseteq \{a\}$$

$$a \in \{a\} \subseteq \{a, b\}$$

$$a \in \{a\} \subseteq \{a, c\}$$

$$a \in \{a\} \subseteq \{a, d\}$$

$$a \in \{a\} \subseteq \{a, b, c\}$$

$$a \in \{a\} \subseteq \{a, b, d\}$$

$$a \in \{a\} \subseteq \{a, c, d\}$$

$$a \in \{a\} \subseteq X$$

This shows that

$$\{a\}, \{a, b\}, \{a, c\}, \{a, d\}, \{a, b, c\}, \{a, b, d\}, \{a, c, d\}, X$$

are all possible neighbourhoods of a .

- (viii) Show that the topological space itself is a neighbourhood of each of its points.

Sol: Let (X, τ) be a topological space. Let x be any point of X . There exists at least one open set, i.e., X itself satisfying the condition $x \in X \subseteq X$. This shows that X is a neighbourhood of x . Hence, X is a neighbourhood of each of its points.

- (ix) Show that the intersection of two neighbourhoods of a point is also its neighbourhood in a topological space.

Sol: Let A and B be two neighbourhoods of x , then there exist open sets U and V such that

$$\begin{aligned} x \in U \subseteq A \text{ and } x \in V \subseteq B \\ \Rightarrow x \in U \cap V \subseteq A \cap B \end{aligned}$$

Since $U \cap V$, being the intersection of two open sets U and V , is open, so by the last expression $A \cap B$ is a neighbourhood of x .

- (x) Show that the union of two neighbourhoods of a point is also its neighbourhood in a topological space.

Sol: Let A and B be two neighbourhoods of x , then there exist open sets U and V such that

$$\begin{aligned} x \in U \subseteq A \text{ and } x \in V \subseteq B \\ \Rightarrow x \in U \cup V \subseteq A \cup B \end{aligned}$$

Since $U \cup V$, being the union of two open sets U and V , is open, so by the last expression $A \cup B$ is a neighbourhood of x .

- (xi) If A is a neighbourhood of x , and $A \subseteq B$, then show that B is also neighbourhood of x .

Sol: Since A is a neighbourhood of x , so there exists an open set U such that

$$x \in U \subseteq A \quad \dots(1)$$

$$\text{Since } A \subseteq B \quad \dots(2)$$

Combining (1) and (2), we have

$$\begin{aligned} x \in U \subseteq A \subseteq B \\ \Rightarrow x \in U \subseteq B \end{aligned}$$

This shows that B is also a neighbourhood of x .

- (xii) Define the closure of a subset A of a topological space X .

Sol: Let A be a subset of a topological space X , then the closure of A is the intersection of all closed supersets of A . The closure of A is denoted by $\bar{A}$.

- (xiii) Define an isolated point of a subset of a topological space.

Sol: A point $x \in A$ is said to be an isolated point of A if there exists an open set U containing x such that $A \cap U = \{x\}$.

- (xiv) Show that each point of a nonempty subset of a discrete topological space is its interior point.

Sol: Let (X, τ) be a discrete topological space, and let A be a subset of X . If x is any point of A , then $\{x\}$ is an open set such that

$$x \in \{x\} \subseteq A$$

This shows that x is an interior point of A , so each point of A is its interior point.

- (xv) **Show that the interior of a subset of a discrete topological space is the set itself.**

Sol: Let (X, τ) be a discrete topological space, and let A be a subset of X . Let $x \in A$, then $\{x\}$ is an open set such that

$$x \in \{x\} \subseteq A$$

This shows that x is an interior point of A , i.e.

$$x \in A^\circ$$

$$\Rightarrow A \subseteq A^\circ$$

But

$$A^\circ \subseteq A$$

...(1)

Combining (1) and (2), we have

...(2)

$$A^\circ = A$$

- (xvi) **Give an example of a topological space having two disjoint dense subsets.**

Sol: Let (R, τ) be a topological space with usual topology, then Q , set of rational numbers and Q^c , set of irrational numbers are disjoint dense subsets of R , because,

$$\overline{Q} = X, \overline{Q^c} = X, Q \cap Q^c = \emptyset$$

- (xvii) **If A and B are two subsets of a topological space X , then show that $\overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$.**

Sol:

$$\because A \cap B \subseteq A, A \cap B \subseteq B$$

$$\therefore \overline{A \cap B} \subseteq \overline{A}, \overline{A \cap B} \subseteq \overline{B}$$

$$\Rightarrow \overline{A \cap B} \subseteq \overline{A} \cap \overline{B}$$

University of Gujrat, 2013 (M.Sc. Math)

- (xviii) **Consider the topological space $X = \{a, b\}$ with the topology $\tau = \{\emptyset, X, \{a\}\}$. Find the interior and limit points of the set $A = X$.**

Sol: a is the interior point of $A = X$, because there exists an open set X , itself, such that $a \in X \subseteq X$. PU, 2013 (BS Math)

Similarly, b is also the interior point of $A = X$, because there exists an open set X , itself, such that $b \in X \subseteq X$.

b is the limit point of $A = X$, because the open set X containing b also contains a point a different from b .

- (xix) **Write three different topologies on $X = \{2, 5, 4, 7\}$ each having exactly four members.**

Sol: Each topology τ on X with four members is of the form PU, 2013 (BS Math)

$\tau = \{\phi, A, B, X\}$, where $\{A, B\}$ is a partition of X or the members of topology are totally ordered by inclusion.

Hence three different topologies on $X = \{2, 5, 4, 7\}$ each having exactly four members are

$$\tau_1 = \{\phi, \{2\}, \{4, 5, 7\}, X\},$$

$$\tau_2 = \{\phi, \{4\}, \{2, 5, 7\}, X\},$$

$$\tau_3 = \{\phi, \{7\}, \{2, 4, 5\}, X\}$$

(xx) Give an example of two subsets A, B of R such that $\text{Int}(A \setminus B) \subset \text{Int}(A) \setminus \text{Int}(B)$.

Sol: Let $A = [-2, 2]$, $B = [-1, 1]$, then

PU, 2013 (BS Math)

$$A \setminus B = [-2, 2] \setminus [-1, 1] = [-2, -1[\cup]1, 2]$$

$$\begin{aligned} \text{Int}(A \setminus B) &= \text{Int}([-2, -1[\cup]1, 2]) \\ &=]-2, -1[\cup]1, 2[\end{aligned}$$

Now

$$\text{Int}(A) = \text{Int}([-2, 2]) =]-2, 2[$$

$$\text{Int}(B) = \text{Int}([-1, 1]) =]-1, 1[$$

$$\text{Int}(A) \setminus \text{Int}(B) =]-2, 2[\setminus]-1, 1[=]-2, -1[\cup]1, 2[$$

Since $] -2, -1[\cup]1, 2[\subset]-2, -1[\cup]1, 2[$, so

$$\text{Int}(A \setminus B) \subset \text{Int}(A) \setminus \text{Int}(B)$$

Q.5 Solve / answer the following short questions:

(i) Define a limit point of a set in a topological space.

Sol: A point x of a topological space X is said to be the limit point of a subset A of X if for every open set U containing x , we have

$$\{A \cap U\} \setminus \{x\} \neq \phi$$

(ii) If A and B are subsets of a topological space X , then show that $A \subseteq B$ implies $A^d \subseteq B^d$.

Sol: Let $x \in A^d$, then x is a limit point of A , so there exists an open set U such that

$$\begin{aligned} \{A \cap U\} \setminus \{x\} &\neq \phi \\ \Rightarrow \{B \cap U\} \setminus \{x\} &\neq \phi \quad \because A \subseteq B \end{aligned}$$

This shows that x is a limit point of B , so $x \in B^d$.

Hence

$$A^d \subseteq B^d$$

(iii) Define an interior point of a subset of a topological space.

Sol: A point $x \in A$ is said to be an interior point of A if there exists an open set U such that $U \subseteq A$.

(iv) Define a discrete topological space.

Sol: The power set $P(X)$ of a nonempty set X is called the discrete topology on X and the space $(X, P(X))$ is called the discrete topological space or simply a discrete space.

(v) Define Sierpinski space with an example.

Sol: For a nonempty subset A of X , $\tau_A = \{\emptyset, A, X\}$ is always a topology on X . The topological space (X, τ_A) is called the Sierpinski space.

(vi) If A and B are two subsets of a topological space, then show that $A \subseteq B \Rightarrow A^\circ \subseteq B^\circ$.

Sol: Let $x \in A^\circ$, then there exists an open set U containing x such that $U \subseteq A$, since $A \subseteq B$, so $U \subseteq B$. This shows that x is also an interior point of B , i.e.

$$\begin{aligned} x &\in B^\circ \\ \Rightarrow A^\circ &\subseteq B^\circ \end{aligned}$$

Long Questions

Q.6 Let (X, τ) be a topological space and A be any arbitrary subset of X then show that $\bar{A} = A \cup A^d$.

University of Sargodha, 2010 (M.Sc. Math)

Sol: Let $x \in \bar{A}$. Further suppose that $x \notin A \cup A^d$, then x is neither in A nor a limit point of A , so there exists an open set U containing x such that

$$\begin{aligned} U \cap A &= \emptyset \\ \Rightarrow A &\subseteq U^c \end{aligned} \quad \dots(1)$$

Since U is open, so U^c is closed and from (1) it is clear that U^c is a closed superset of A . But $\bar{A}$ is the smallest closed superset of A , so

$$\begin{aligned} \bar{A} &\subseteq U^c \\ \Rightarrow \bar{A} \cap U &= \emptyset \end{aligned}$$

which is a contradiction, because x is in both $\bar{A}$ and U . Hence

$$\begin{aligned} x &\in A \cup A^d \\ \Rightarrow \bar{A} &\subseteq A \cup A^d \end{aligned} \quad \dots(2)$$

Conversely, let $x \in A \cup A^d$, then x is either in A or a limit point of A , so in both cases, each open set containing x intersects A .

Let $x \notin \bar{A}$, then $x \in (\bar{A})^c$. Since $\bar{A}$ is closed, so $(\bar{A})^c$ is an open set containing x . Since $\bar{A} \cap (\bar{A})^c = \emptyset$, so $\bar{A} \cap (\bar{A})^c = \emptyset$ ($\because A \subseteq \bar{A}$).

Thus, the open set $(\bar{A})^c$ containing x does not intersect A , which is a contradiction, so

$$\begin{aligned} x &\in \bar{A} \\ \Rightarrow A \cup A^d &\subseteq \bar{A} \end{aligned} \quad \dots(3)$$

Combining (2) and (3), we have $\bar{A} = A \cup A^d$.

- Q.7** Define usual topology on the real line R and prove that the set
- (i) $R \setminus \{5, 10, 15\}$ is open in R
 - (ii) The set Z is closed in R
 - (iii) Q is neither open nor closed in R (with the usual topology on R)

Sol: The collection, τ , of all open intervals is called usual topology on the real line R . PU, 2012 (M.Sc. Math)

(i) Let $A = R \setminus \{5, 10, 15\}$, then $A^c = \{5, 10, 15\}$ has no limit points and so A^c is closed. Hence, $A = R \setminus \{5, 10, 15\}$ is an open set in R .

(ii) Since Z has no limit points and so Z is closed.

(iii) Since every real number is a limit point of Q , so Q is not closed.

Let $q \in Q$, then any open interval $]q-r, q+r[$ centred at q also contains irrational numbers, so $]q-r, q+r[\not\subseteq Q$. This shows that Q is not open.

Hence Q is neither open nor closed in R (with the usual topology on R).

- Q.8** Let (X, τ) be a topological space and $A \subseteq X$. Let A^d and $\bar{A}$ denote the derived set and closure of A respectively such that $\bar{A} = A \cup A^d$. Show that $\bar{A}$ is a closed subset of X .

Sol: Suppose on contrary that $\bar{A}$ is not closed, then there exists a limit point x of $\bar{A}$ such that PU, 2008 (M.Sc. Math)

$$x \notin \bar{A}$$

$$\Rightarrow x \notin A \cup A^d$$

$$\Rightarrow x \notin A \text{ and } x \notin A^d$$

This shows that x is neither in A nor a limit point of A , so there is an open set containing x such that

$$U \cap A = \phi$$

$$\Rightarrow A \subseteq U^c$$

This shows that U^c is a closed superset of A . Since $\bar{A}$ is the smallest closed superset of A , so

$$\bar{A} \subseteq U^c$$

$$\Rightarrow U \cap \bar{A} = \phi$$

This shows that x is not a limit point of $\bar{A}$ which is a contradiction, so $\bar{A}$ is closed.

Q.9 Consider R , the set of real numbers along with the usual topology defined on it. Determine and justify the set of limit points of

- (i) Z , the set of integers.
- (ii) Q , the set of rational numbers.
- (iii) $[0, 1]$
- (iv) $]0, 1[$
- (v) $\left\{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{100}\right\}$

PU, 2005 (M.Sc. Math)

Sol:(i) Since Z has no limit points and so Z is closed. Thus, $Z^d = \phi$.

(ii) Since every real number is a limit point of Q , so Q is not closed. Thus, $Q^d = \phi$.

(iii) Since all real numbers between 0 and 1 including 0 and 1 are limit points of $[0, 1]^d = [0, 1]$.

(iv) Since all real numbers between 0 and 1 including 0 and 1 are limit points of $]0, 1[^d = [0, 1]$.

(v) Since the elements of the set $\left\{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{100}\right\}$ have gaps, so no real numbers is its limit points.

$$\text{Thus, } \left\{\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots, \frac{1}{100}\right\}^d = \phi$$

Q.10

If A and B are subsets of a topological space X then justify whether

- (i) $\text{Ext}(A \cup B) = \text{Ext}(A) \cap \text{Ext}(B)$.
- (ii) $\text{Ext}(A) \cup \text{Ext}(B) \subseteq \text{Ext}(A \cap B)$.

PU, 2003 (M.Sc. Math)

Sol:(i) Let

$$\begin{aligned} x &\in \text{Ext}(A \cup B) \\ &\Rightarrow x \in \text{Int}(A \cup B)^c \\ &\Rightarrow x \in \text{Int}(A^c \cap B^c) \\ &\Rightarrow x \in \text{Int}(A^c) \cap \text{Int}(B^c) \\ &\Rightarrow x \in \text{Ext}(A) \cap \text{Ext}(B) \\ &\Rightarrow \text{Ext}(A \cup B) \subseteq \text{Ext}(A) \cap \text{Ext}(B) \end{aligned}$$

Conversely, let

$$\begin{aligned} x &\in \text{Ext}(A) \cap \text{Ext}(B) \\ &\Rightarrow x \in \text{Int}(A^c) \cap \text{Int}(B^c) \\ &\Rightarrow x \in \text{Int}(A^c \cap B^c) \\ &\Rightarrow x \in \text{Int}(A \cup B)^c \\ &\Rightarrow x \in \text{Ext}(A \cup B) \end{aligned}$$

$$\Rightarrow \text{Ext}(A) \cap \text{Ext}(B) \subseteq \text{Ext}(A \cup B)$$

Hence $\text{Ext}(A \cup B) = \text{Ext}(A) \cap \text{Ext}(B)$.

- (ii) Let $x \in \text{Ext}(A) \cup \text{Ext}(B)$, then $x \in \text{Int}(A^c) \cup \text{Int}(B^c)$
 $\Rightarrow x \in \text{Int}(A^c \cup B^c) \quad \because \text{Int}(A^c) \cup \text{Int}(B^c) \subseteq \text{Int}(A^c \cup B^c)$
 $\Rightarrow x \in \text{Int}(A \cap B)^c$
 $\Rightarrow x \in \text{Ext}(A \cap B)$
 $\Rightarrow \text{Ext}(A) \cup \text{Ext}(B) \subseteq \text{Ext}(A \cap B)$

Q.11 Show that a subset A of a topological space (X, τ) has empty frontier if and only if A is both open and closed.

Sol: Let A be a subset of a topological space X . Let A have empty boundary, i.e. PU, 2013 (BS Math)

$$F_r(A) = \phi$$

$$\Rightarrow \bar{A} \cap \bar{A}^c = \phi$$

$$\Rightarrow \bar{A} \subseteq (\bar{A}^c)^c \quad \dots(1)$$

and

$$\bar{A}^c \subseteq (\bar{A})^c \quad \dots(2)$$

Since $A^\circ = (\bar{A}^c)^c$, so (1) takes the form

$$\bar{A} \subseteq (\bar{A}^c)^c = A^\circ$$

$$\Rightarrow \bar{A} \subseteq A^\circ$$

$$\Rightarrow \bar{A} \subseteq A \quad \because A^\circ \subseteq A$$

But $A \subseteq \bar{A}$, so $\bar{A} = A$. This shows that A is closed.

Since $(A^c)^\circ = (\bar{A})^c$, so (2) takes the form

$$\bar{A}^c \subseteq (\bar{A})^c = (A^c)^\circ$$

$$\Rightarrow \bar{A}^c \subseteq (A^c)^\circ$$

$$\Rightarrow \bar{A}^c \subseteq A^c \quad \because (A^c)^\circ \subseteq A^c$$

But $A^c \subseteq \bar{A}^c$, so $\bar{A}^c = A^c$. This shows that A^c is closed, so A is open. Hence, A is both open and closed.

Conversely, let A both open and closed, then

$$F_r(A) = \bar{A} \cap \bar{A}^c = A \cap A^c = \phi \quad \because \bar{A} = A, \bar{A}^c = A^c$$

Q.12 Let (X, τ_d) be a discrete topological space and $A \subseteq X$. Find the derived set of A . Also show that

$$\bar{A} = \{x \in X : A \cap U_x \neq \phi, \text{ for all opensets } U_x \text{ containing } x\}$$

PU, 2008 (M.Sc. Math)

Sol: Since (X, τ_d) is a discrete topological space, so for any $x \in X$, $\{x\}$ is an open set. Clearly, this open set $\{x\}$ does not contain any point of A different from x , so x is an isolated point of A , i.e. x is not a limit point of A . This shows that no point of X is the limit point of A , so $A^d = \phi$.

Let $B = \{x \in X : A \cap U_x \neq \phi, \text{ for all open sets } U_x \text{ containing } x\}$, then we have to show that $\bar{A} = B$. For this, let $x \in \bar{A}$. Further suppose that $x \notin B$, then there exists an open set U_x containing x such that $A \cap U_x = \phi$, then

$$A \subseteq U_x^c \quad \dots(1)$$

Since U_x is open, so U_x^c is closed and from (1) it is clear that U_x^c is a closed superset of A . But $\bar{A}$ is the smallest closed superset of A , so

$$\begin{aligned} \bar{A} &\subseteq U_x^c \\ \Rightarrow \bar{A} \cap U_x &= \phi \end{aligned}$$

which is a contradiction, because x is in both $\bar{A}$ and U_x . Hence, $x \in B$, and so $\bar{A} \subseteq B$.

Conversely, let $x \in B$, then each open set containing x intersects A . Let $x \notin \bar{A}$, then $x \in (\bar{A})^c$. Since $\bar{A}$ is closed, so $(\bar{A})^c$ is an open set containing x .

Since
$$\bar{A} \cap (\bar{A})^c = \phi$$

so
$$A \cap (\bar{A})^c = \phi \quad (\because A \subset \bar{A})$$

Thus, the open set $(\bar{A})^c$ containing x does not intersect A , which is a contradiction, so

$$\begin{aligned} x &\in \bar{A} \\ \Rightarrow B &\subseteq \bar{A} \end{aligned}$$

This shows that $\bar{A} = B$, that is

$$\bar{A} = \{x \in X : A \cap U_x \neq \phi, \text{ for all open sets } U_x \text{ containing } x\}$$

Q.13 Let (R, τ_c) be the cofinite topological space and $A \subseteq R$. Show that A^d is always a closed set.

Sol: Case-I:

PU, 2008 (M.Sc. Math)

If A is finite, then by the definition of cofinite topological space, it is closed, so it contains all of its limit points. Then derived set of A is a subset of A , i.e.

$$A^d \subseteq A$$

Since A is finite, so A^d , being the subset of a finite set, is finite. Consequently, A^d , being the finite subset of a cofinite topological space, is closed.

Case-II:

If A is infinite, then for any $x, y \in X$, $y \neq x$, and $U = X - \{y\}$

$$U \cap A \neq \phi$$

Since $U^c = \{y\}$ = finite, so $U = X - \{y\}$ is an open set containing x , because $y \neq x$.

Since A is infinite, and U contains all points of X except y , so

$$(U \cap A) \setminus \{x\} \neq \phi$$

This shows that x is a limit point of A , i.e. $x \in A^d$.

Similarly, we can show that each point of X is in A^d , so

$$A^d = X$$

This shows that A^d is closed.

Q.14 Define limit point of a set A in a topological space.

Let A be a subset of (X, τ) and $\bar{A}$ be the closure of A and A^d is the derived set of A , then show that

$$(i) \quad \bar{A} = A \cup A^d$$

$$(ii) \quad A \text{ is closed if and only if } A \supset A^d.$$

Sol:(i) Let $x \in \bar{A}$. Further suppose that $x \notin A \cup A^d$, then x is neither in A nor a limit point of A , so there exists an open set U containing x such that

$$U \cap A = \phi$$

$$\Rightarrow A \subseteq U^c$$

...(1)

Since U is open, so U^c is closed and from (1) it is clear that U^c is a closed superset of A . But $\bar{A}$ is the smallest closed superset of A , so

$$\bar{A} \subseteq U^c$$

$$\Rightarrow \bar{A} \cap U = \phi$$

which is a contradiction, because x is in both $\bar{A}$ and U . Hence

$$x \in A \cup A^d$$

$$\Rightarrow \bar{A} \subseteq A \cup A^d$$

...(2)

Conversely, let $x \in A \cup A^d$, then x is either in A or a limit point of A , so in both cases, each open set containing x intersects A .

Let $x \notin \bar{A}$, then $x \in (\bar{A})^c$. Since $\bar{A}$ is closed, so $(\bar{A})^c$ is an open set containing x . Since $\bar{A} \cap (\bar{A})^c = \phi$, so $\bar{A} \cap (\bar{A})^c = \phi$ ($\because A \subseteq \bar{A}$).

Thus, the open set $(\bar{A})^c$ containing x does not intersect A , which is a contradiction, so

$$\begin{aligned} x &\in \bar{A} \\ \Rightarrow A \cup A^d &\subseteq \bar{A} \end{aligned} \quad \dots(3)$$

Combining (2) and (3), we have $\bar{A} = A \cup A^d$.

- (ii) Let A be closed, then A contains all of its limit points, so $A^d \subset A$, i.e. $A \supset A^d$.

Conversely, let $A \supset A^d$, then A contains all of its limit points, so A is closed.

Q.15 If τ_1 and τ_2 are topologies on a set X , show that $\tau_1 \cap \tau_2$ is also a topology but $\tau_1 \cup \tau_2$ need not be a topology.

PU, 2004 (M.Sc. Math)

Sol: Let X be a nonempty set and τ_1, τ_2 be two topologies on X .

First of all we show that $\tau_1 \cap \tau_2$ is also a topology on X .

- (i) Let, for any indexing set I and $\alpha \in I$, $A_\alpha \in \tau_1 \cap \tau_2$.

Then $A_\alpha \in \tau_1$ and $A_\alpha \in \tau_2$

Since τ_1 is a topology, so $A_\alpha \in \tau_1$ gives

$$\bigcup_{\alpha \in I} A_\alpha \in \tau_1 \quad \dots(1)$$

Since τ_2 is a topology, so $A_\alpha \in \tau_2$ gives

$$\bigcup_{\alpha \in I} A_\alpha \in \tau_2 \quad \dots(2)$$

From (1) and (2), we have

$$\bigcup_{\alpha \in I} A_\alpha \in \tau_1 \cap \tau_2$$

This shows that $\tau_1 \cap \tau_2$ is closed under arbitrary unions.

- (ii) Let, for $1 \leq i \leq n$, $A_i \in \tau_1 \cap \tau_2$.

Then $A_i \in \tau_1$ and $A_i \in \tau_2$

Since τ_1 is a topology, so $A_i \in \tau_1$ gives

$$\bigcap_{i=1}^n A_i \in \tau_1 \quad \dots(3)$$

Since τ_2 is a topology, so $A_i \in \tau_2$ gives

$$\bigcap_{i=1}^n A_i \in \tau_2 \quad \dots(4)$$

From (1) and (2), we have

$$\bigcap_{i=1}^n A_i \in \tau_1 \cap \tau_2$$

This shows that $\tau_1 \cap \tau_2$ is closed under finite intersections.

- (iii) Since τ_1 and τ_2 are topologies on X , so

$$\phi, X \in \tau_1 \text{ and } \phi, X \in \tau_2$$

$$\Rightarrow \phi, X \in \tau_1 \cap \tau_2$$

This shows that $\tau_1 \cap \tau_2$ is also a topology on X .

Let $X = \{a, b, c\}$ with topologies

$$\tau_1 = \{\phi, \{a\}, \{b, c\}, X\} \text{ and } \tau_2 = \{\phi, \{a, b\}, X\}$$

Now $\tau_1 \cup \tau_2 = \{\phi, \{a\}, \{b, c\}, \{a, b\}, X\}$

Since $\{b, c\}, \{a, b\} \in \tau_1 \cup \tau_2$ and $\{b, c\} \cap \{a, b\} = \{b\} \notin \tau_1 \cup \tau_2$, so $\tau_1 \cup \tau_2$ is not a topology on X .

This shows that $\tau_1 \cup \tau_2$ need not be a topology.

Q.16 Let (R, τ) be a cofinite topological space. Justify and determine whether the sets $\{\pi, e\}$ and $[0, 1]$ closed in R .

Sol: Since (R, τ) is a cofinite topological space and $\{\pi, e\}$ is a finite subset of R , so $\{\pi, e\}$ is closed in R . PU, 2004 (M.Sc. Math)

Since (R, τ) is a cofinite topological space and $[0, 1]$ is not a finite subset of R , so $[0, 1]$ is not closed in R .

Q.17 Give an example of a cofinite topological space under which each subset is both open and closed. Also show that in a topological space, union of any finite number of closed sets is closed.

Sol: Let X be a finite set and (X, τ_c) be a cofinite topological space. PU, 2004 (M.Sc. Math)
Let A be any subset of X . Since X is finite, so both A and A^c are finite. Since τ_c is a cofinite topology, so both A and A^c , being finite subsets, are closed. Consequently A is both open and closed. Hence, each subset is both open and closed.

Let (X, τ) be a topological space. Let $\{U_i : 1 \leq i \leq n\}$ be any finite collection of closed subsets of X , then each U_i is a closed set, so each U_i^c is a member of τ .

Therefore, $\bigcap_{i=1}^n U_i^c \in \tau$ and so $\left(\bigcap_{i=1}^n U_i^c\right)^c = \bigcup_{i=1}^n U_i$ is closed.

This shows that the union of any finite number of closed sets is closed.

Q.18 Let (X, τ) be a topological space and $A, B \subseteq X$. Show that

(i) A is open if and only if $A^\circ = A$.

(ii) $(A^\circ)^\circ = A^\circ$

(iii) $(A \cup B)^\circ \supseteq A^\circ \cup B^\circ$

Sol:(i) Let A be an open subset of a topological space X , then A , being the largest open subset of A , is itself own interior, i.e. $A = A^\circ$. PU, 2004 (M.Sc. Math)

Conversely, let $A = A^\circ$, then A is open, because A° is open.

(ii) Since A° is an open set, so $(A^\circ)^\circ = A^\circ$.

(iii) Since $A \subseteq A \cup B$, and $B \subseteq A \cup B$, so

$$\begin{aligned} A^\circ &\subseteq (A \cup B)^\circ, B^\circ \subseteq (A \cup B)^\circ \\ \Rightarrow A^\circ \cup B^\circ &\subseteq (A \cup B)^\circ \\ \Rightarrow (A \cup B)^\circ &\supseteq A^\circ \cup B^\circ \end{aligned}$$

Q.19 Give an example to justify that the intersection of countable number of open sets may not be open.

PU, 2003 (M.Sc. Math)

Sol: Let (R, τ) be a usual topological space, then for $n \in \mathbb{N}$, open intervals

$$I_n = \left(-\frac{1}{n}, \frac{1}{n}\right)$$

are open sets in R .

Now

$$\begin{aligned} \bigcap_{n \in \mathbb{N}} I_n &= \bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right) \\ &= (-1, 1) \cap \left(-\frac{1}{2}, \frac{1}{2}\right) \cap \left(-\frac{1}{3}, \frac{1}{3}\right) \cap \dots \\ &= \{0\} \end{aligned}$$

Since $\{0\}$, being the singleton subset of R , is not open. This shows that the intersection of countable number of open sets may not be open.

Q.20 If A is a subset of a topological space X , then show that A is open iff A is the neighbourhood of each of its points.

PU, 2003 (M.Sc. Math)

Sol: Let (X, τ) be a topological space. Let A be an open subset of X . Let x be any arbitrary point of A , then A itself satisfies the condition of the neighbourhood, i.e.

$$x \in A \subseteq A$$

This shows that A is a neighbourhood of x . Since x was an arbitrary point of A , so A is a neighbourhood of each of its points.

Conversely, suppose that A is a neighbourhood of each of its points, then we have to show that A is open. For this, let x be an arbitrary point of A . Since A is a neighbourhood of each of its points, so there exists an open set U_x such that

$$\begin{aligned} x \in U_x &\subseteq A \\ \Rightarrow \bigcup_{x \in A} \{x\} &\subseteq \bigcup_{x \in A} U_x \subseteq A \\ \Rightarrow A &\subseteq \bigcup_{x \in A} U_x \subseteq A \end{aligned}$$

$$\Rightarrow A = \bigcup_{x \in A} U_x$$

Thus, A , being the union of open sets, is open.

Q.21 If A is a subset of a topological space X , then justify that $x \in \bar{A}$ iff for each neighbourhood N of x , $N \cap A \neq \emptyset$.

PU, 2003 (M.Sc. Math)

Sol: Let $x \in \bar{A}$. Suppose on contrary that there exists a neighbourhood N of x such that

$$N \cap A = \emptyset$$

$$\Rightarrow A \subseteq N^c$$

Since N is a neighbourhood of x , so there must exist an open set U such that

$$x \in U \subseteq N$$

$$\Rightarrow N^c \subseteq U^c$$

$$\Rightarrow A \subseteq N^c \subseteq U^c \quad \because A \subseteq N^c$$

$$\Rightarrow A \subseteq U^c$$

Thus, U^c , being the complement of an open set, is a closed superset of A . But the smallest closed superset of A is $\bar{A}$, so

$$\bar{A} \subseteq U^c$$

$$\Rightarrow \bar{A} \cap U = \emptyset$$

which is a contradiction, because x is in both $\bar{A}$ and U . This shows that $N \cap A \neq \emptyset$.

Conversely, let for each neighbourhood N of x , $N \cap A \neq \emptyset$. If possible, let $x \notin \bar{A}$, then $x \in (\bar{A})^c$. Since $(\bar{A})^c$, being the complement of a closed set, is open. Since $(\bar{A})^c$ is an open set containing x , so $(\bar{A})^c$ is a neighbourhood of x . By assumption

$$(\bar{A})^c \cap A \neq \emptyset$$

$$\Rightarrow (\bar{A})^c \cap \bar{A} \neq \emptyset \quad \because A \subseteq \bar{A}$$

Which is not possible, so $x \in \bar{A}$.

Q.22 Give an example to show that an arbitrary intersection of open sets in a topological space is not open.

PU, 2002 (M.Sc. Math)

Sol: Let (R, τ) be a usual topological space, then for $n \in \mathbb{N}$, open intervals

$$I_n = \left(-\frac{1}{n}, \frac{1}{n} \right)$$

are open sets in R .

Now

$$\begin{aligned}
 \bigcap_{n \in \mathbb{N}} I_n &= \bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n} \right) \\
 &= (-1, 1) \cap \left(-\frac{1}{2}, \frac{1}{2} \right) \cap \left(-\frac{1}{3}, \frac{1}{3} \right) \cap \dots \\
 &= \{0\}
 \end{aligned}$$

Since $\{0\}$, being the singleton subset of R , is not open. This shows that an arbitrary intersection of open sets in a topological space is not open.

Q.23 For a subset A of a topological space, show that $\overline{A^c} = (A^\circ)^c$.

PU, 2002 (M.Sc. Math)

Sol: Let $\{V_\alpha : \alpha \in I\}$ be a collection of all open subsets of A , then

$$\begin{aligned}
 A^\circ &= \bigcup_{\alpha \in I} V_\alpha \\
 \Rightarrow (A^\circ)^c &= \left[\bigcup_{\alpha \in I} V_\alpha \right]^c = \bigcap_{\alpha \in I} V_\alpha^c
 \end{aligned}$$

Since each V_α is an open subset of A , so each V_α^c is a closed superset of A^c .

Thus, $\{V_\alpha^c : \alpha \in I\}$ is a collection of all closed supersets of A^c .

Since $\overline{A^c}$, being the smallest closed superset of A^c , is the intersection of all of its closed supersets, so

$$\begin{aligned}
 \overline{A^c} &= \bigcap_{\alpha \in I} V_\alpha^c \\
 \Rightarrow \overline{A^c} &= (A^\circ)^c \quad \because \bigcap_{\alpha \in I} V_\alpha^c = (A^\circ)^c
 \end{aligned}$$

Q.24 Let A be a subset of a topological space X . A point $x \in X$ is in $\overline{A}$ if and only if for every open set U containing x , $U \cap A \neq \emptyset$.

PU, 2002 (M.Sc. Math)

Sol: Let $x \in \overline{A}$. Suppose on contrary that there exists an open set U containing x such that

$$\begin{aligned}
 U \cap A &= \emptyset \\
 \Rightarrow A &\subseteq U^c
 \end{aligned}$$

Thus, U^c , being the complement of an open set, is a closed superset of A . But the smallest closed superset of A is $\overline{A}$, so

$$\begin{aligned}
 \overline{A} &\subseteq U^c \\
 \Rightarrow \overline{A} \cap U &= \emptyset
 \end{aligned}$$

which is a contradiction, because x is in both $\overline{A}$ and U . This shows that $U \cap A \neq \emptyset$.

Conversely, let for every open set U containing x ,

$$U \cap A \neq \emptyset$$

If possible, let $x \notin \bar{A}$, then $x \in (\bar{A})^c$. Since $(\bar{A})^c$, being the complement of a closed set, is open. By assumption

$$\begin{aligned} (\bar{A})^c \cap A &\neq \phi \\ \Rightarrow (\bar{A})^c \cap \bar{A} &\neq \phi \quad \because A \subseteq \bar{A} \end{aligned}$$

Which is not possible, so $x \in \bar{A}$.

Q.25 Let A be a subset of (X, τ) . Then prove that

(i) $\bar{A} = A \cup F_r(A)$

(ii) $A^\circ = A \setminus F_r(A)$

PU, 2001 (M.Sc. Math)

Sol:(i) $X = A \cup A^c \subseteq A \cup \bar{A}^c \quad \because A^c \subseteq A \cup \bar{A}^c$

But $A \cup \bar{A}^c \subseteq X$

Therefore, $X = A \cup \bar{A}^c \quad \dots(1)$

$$\begin{aligned} A \cup F_r(A) &= A \cup (\bar{A} \cap \bar{A}^c) \\ &= (A \cup \bar{A}) \cap (A \cup \bar{A}^c) \\ &= \bar{A} \cap (A \cup \bar{A}^c) \\ &= \bar{A} \cap X = \bar{A} \quad (\text{using (1)}) \end{aligned}$$

(ii) Let $x \in A^\circ$, the x is an interior point of A , so by the definition of an interior point, there is an open set U such that

$$x \in U \subseteq A$$

This shows that

$$U \cap A^c = \phi$$

This shows that x is not a boundary point of A , i.e.

$$x \notin F_r(A)$$

Hence

$$\begin{aligned} x &\in A \text{ and } x \notin F_r(A) \\ \Rightarrow x &\in A \setminus F_r(A) \\ \Rightarrow A^\circ &\subseteq A \setminus F_r(A) \quad \dots(1) \end{aligned}$$

Conversely, let

$$\begin{aligned} y &\in A \setminus F_r(A) \\ y &\in A \text{ and } y \notin F_r(A) \end{aligned}$$

This shows that there must exist at least one open set V containing y such that

$$\begin{aligned} V \cap A^c &= \phi \\ \Rightarrow V &\subseteq A \\ \Rightarrow y &\in V \subseteq A \quad \because y \in V \end{aligned}$$

$$\Rightarrow y \in A^\circ$$

$$\Rightarrow A \setminus F_r(A) \subseteq A^\circ \quad \dots(2)$$

From (1) and (2), we have

$$A^\circ = A \setminus F_r(A)$$

Q.26 If A and B are two subsets of a topological space (X, τ) , then show that

$$(i) \quad \text{Ext}(A) = (\overline{A})^c$$

$$(ii) \quad \text{Ext}(A) \cup \text{Ext}(B) \subseteq \text{Ext}(A \cap B)$$

$$(iii) \quad \text{Ext}(A \cup B) = \text{Ext}(A) \cap \text{Ext}(B)$$

PU, 2001 (M.Sc. Math)

Sol:(i) We know that

$$\text{Ext}(A) = \text{Int}(A^c) \quad \dots(1)$$

Let $\{V_\alpha : \alpha \in I\}$ be a collection of all open subsets of A^c , then

$$\text{Int}(A^c) = \bigcup_{\alpha \in I} V_\alpha$$

$$\Rightarrow [\text{Int}(A^c)]^c = [\bigcup_{\alpha \in I} V_\alpha]^c = \bigcap_{\alpha \in I} V_\alpha^c$$

Since each V_α is an open subset of A^c , so each V_α^c is a closed superset of A .

Thus, $\{V_\alpha^c : \alpha \in I\}$ is a collection of all closed supersets of A . Since $\overline{A}$, being the smallest closed superset of A , is the intersection of all of its closed supersets, so

$$\overline{A} = \bigcap_{\alpha \in I} V_\alpha^c$$

$$\Rightarrow \overline{A} = [\text{Int}(A^c)]^c \quad \because [\text{Int}(A^c)]^c = \bigcap_{\alpha \in I} V_\alpha^c$$

Taking complements on both sides, we have

$$\text{Int}(A^c) = (\overline{A})^c \quad \dots(2)$$

Comparing (1) and (2), we have

$$\text{Ext}(A) = (\overline{A})^c$$

(ii) Let $x \in \text{Ext}(A) \cup \text{Ext}(B)$,

then $x \in \text{Int}(A^c) \cup \text{Int}(B^c)$

$$\Rightarrow x \in \text{Int}(A^c \cup B^c) \quad \because \text{Int}(A^c) \cup \text{Int}(B^c) \subseteq \text{Int}(A^c \cup B^c)$$

$$\Rightarrow x \in \text{Int}(A \cap B)^c$$

$$\Rightarrow x \in \text{Ext}(A \cap B)$$

$$\Rightarrow \text{Ext}(A) \cup \text{Ext}(B) \subseteq \text{Ext}(A \cap B)$$

(iii) Let

$$x \in \text{Ext}(A \cup B)$$

$$\Rightarrow x \in \text{Int}(A \cup B)^c$$

$$\Rightarrow x \in \text{Int}(A^c \cap B^c)$$

$$\Rightarrow x \in \text{Int}(A^c) \cap \text{Int}(B^c)$$

$$\Rightarrow x \in \text{Ext}(A) \cap \text{Ext}(B)$$

$$\Rightarrow \text{Ext}(A \cup B) \subseteq \text{Ext}(A) \cap \text{Ext}(B) \quad \dots(1)$$

Conversely, let

$$x \in \text{Ext}(A) \cap \text{Ext}(B)$$

$$\Rightarrow x \in \text{Int}(A^c) \cap \text{Int}(B^c)$$

$$\Rightarrow x \in \text{Int}(A^c \cap B^c)$$

$$\Rightarrow x \in \text{Int}(A \cup B)^c$$

$$\Rightarrow x \in \text{Ext}(A \cup B)$$

$$\Rightarrow \text{Ext}(A) \cap \text{Ext}(B) \subseteq \text{Ext}(A \cup B) \quad \dots(2)$$

Combining (1) and (2), we have

$$\text{Ext}(A \cup B) = \text{Ext}(A) \cap \text{Ext}(B)$$

Q.27 Define subspace. Prove that the set Z of integers as subspace of R has discrete topology.

Sol: Subspace: Let (X, τ_X) be a topological space and $Y \subset X$ a nonempty subset, then

$$\tau_Y = \{V \cap Y : V \in \tau_X\}$$

is a topology on Y , called the topology induced by τ_X on Y or relative topology on Y . The pair (Y, τ_Y) is called the subspace of X .

Let $n \in Z$, then $(n-1, n+1)$, being an open interval, an open set in R , so by the definition of relative topology,

$$(n-1, n+1) \cap Z = \{n\}$$

is an open set. This shows that each singleton subset of Z is an open set in Z .

Let A be any subset of Z , then A , being the union of singleton subsets of Z , is an open subset of Z . This shows that every subset of Z is an open set, so the set Z of integers as subspace of R has discrete topology.

Q.28 Let X be a cofinite topological space. Show that A^d is closed for any subset A of X .

Sol: Case-I:

If A is finite, then by the definition of cofinite topological space, it is closed, so it contains all of its limit points. Then derived set of A is a subset of A , i.e.

$$A^d \subseteq A$$

Since A is finite, so A^d , being the subset of a finite set, is finite. Consequently, A^d , being the finite subset of a cofinite topological

space, is closed.

Case-II:

If A is infinite, then for any $x, y \in X$, $y \neq x$, and $U = X - \{y\}$

$$U \cap A \neq \emptyset$$

Since

$$U^c = \{y\} = \text{finite}$$

therefore,

$$U = X - \{y\}$$

is an open set containing x , because $y \neq x$.

Since A is infinite, and U contains all points of X except y , so

$$(U \cap A) \setminus \{x\} \neq \emptyset$$

This shows that x is a limit point of A , i.e. $x \in A^d$.

Similarly, we can show that each point of X is in A^d , so

$$A^d = X$$

This shows that A^d is closed.

Chapter 2

CONTINUITY AND COUNTABILITY

EXERCISE 2

Short Questions

Q.2 Solve / answer the following short questions:

(i) Define continuous function at a point in topological spaces.

Sol: Let (X, τ_X) and (Y, τ_Y) be topological spaces. A function $f: X \rightarrow Y$ is said to be a continuous function at a point x_0 of X if for any neighbourhood N_Y of $f(x_0)$ in Y , there is a neighbourhood N_X of x_0 in X such that

$$f(N_X) \subseteq N_Y$$

(ii) Define continuous function on a topological space.

Sol: Let (X, τ_X) and (Y, τ_Y) be topological spaces. A function $f: X \rightarrow Y$ is said to be a continuous function at a point x_0 of X if for any neighbourhood N_Y of $f(x_0)$ in Y , there is a neighbourhood N_X of x_0 in X such that

$$f(N_X) \subseteq N_Y$$

The function $f: X \rightarrow Y$ is said to be a continuous function on X if it is continuous at each point of X .

(iii) If X is an arbitrary topological space and Y is an indiscrete topological space, then show that every function $f: X \rightarrow Y$ is a continuous function on X .

Sol: Since Y is an indiscrete topological space, so the only possible open subsets of Y are $\emptyset$ and Y itself, then $f^{-1}(\emptyset) = \emptyset$ and $f^{-1}(Y) = X$ both are open subsets of X , so f is continuous.

(iv) If X is a discrete topological space and Y is an arbitrary

topological space, then every function $f: X \rightarrow Y$ is a continuous function on X .

Sol: Let v be an open subset of Y , then $f^{-1}(v)$ is a subset of X . Since X is a discrete topological space, so its each subset will be an open set.

Thus, $f^{-1}(v)$ is an open subset of X .

This shows that f is continuous on X .

(v) **Define an open mapping in topological spaces.**

Sol: A mapping f from one topological space X into another topological space Y is said to be an open mapping if for every open set G in X , $f(G)$ is open in Y .

(vi) **Give an example of mapping in topological spaces which is continuous but not open.**

Sol: For example, let

$$X = \{x, y, z\}, \quad \tau_X = \{\emptyset, \{x\}, \{x, y\}, \{y, z\}, X\}$$

$$\text{and } Y = \{1, 2, 3\}, \quad \tau_Y = \{\emptyset, \{1\}, Y\}, \text{ then}$$

$f: X \rightarrow Y$ defined by $f(x) = 2, f(y) = 1, f(z) = 3$ is continuous but not open, because $A = \{x, y\}$ is open in X but $f(A) = \{1, 2\}$ is not open in Y .

(vii) **Define a closed mapping in topological spaces.**

Sol: A mapping f from one topological space X into another topological space Y is said to be a closed mapping if for every closed set F in X , $f(F)$ is closed in Y .

(viii) **Let X and Y be topological spaces. If a bijective mapping $f: X \rightarrow Y$ is open, then show that $f^{-1}: Y \rightarrow X$ is continuous.**

Sol: Let $f: X \rightarrow Y$ be an open mapping, then we have to show that $f^{-1}: Y \rightarrow X$ is continuous. For this, let G be an open subset of X , then $f(G)$ is open in Y , because f is open. Since $(f^{-1})^{-1}(G) = f(G)$, so $(f^{-1})^{-1}(G)$ is open in Y . This shows that $f^{-1}: Y \rightarrow X$ is continuous.

(ix) **Let X and Y be topological spaces. If a bijective mapping $f^{-1}: Y \rightarrow X$ is continuous, then show that $f: X \rightarrow Y$ is open.**

Sol: Let $f^{-1}: Y \rightarrow X$ be continuous, then we have to show that $f: X \rightarrow Y$ is open. For this let G be an open subset of X , then $(f^{-1})^{-1}(G) = f(G)$ is open in Y , because f^{-1} is continuous. This shows that $f: X \rightarrow Y$ is open.

(x) **Let X and Y be topological spaces. If a bijective mapping $f: X \rightarrow Y$ is closed, then show that $f^{-1}: Y \rightarrow X$ is continuous.**

Sol: Let $f: X \rightarrow Y$ be a closed mapping, then we have to show that $f^{-1}: Y \rightarrow X$ is continuous. For this, let F be a closed subset of X , then $f(F)$ is closed in Y , because f is closed.

Since $(f^{-1})^{-1}(f(F)) = f(F)$, so $(f^{-1})^{-1}(f(F))$ is closed in X .

This shows that $f^{-1}: Y \rightarrow X$ is continuous.

(xi) Let X and Y be topological spaces. If a bijective mapping $f^{-1}: Y \rightarrow X$ is continuous, then show that $f: X \rightarrow Y$ is closed.

Sol: Let $f^{-1}: Y \rightarrow X$ be continuous, then we have to show that $f: X \rightarrow Y$ is closed. For this let F be a closed subset of X , then $(f^{-1})^{-1}(f(F)) = f(F)$ is closed in Y , because f^{-1} is continuous. This shows that $f: X \rightarrow Y$ is closed.

(xii) Let X and Y be topological spaces. If $f: X \rightarrow Y$ is an open mapping, then show that $f(A^\circ) = [f(A)]^\circ$ for every subset A of X .

Sol: Let $f: X \rightarrow Y$ be an open mapping. Let A be any subset of X , then A° is an open subset of X , so by the definition of open mapping, $f(A^\circ)$ is an open subset of Y .

Since $A^\circ \subset A$, so $f(A^\circ) \subset f(A)$.

This shows that $f(A^\circ)$ is an open subset of $f(A)$.

But $[f(A)]^\circ$ is the largest open subset of $f(A)$, so $f(A^\circ) \subset [f(A)]^\circ$.

(xiii) Let X and Y be topological spaces. If $f(A^\circ) = [f(A)]^\circ$ for every subset A of X , then show that the mapping $f: X \rightarrow Y$ is open.

Sol: Let $f(A^\circ) = [f(A)]^\circ$ for every subset A of X , then we have to show that f is open. For this, let A be an open subset of X . Then, by the given condition,

$$f(A^\circ) = [f(A)]^\circ$$

$$f(A) = [f(A)]^\circ \because A^\circ \subset A \quad \dots(1)$$

$$\text{But} \quad [f(A)]^\circ \subset f(A) \quad \dots(2)$$

Combining (1) and (2), we have

$$[f(A)]^\circ = f(A)$$

This shows that $f(A)$ is open, so f is an open mapping.

(xiv) Let X and Y be topological spaces. If a mapping $f: X \rightarrow Y$ is open, then show that $[f(A)]^\circ = f(A)$ for every open subset A of X .

Sol: Let $f: X \rightarrow Y$ be an open mapping. Let A be an open subset of X ,

then, by the definition of open mapping, $f(A)$ is an open subset of Y and so

$$f(A)^\circ \subseteq f(A)$$

- (xv) Let X and Y be topological spaces. If $[f(A)]^\circ \subseteq f(A)$ for every open subset A of X , then show that the mapping $f: X \rightarrow Y$ is open.

Sol: Let $[f(A)]^\circ \subseteq f(A)$ for every open subset A of X , then we have to show that f is open. For this, let A be an open subset of X . Then, by the given condition

$$[f(A)]^\circ \subseteq f(A)$$

This shows that $f(A)$ is open, so f is an open mapping.

- (xvi) Let X and Y be topological spaces. If a mapping $f: X \rightarrow Y$ is closed, then show that $\overline{f(A)} \subseteq f(\overline{A})$ for every subset A of X .

Sol: Let $f: X \rightarrow Y$ be a closed mapping. Let A be any subset of X , then $\overline{A}$ is a closed subset of X , so by the definition of closed mapping, $f(\overline{A})$ is a closed subset of Y . Since $A \subseteq \overline{A}$, so $f(A) \subseteq f(\overline{A})$. This shows that $f(\overline{A})$ is a closed superset of $f(A)$. But $\overline{f(A)}$ is the smallest closed superset of $f(A)$, so

$$\overline{f(A)} \subseteq f(\overline{A})$$

- (xvii) Let X and Y be topological spaces. If $\overline{f(A)} \subseteq f(\overline{A})$ for every subset A of X , then show that the mapping $f: X \rightarrow Y$ is closed.

Sol: Let $\overline{f(A)} \subseteq f(\overline{A})$ for every subset A of X , then we have to show that f is closed. For this, let A be a closed subset of X . Then, by the given condition,

$$\overline{f(A)} \subseteq f(\overline{A})$$

$$\overline{f(A)} \subseteq f(A) \quad \because \overline{A} = A \quad \dots(1)$$

But

$$f(A) \subseteq \overline{f(A)} \quad \dots(2)$$

Combining (1) and (2), we have $\overline{f(A)} = f(A)$.

This shows that $f(A)$ is closed, so f is a closed mapping.

- (xviii) Let X and Y be topological spaces. If a mapping $f: X \rightarrow Y$ is closed, then show that $\overline{f(A)} \subseteq f(\overline{A})$ for every closed subset A of X .

Sol: Let $f: X \rightarrow Y$ be a closed mapping. Let A be a closed subset of X , then, by the definition of closed mapping, $f(A)$ is a closed subset

of Y and so

$$\overline{f(A)} = f(A)$$

- (xix) Let X and Y be topological spaces. If $\overline{f(A)} = f(A)$ for every closed subset A of X , then show that the mapping $f: X \rightarrow Y$ is closed.

Sol: Let $\overline{f(A)} = f(A)$ for every closed subset A of X , then we have to show that f is closed. For this, let A be a closed subset of X . Then, by the given condition, $\overline{f(A)} = f(A)$. This shows that $f(A)$ is closed, so f is a closed mapping.

- (xx) Give an example of a real function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that f is continuous and closed, but not open.

Sol: Let f be constant function, say

$$f(x) = 2 \text{ for all } x \in \mathbb{R}$$

For any subset A of $\mathbb{R}$, $f(A) = \{2\}$.

Since $\{2\}$ is a closed subset of $\mathbb{R}$, so the function f is closed function.

Since $\{2\}$ is not an open subset of $\mathbb{R}$, so the function f is not open function.

Furthermore, f , being a constant function, is continuous function.

Q.3 Solve / answer the following short questions:

- (i) Show that the real function $f: \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$ is not open.

Sol: $U = (-1, 1)$ be an open set, then $f(U) = [0, 1)$, which is not open; hence f is not an open function.

- (ii) Define homeomorphism.

Sol: Let X and Y be topological spaces. A function $f: X \rightarrow Y$ is said to be homeomorphism if

- (i) f is bijective
- (ii) f is continuous
- (iii) f^{-1} is continuous

- (iii) Define homeomorphic spaces.

Sol: Two topological spaces X and Y are said to be homeomorphic if there exists a homeomorphism of X onto Y and in this case, Y is called a homeomorphic image of X . If X and Y are homeomorphic topological spaces then we write $X \sim Y$.

- (iv) If X and Y are topological spaces, let $X \sim Y$ mean that X and Y are homeomorphic. Show that this relation is reflexive.

Sol: Let us consider the identity mapping $I: X \rightarrow X$ defined by

$$I(x) = x \text{ for all } x \in X$$

Then obviously I is bijective. Let V be any open subset of X , then

$I^{-1}(V)$ V is open in X , so I is continuous. Let U be an open subset of X , then $I(U)$ U is open in X , so I is open. Thus, I , being bijective, continuous, and open, is a homeomorphism.

This shows that $X \sim X$, i.e., $\sim$ is reflexive.

- (v) **If X and Y are topological spaces, let $X \sim Y$ mean that X and Y are homeomorphic. Show that this relation is symmetric.**

Sol: Let $X \sim Y$, then there exists a homeomorphism $f: X \rightarrow Y$.

Now $f^{-1}: Y \rightarrow X$ is bijective, because $f: X \rightarrow Y$, being a homeomorphism, is bijective.

$f^{-1}: Y \rightarrow X$ is continuous, because $f: X \rightarrow Y$, being a homeomorphism, is open.

Now $f^{-1}: Y \rightarrow X$ is open, because $f: X \rightarrow Y$, being a homeomorphism, is continuous. This shows that $Y \sim X$, i.e., $\sim$ is symmetric.

- (vi) **If X and Y are topological spaces, let $X \sim Y$ mean that X and Y are homeomorphic. Show that this relation is transitive.**

Sol: Let $X \sim Y$ and $Y \sim Z$, then there exist homeomorphisms $f: X \rightarrow Y$ and $g: Y \rightarrow Z$, i.e., both f and g are bijective, continuous, and open mappings.

Since the composition of two bijective, continuous, and open mappings are bijective, continuous, and open respectively, so $g \circ f: X \rightarrow Z$ is bijective, continuous, and open, i.e., $g \circ f: X \rightarrow Z$ is a homeomorphism.

This shows that $X \sim Z$, i.e., $\sim$ is transitive.

- (vii) **Give an example to show that a one-to-one continuous mapping need not be homeomorphism.**

Sol: Let (X, τ_X) and (Y, τ_Y) be discrete and indiscrete topological spaces respectively. Let $f: X \rightarrow Y$ be a bijective continuous mapping. Let U be an open subset of X , but then $f(U)$ need not be an open subset of Y , because Y is an indiscrete topological space. In fact, every subset of X is open and its image need not be open in Y . Thus, f is not an open mapping. This shows that f is not a homeomorphism.

- (viii) **Show that the real line R and the open interval $(-2, 2)$ are homeomorphic.**

Sol: Let $X = (-2, 2)$ and consider the function $f: X \rightarrow R$ defined by

$$f(x) = \tan \frac{x}{4} \quad \dots(1)$$

Obviously, the function f is bijective and continuous. The function f^{-1} is also continuous, so the real line R and the open interval

- (ix) **Show that the closed interval $A = [2, 3]$ is homeomorphic to the closed unit interval $I = [0, 1]$.**

Sol: Let us consider the function $f : I \rightarrow A$ defined by

$$f(x) = [3 - (2)]x + (2) \\ f(x) = 5x - 2$$

The function f , being a linear function, is bijective and continuous. The function f^{-1} is also continuous, so the function f is homeomorphism.

This shows that closed interval $A = [2, 3]$ is homeomorphic to the closed unit interval $I = [0, 1]$.

- (x) **Show that the intervals $(0, 1)$ and (a, b) are homeomorphic.**

Sol: Let us consider the function $f : I \rightarrow A$ defined by

$$f(x) = (b - a)x + a \quad \dots(1)$$

The function f , being a linear function, is bijective and continuous. The function f^{-1} is also continuous, so the function f is homeomorphism.

This shows that the intervals $(0, 1)$ and (a, b) are homeomorphic.

- (xi) **Define an open base in a topological space.**

Sol: Let (X, τ) be a topological space. A sub-collection of τ is said to be an open base for X if each set in τ is a union of members of . The member of an open base is called the basic open set.

- (xii) **Let $X = \{a, b, c\}$. Show that $\{\{a, b\}, \{b, c\}\}$ cannot be a base for any topology on X .**

Sol: If $\{\{a, b\}, \{b, c\}\}$ is a base, then $\{a, b\}, \{b, c\}$ are open sets, so

$$\{a, b\} \cup \{b, c\} = \{a, b, c\}$$

is also an open set. But $\{b\}$ is not the union of members of $\{\{a, b\}, \{b, c\}\}$, so cannot be a base for any topology on X .

- (xiii) **Define relative open base.**

Sol: If (Y, τ_Y) is a subspace of a topological space (X, τ_X) and $\mathcal{B}$ is an open base for X , then $\mathcal{B}_Y$ is called the *relative open base* for Y or *relative base* for the relative topology τ_Y of Y . In set builder form, we write it as

$$\mathcal{B}_Y = \{B \cap Y : B \in \mathcal{B}\}$$

- (xiv) **Define an open subbase.**

Sol: Let (X, τ) be a topological space. A collection S of subsets of X is said to be an open subbase for X or a subbase for topology τ if all finite intersections of members of S forms a base for τ .

- (xv) Show that the class of all infinite open intervals is a subbase for R .

Sol: Since the class of open intervals form an open base for the usual topology on the real line R , so

$$\{(a,b) : a, b \in R, a < b\}$$

is a base for R .

Since every open interval (a,b) in the real line R is the intersection of two of two infinite intervals $(-\infty, b)$ and (a, ∞) , i.e.

$$(-\infty, b) \cap (a, \infty) = (a,b)$$

This shows that the class of all infinite open intervals is a subbase for R .

- (xvi) Define a local base at a point.

Sol: Let (X, τ) be a topological space and $x \in X$. A sub-collection of τ is said to be a neighbourhood base at x or local base at x or simply a base at a point x if, for any open set U containing x , there is a B such that $x \in B \subseteq U$.

- (xvii) Show that every discrete topological space has a countable neighbourhood base at each of its points.

Sol: Let (X, τ) be a discrete topological space and $x \in X$ be an arbitrary point of X . Since τ is a discrete topology on X , so each subset of X is an open set. Thus, $\{x\}$, being a subset of X , is open in X . The collection $\{\{x\}\}$ is a neighbourhood base at x , because for any open set U containing x , we have a member $\{x\}$ of $\{\{x\}\}$ such that $x \in \{x\} \subseteq U$. Obviously $\{\{x\}\}$ is countable. Since x is an arbitrary point of X . This shows that X has a countable neighbourhood base at each of its points.

Long Questions

- Q.4 Let X and Y be topological spaces. Prove that a function $f: X \rightarrow Y$ is continuous on X if and only if for each open subset V of Y , $f^{-1}(V)$ is open in X .

PU, 2013; 2008 (M.Sc. Math)

Sol: Let $f: X \rightarrow Y$ be a continuous function on X . Let V be an open subset of Y , then we have to show that $f^{-1}(V)$ is open in X . For this, let $x \in f^{-1}(V)$, then $f(x) \in V$, and V , being an open set, is a neighbourhood of $f(x)$, so by the definition of continuity at x , there exists an open subset U_x of X containing x such that

$$\begin{aligned} f(U_x) &\subseteq V \\ U_x &\subseteq f^{-1}(V) \end{aligned} \quad \dots(1)$$

Since $x \in U_x$, so using this in (1), we have

$$\begin{aligned} x &\in U_x \cap f^{-1}(V) \\ U_x \cap f^{-1}(V) &\cap \{x\} = U_x \cap f^{-1}(V) \cap U_x \cap f^{-1}(V) \\ f^{-1}(V) &\cap U_x \cap f^{-1}(V) \cap U_x \cap f^{-1}(V) \\ f^{-1}(V) &\cap U_x \cap f^{-1}(V) \cap U_x \end{aligned}$$

This shows that $f^{-1}(V)$, being the union of open sets, is an open set in X .

Conversely, let for every open set V in Y , $f^{-1}(V)$ be open in X , then we have to show that f is continuous. For this let x be an arbitrary point of X and N_Y a neighbourhood of $f(x)$, then there must exist an open subset V of Y such that

$$f(x) \in V \subset N_Y \quad \dots(2)$$

$$\begin{aligned} f(x) &\in V \\ x &\in f^{-1}(V) \end{aligned} \quad \dots(3)$$

Since V is an open subset of Y , so by given condition, $f^{-1}(V)$ is an open subset of X . Since, by (3), $x \in f^{-1}(V)$, so $f^{-1}(V)$, being an open set, is a neighbourhood of x . Let $N_X = f^{-1}(V)$, then $f(N_X) \subset V$. Using this in (2), we have

$$f(N_X) \subset N_Y$$

This shows that f is continuous at x . Since x was an arbitrary point of X , so f is continuous at each point of x , i.e. f is continuous on X .

Q.5 Show that a function f from one topology space X into another topological space Y is continuous if and only if for every closed set C in Y , $f^{-1}(C)$ is closed in X .

University of Gujrat, 2013; PU, 2006 (M.Sc. Math)

Sol: Let $f: X \rightarrow Y$ be a continuous function on X . Let C be a closed subset of Y , then we have to show that $f^{-1}(C)$ is closed in X . For this it is enough to show that $[f^{-1}(C)]^c$ is open. Since C is a closed subset of Y , so $C^c = Y \setminus C$ is open subset of Y . Let $x \in f^{-1}(C^c)$, then $f(x) \in C^c$, and C^c , being an open set, is a neighbourhood of $f(x)$, so by the definition of continuity at x , there exists an open subset U_x of X containing x such that

$$\begin{aligned} f(U_x) &\subset C^c \\ U_x &\subset f^{-1}(C^c) \end{aligned} \quad \dots(1)$$

Since $x \in U_x$, so using this in (1), we have

$$\begin{aligned}
 & x \in \bigcup_{x \in f^{-1}(C^c)} f^{-1}(C^c) \\
 & \bigcup_{x \in f^{-1}(C^c)} \{x\} \subseteq \bigcup_{x \in f^{-1}(C^c)} U_x \subseteq f^{-1}(C^c) \\
 & f^{-1}(C^c) \subseteq \bigcup_{x \in f^{-1}(C^c)} U_x \subseteq f^{-1}(C^c) \\
 & f^{-1}(C^c) = \bigcup_{x \in f^{-1}(C^c)} U_x
 \end{aligned}$$

This shows that $f^{-1}(C^c)$, being the union of open sets, is an open set in X . Thus,

$$\begin{aligned}
 f^{-1}(C^c) &= f^{-1}(Y \setminus C) \\
 &= f^{-1}(Y) \setminus f^{-1}(C) \\
 &= X \setminus f^{-1}(C)
 \end{aligned}$$

is open in X , i.e., $[f^{-1}(C)]^c$ is open in X , so $f^{-1}(C)$ is closed in X .

Conversely, let for every closed set C in Y , $f^{-1}(C)$ be closed in X , then we have to show that f is continuous. For this let x be an arbitrary point of X and N_Y a neighbourhood of $f(x)$, then there must exist an open subset V of Y such that

$$f(x) \in V \subseteq N_Y \quad \dots(2)$$

$$\begin{aligned}
 & f(x) \in V \\
 & x \in f^{-1}(V) \quad \dots(3)
 \end{aligned}$$

Since V is an open subset of Y , so V^c is a closed subset of Y and, therefore, by given condition $f^{-1}(V^c)$ is closed in X .

Thus

$$\begin{aligned}
 f^{-1}(V^c) &= f^{-1}(Y \setminus V) \\
 &= f^{-1}(Y) \setminus f^{-1}(V) \\
 &= X \setminus f^{-1}(V)
 \end{aligned}$$

is closed in X , so, $f^{-1}(V)$ is an open subset of X . Since, by (3), $x \in f^{-1}(V)$, so $f^{-1}(V)$, being an open set, is a neighbourhood of x .

Let $N_x = f^{-1}(V)$, then $f(N_x) \subseteq V$. Using this in (2), we have

$$f(N_x) \subseteq V \subseteq N_Y$$

This shows that f is continuous at x . Since x was an arbitrary point of X , so f is continuous at each point of X , i.e. f is continuous on X .

Q.6 Prove that a function defined on a topological space (X, τ) is continuous if and only if $f(\overline{A}) \subseteq \overline{f(A)}$.

Sol: University of Sargodha, 2010; PU, 2005; 2003; 2000 (M.Sc. Math)

Let f be continuous on X and A be an arbitrary subset of X , then $f(A)$ is a subset of Y and $\overline{f(A)}$ is a closed subset of Y . Since f is

continuous on X , so for a closed subset $\overline{f(A)}$ of Y , $f^{-1}\overline{f(A)}$ is a closed subset of X .

Since $f(A) \subseteq \overline{f(A)}$

So $A \subseteq f^{-1}\overline{f(A)}$

This shows that $f^{-1}\overline{f(A)}$ is a closed superset of A . But $\overline{A}$ is the smallest closed superset of A , so

$$\overline{A} \subseteq f^{-1}\overline{f(A)}$$

$$f(\overline{A}) \subseteq \overline{f(A)}$$

Conversely, let for any subset A of X , $f(\overline{A}) \subseteq \overline{f(A)}$, then we have to show that f is continuous on X . For this let C be a closed subset of Y , then $f^{-1}(C)$ is a subset of X , so by given condition

$$\overline{f f^{-1}(C)} \subseteq \overline{f f^{-1}(C)}$$

$$\overline{f f^{-1}(C)} \subseteq \overline{C}$$

$$\overline{f f^{-1}(C)} \subseteq C \quad \because C \text{ is closed}$$

$$\overline{f^{-1}(C)} \subseteq f^{-1}(C) \quad \dots(1)$$

$$\text{But } f^{-1}(C) \subseteq \overline{f^{-1}(C)} \quad \dots(2)$$

From (1) and (2), we have

$$\overline{f^{-1}(C)} = f^{-1}(C)$$

This shows that $f^{-1}(C)$ is closed in X . Hence, f is continuous on X .

Q.7 Let $X = \{1,3,5,9,11\}$ and $S = \{\{1,3\}, \{5,9\}\}$ be a class of subsets of X . Determine the topology generated by S . Moreover, if $Y = \{1,3,9\}$ is a subspace of X , then using S determine the subbase for the relative topology τ_Y .

PU, 2004 (M.Sc. Math)

Sol: $X = \{1,3,5,9,11\}$, $S = \{\{1,3\}, \{5,9\}\}$.

Taking all finite intersections of sets in S , we have

$$\begin{aligned} &\{1,3\} \cap \{1,3\} = \{1,3\} \\ &\{5,9\} \cap \{5,9\} = \{5,9\} \\ &\{1,3\} \cap \{5,9\} = \emptyset \\ &\bigcap A = X, \quad A \in S \end{aligned}$$

Note: X is the empty intersection of members of S .

Since τ , being the open base, is the class of all finite intersections of sets in S , so

$$\tau = \{\emptyset, \{1,3\}, \{5,9\}, X\}$$

Next taking the unions of members of τ , we have

$$\begin{array}{ccc} \{1,3\} & \{1,3\} & \{1,3\} \\ \{5,9\} & \{5,9\} & \{5,9\} \\ \{1,3\} & \{5,9\} & \{1,3,5,9\} \end{array}$$

$$X \quad X$$

Since τ_X , being the topology on X , is the class of unions of members of τ_X , so

$$\tau_X = \{ \emptyset, \{1,3\}, \{5,9\}, \{1,3,5,9\}, X \}$$

Which is the topology on X generated by S .

Next taking the intersections of members of S with Y , we have

$$\begin{array}{ccc} \{1,3\} \cap Y & = & \{1,3\} \\ \{5,9\} \cap Y & = & \{5,9\} \\ \{1,3,5,9\} \cap Y & = & \{1,3,9\} \end{array}$$

This shows that the subbase for the relative topology τ_Y determined by S is

$$S_Y = \{\{1,3\}, \{9\}\}$$

Q.8 Let S be a non-empty collection of subsets of X . Suppose that $\bigcup S = X$, then show that S is a subbase for some topology on X .

PU, 2011; 2003 (M.Sc. Math)

Sol: Let $\mathcal{B}$ be a collection of all those subsets of X which are finite intersections of the members of S , i.e.

$$B \in \mathcal{B} \iff B = \bigcap_{i=1}^n S_i, S_i \in S \text{ for some natural number } n$$

Then, in order to show that S is an open subbase, it is enough to show that $\mathcal{B}$ is an open base.

(i) Clearly the union of members of S is a subset of the union of members of $\mathcal{B}$, i.e.

$$\bigcup S \subseteq \bigcup \mathcal{B}$$

Since

$$\bigcup S = X$$

So

$$X \subseteq \bigcup \mathcal{B}$$

But

$$\bigcup \mathcal{B} \subseteq X$$

So

$$\bigcup \mathcal{B} = X$$

(ii) Next suppose that $B_1, B_2 \in \mathcal{B}$ and $x \in B_1 \cap B_2$. Since both B_1 and B_2 are finite intersections of members of S , so $B_1 \cap B_2$ is also the finite intersection of members of S and so $B_1 \cap B_2$ is a member of $\mathcal{B}$.

Thus, for B_1, B_2 and $x \in B_1 \cap B_2$, there is a $B \in \mathcal{B}$ such that

$$x \in B \subseteq B_1 \cap B_2$$

From (i) and (ii) it is clear that $\mathcal{B}$ is a base for some topology on X . Hence $\mathcal{S}$ is an open subbase.

Q.9 Prove that a closed subspace of Lindelöf space is Lindelöf.

University of Gujrat, 2013; PU, 2005; 2003; 2000 (M.Sc. Math)

Sol: Let X be a Lindelöf space. Let Y be a closed subspace of X , then we have to show that Y is also a Lindelöf space. For this, let $\{U_\alpha : \alpha \in I\}$ be an open cover for Y , i.e.

$$Y \subseteq \bigcup_{\alpha \in I} U_\alpha \quad \dots(1)$$

Since each U_α is an open set in Y , so open sets V_α in X are defined as

$$U_\alpha = V_\alpha \cap Y \quad \dots(2)$$

from which we have

$$\begin{aligned} Y &= \bigcup_{\alpha \in I} U_\alpha = \bigcup_{\alpha \in I} (V_\alpha \cap Y) \\ &= \bigcup_{\alpha \in I} V_\alpha \cap Y \end{aligned}$$

Using this in (1), we have

$$Y \subseteq \bigcup_{\alpha \in I} V_\alpha \cap Y \quad \dots(3)$$

This is only possible if

$$Y \subseteq \bigcup_{\alpha \in I} V_\alpha \quad \dots(4)$$

Since $Y^c \subseteq X$, so from (4), we have

$$Y^c \subseteq \bigcup_{\alpha \in I} V_\alpha \cap X$$

This shows that $\{Y^c, V_\alpha : \alpha \in I\}$ is an open cover for X . But X is Lindelöf space, so this open cover has a countable subcover.

Let $\{Y^c, V_n : n \in \mathbb{N}\}$ be a countable subcover for X , i.e.

$$\begin{aligned} X &= Y^c \cup \bigcup_{n=1}^{\infty} V_n \\ Y &= X \cap Y = Y^c \cap Y \cup \bigcup_{n=1}^{\infty} V_n \cap Y \\ &= Y^c \cap Y \cup \bigcup_{n=1}^{\infty} V_n \cap Y \\ &= Y^c \cap Y \cup \bigcup_{n=1}^{\infty} V_n \cap Y \\ &= Y^c \cap Y \cup \bigcup_{n=1}^{\infty} V_n \cap Y \\ &= Y^c \cap Y \cup \bigcup_{n=1}^{\infty} V_n \cap Y \end{aligned} \quad \dots(5)$$

$$(2) \quad \bigcup_{n=1}^{\infty} U_n = Y \cap \bigcup_{n=1}^{\infty} V_n$$

Putting this value in (5), we have

$$Y = \bigcup_{n=1}^{\infty} U_n$$

This shows that $\{U_n : n \in \mathbb{N}\}$ is a countable subcover of $\{U : \dots\}$. Hence Y is a Lindelöf space.

Q.10 Prove that every second countable space is Lindelöf space.

PU, 2012; 2007; 2001 (M.Sc. Math)

Sol: Let X be a second countable space, then we have to show that X is Lindelöf space. For this, let $\{U : \dots\}$ be an open cover for X , i.e.

$$X \subseteq \bigcup_i U_i \quad \dots(1)$$

Since X is second countable, so it has a countable open base.

Let $\{B_n : n \in \mathbb{N}\}$ be a countable open base for X . Let $x \in X$, then $x \in U$ for some U . Since each U is an open set, so by the definition of open base, there is some B_{n_x} in such that

$$\begin{aligned} x &\in B_{n_x} \subseteq U \\ \bigcup_{x \in X} \{x\} &\subseteq \bigcup_{x \in X} B_{n_x} \subseteq \bigcup_{x \in X} U \\ X &\subseteq \bigcup_{x \in X} B_{n_x} \subseteq X \\ X &\subseteq \bigcup_{x \in X} B_{n_x} \end{aligned} \quad \dots(2)$$

Since each B_{n_x} is some B_{n_i} for some natural number i , therefore

$$(2) \text{ takes the form } X \subseteq \bigcup_{i=1}^{\infty} B_{n_i}.$$

Now for each B_{n_i} , select a corresponding U_i containing B_{n_i} , i.e.

$$\begin{aligned} B_{n_i} &\subseteq U_i \\ \bigcup_{i=1}^{\infty} B_{n_i} &\subseteq \bigcup_{i=1}^{\infty} U_i \\ X &\subseteq \bigcup_{i=1}^{\infty} U_i \end{aligned}$$

But $\bigcup_{i=1}^{\infty} U_i \subseteq X$, so $X = \bigcup_{i=1}^{\infty} U_i$.

This shows that $\{U_i : i = 1, 2, \dots\}$ is a countable subcover of $\{U : \dots\}$. Hence X is Lindelöf space.

Q.11 Show that a collection of subsets of a topological space (X, τ) is a base for τ if and only if

- (i) every point of X is in some B
- (ii) for B_1, B_2 and $x \in B_1 \cap B_2$, there is a B such that $x \in B \subseteq B_1 \cap B_2$.

University of Gujrat, 2013; PU, 2009; 2006; 2001 (M.Sc. Math)

Sol: Let $\mathcal{B}$ be a base for the topology τ .

- (i) Let x be an arbitrary point of X . Since X is an open set, so there exists a B_x such that

$$\begin{aligned} x &\in B_x \subseteq X \\ \bigcup_{x \in X} \{x\} &\subseteq \bigcup_{x \in X} B_x \subseteq X \end{aligned}$$

$$X = \bigcup_{x \in X} B_x \quad X$$

$$X = \bigcup_{x \in X} B_x$$

This shows that each point of X is in some B .

- (ii) Let B_1, B_2 and $x \in B_1 \cap B_2$. Since B_1, B_2 are open sets, so $B_1 \cap B_2$ is also an open set, and therefore, there exists B such that

$$x \in B \subseteq B_1 \cap B_2$$

Conversely, let conditions (i) and (ii) be satisfied for a collection of subsets of X , then we have to show that $\mathcal{B}$ is a base for some topology. For this, let $\mathcal{B}$ be a collection of those subsets of X which are obtained by the unions of members of $\mathcal{B}$. In order to show that $\mathcal{B}$ is a base for τ , it is enough to show that $\mathcal{B}$ is a topology on X . In the following we prove that $\mathcal{B}$ is a topology on X .

- T_1 : The empty set $\emptyset$, being the union of empty collection of subsets in $\mathcal{B}$, is in $\mathcal{B}$. Since X is open, so for any $x \in X$ the condition (i), gives a B_x such that $x \in B_x \subseteq X$.

Then

$$\bigcup_{x \in X} \{x\} \subseteq \bigcup_{x \in X} B_x \subseteq X$$

$$X \subseteq \bigcup_{x \in X} B_x \subseteq X$$

$$X = \bigcup_{x \in X} B_x$$

This shows that X , being the union of members of $\mathcal{B}$, is in $\mathcal{B}$.

- T_2 : The union of any number of members of $\mathcal{B}$, being the union of members of $\mathcal{B}$ is in $\mathcal{B}$.

- T_3 : Let U_1, U_2 , then by the definition of

$$U_1 = \bigcup_{B \in \mathcal{B}} B, U_2 = \bigcup_{B \in \mathcal{B}} B$$

for some $\mathcal{B}$, ranging over sub-collection of $\mathcal{B}$. Therefore,

$$U_1 \cap U_2 = \left(\bigcup_{B \in \mathcal{B}} B \right) \cap \left(\bigcup_{B \in \mathcal{B}} B \right)$$

$$= \bigcup_{B \in \mathcal{B}} (B \cap B) \quad \dots(1)$$

By the condition (ii), for any $x \in B \cap B$, there is a B_x such that

$$x \in B_x \subseteq B \cap B$$

$$\bigcup_{x \in B \cap B} \{x\} \subseteq \bigcup_{x \in B \cap B} B_x \subseteq B \cap B$$

$$B \cap B \subseteq \bigcup_{x \in B \cap B} B_x \subseteq B \cap B$$

$$B \cap B = \bigcup_{x \in B \cap B} B_x$$

Putting this value in (1), we have

$$U_1 \cup U_2 \quad (B \cup B)$$

$$\bigcup_{x \in B} B_x$$

This shows that $U_1 \cup U_2$ is the union of members of $\mathcal{B}$, so is in $\mathcal{B}$. In the same way, we can show that the intersection of any finite number of members of $\mathcal{B}$ is in $\mathcal{B}$. Since all the conditions of the topology are satisfied, so $\mathcal{B}$ is a topology on X . Consequently $\mathcal{B}$ is a base for τ .

Q.12 Let (X, τ) be a topological space. Show that a collection $\{B_\alpha : \alpha \in I\}$ of sets in X is a base for τ if and only if, for any open set U and any point x in U , there is a B_α such that $x \in B_\alpha \subseteq U$.

PU, 2005 (M.Sc. Math)

Sol: Let the collection $\{B_\alpha : \alpha \in I\}$ of sets in X be a base for τ . Let U be an open set and x be any point of U . Since U , being an open set, is a member of τ , so by the definition of open base, it should be the union of members of $\mathcal{B}$, i.e. there should exist an indexing set $I_1 \subseteq I$ and B_α such that

$$U = \bigcup_{\alpha \in I_1} B_\alpha$$

This shows that $B_\alpha \subseteq U$ for each $\alpha \in I_1$. Since $x \in U$, so

$$x \in \bigcup_{\alpha \in I_1} B_\alpha \\ x \in B_\alpha \text{ for some } \alpha \in I_1$$

This shows that

$$x \in B_\alpha \subseteq U \text{ for some } \alpha \in I_1 \subseteq I$$

Thus, $x \in B_\alpha \subseteq U$ for some B_α .

Conversely, suppose that for any open set U and any point x in U , there is a B_x such that $x \in B_x \subseteq U$, then

$$\bigcup_{x \in U} \{x\} = \bigcup_{x \in U} B_x \subseteq U \\ U = \bigcup_{x \in U} B_x \subseteq U \\ U = \bigcup_{x \in U} B_x$$

This shows that the open set U , a member of τ , is the union of members of $\mathcal{B}$.

Since U is an arbitrary member of τ , so each member of τ is a union of members of $\mathcal{B}$.

This shows that $\mathcal{B}$ is a base for τ .

Q.13 Given the bases $\{\{a, b\}, \{b\}, \{c\}, \{d\}\}$, find the topology generated by $\mathcal{B}$ on $X = \{a, b, c, d\}$.

PU, 2013 (BS Math)

Sol: $\{\{a,b\}, \{b\}, \{c\}, \{d\}\}, X \quad \{a,b,c,d\}$

First we take the unions of members of , i.e.

 $\{a,b\} \quad \{a,b\} \quad \{a,b\}$ $\{b\} \quad \{b\} \quad \{b\}$ $\{c\} \quad \{c\} \quad \{c\}$ $\{d\} \quad \{d\} \quad \{d\}$ $\{b\} \quad \{c\} \quad \{b,c\}$ $\{b\} \quad \{d\} \quad \{b,d\}$ $\{c\} \quad \{d\} \quad \{c,d\}$ $\{b\} \quad \{c\} \quad \{d\} \quad \{b,c,d\}$ $\{a,b\} \quad \{c\} \quad \{a,b,c\}$ $\{a,b\} \quad \{d\} \quad \{a,b,d\}$ $\{a,b\} \quad \{c\} \quad \{d\} \quad \{a,b,c,d\} \quad X$ $A \quad , \quad A$

Thus, the required topology generated by is

 $\{ \quad , \{b\}, \{c\}, \{d\}, \{a,b\}, \{b,c\}, \{b,d\}, \{c,d\}, \{b,c,d\}, \{a,b,c\}, \{a,b,d\}, X \}$

Chapter 3

SEPARATION

EXERCISE 3

Short Questions

Q.2 Solve / answer the following short questions:

(i) Define T_0 - space.

Sol: A topological space X is said to be a T_0 -space if for any $x, y \in X, x \neq y$, there exists an open set U such that $x \in U$ but $y \notin U$.

(ii) Consider the topology $\tau = \{\phi, \{b\}, \{a, b\}\}$ on the set $X = \{a, b\}$. Whether (X, τ)

(a) is a T_0 - space?

(b) is a T_1 - space?

Sol:(a) Since $a \neq b$, and there is an open set $\{b\}$ such that $b \in \{b\}$ and $a \notin \{b\}$, so (X, τ) is a T_0 - space.

(b) There are only two nonempty open sets, $\{b\}, \{a, b\}$.

The open set $\{a, b\}$ contains both a and b , so (X, τ) is not a T_1 - space.

(iii) Let X be an indiscrete topological space with at least two points, then show that X is not a T_0 -space.

Sol: Suppose on contrary that X is a T_0 -space, then for $x, y \in X, x \neq y$, there exists an open set U such that $x \in U$ and $y \notin U$. This shows that $U \neq \phi$ and $U \neq X$ which is not possible, because the only possible open sets are ϕ and X , so X is not a T_0 -space.

(iv) Let X be a discrete topological space with at least two points, then show that X is a T_0 -space.

Sol: Let $x, y \in X, x \neq y$. Since X is a discrete topological space, so each of its subsets is an open set. Thus, $\{x\}$, being the subset of X , is open such that $x \in \{x\}$ and $y \notin \{x\}$. This shows that X is a T_0 -space.

(v) **Show that the real line R with usual topology is T_0 -space.**

Sol: Let $x, y \in R, x \neq y$. Further assume that $x < y$. Since the usual topology on R consists of open intervals, so we have an open set $]-\infty, y[$, such that $x \in]-\infty, y[$ and $y \notin]-\infty, y[$. This shows that the real line R with usual topology is a T_0 -space.

(vi) **Show that every two point cofinite topological space is a T_0 -space.**

Sol: Let X be a two point cofinite topological space, then we have to show that X is a T_0 -space. For this, let $x, y \in X, x \neq y$. Since X is cofinite topological space and $\{y\}$ is finite, so $U = X \setminus \{y\}$ is an open set. Since $x \in U$ and $y \notin U$, so X is a T_0 -space. This completes the proof.

(vii) **Show that every two point cocountable topological space is a T_0 -space.**

Sol: Let X be a two point cocountable topological space, then we have to show that X is a T_0 -space. For this, let $x, y \in X, x \neq y$. Since X is cocountable topological space and $\{y\}$ is countable, so $U = X \setminus \{y\}$ is an open set. Since $x \in U$ and $y \notin U$, so X is a T_0 -space.

(viii) **If each singleton subset of a two point topological space is closed, then show that it is a T_0 -space.**

Sol: Let X be a two point topological space. Let each singleton subset of X be closed, then we have to show that X is a T_0 -space. For this, let $x, y \in X, x \neq y$. Since each singleton subset of X is closed, so $\{y\}$, being a singleton subset of X , is closed and so $\{y\}^c$ is an open set. Since $x \in \{y\}^c$ and $y \notin \{y\}^c$, so X is a T_0 -space.

(ix) **If each finite subset of a two point topological space is closed, then show that it is a T_0 -space.**

Sol: Let X be a two point topological space. Let each finite subset of X be closed, then we have to show that X is a T_0 -space. For this, let $x, y \in X, x \neq y$. Since each finite subset of X is a closed set, so $\{y\}$, being a finite subset of X , is closed and so $\{y\}^c$ is an open set. Since $x \in \{y\}^c$ and $y \notin \{y\}^c$, so X is a T_0 -space.

(x) **Define T_1 -space.**

Sol: A topological space X is said to be a T_1 -space if for any

$$x, y \in X, x \neq y$$

there exist open sets U and V such that $x \in U, y \notin U$ and $y \in V, x \notin V$.

(xi) Show that every T_1 -space is a T_0 -space.

Sol: Let X be a T_1 -space, then we have to show that X is a T_0 -space.

For this, let $x, y \in X, x \neq y$.

Since X is a T_1 -space, so there exist open sets U and V such that $x \in U, y \notin U$ and $y \in V, x \notin V$.

Thus, we can find an open set U such that $x \in U, y \notin U$, so X is a T_0 -space.

(xii) Show that an indiscrete topological space with at least two points is not a T_1 -space.

Sol: Suppose on contrary that X is a T_1 -space, then for $x, y \in X, x \neq y$, there exist open sets U and V such that

(i) $x \in U, y \notin U$, (ii) $y \in V, x \notin V$

From (i), we conclude that $U \neq \emptyset$ and $U \neq X$ which is not possible, because the only possible open sets are $\emptyset$ and X . This shows that X is not a T_1 -space.

(xiii) Show that the discrete topological space with at least two points is a T_1 -space.

Sol: Let X be a discrete topological space with at least two points.

Let $x, y \in X, x \neq y$. Since X is a discrete topological space, so each of its subsets is an open set. Thus, $\{x\}$ and $\{y\}$, being the subsets of X , are open subsets of X such that $x \in \{x\}, y \notin \{x\}$ and $y \in \{y\}, x \notin \{y\}$. This shows that X is a T_1 -space.

(xiv) Show the real line R with usual topology is T_1 -space.

Sol: Let $x, y \in R, x \neq y$. Further assume that $x < y$. Since the usual topology on R consists of open intervals, so we have open sets $U = (-\infty, y)$ and $V = (x, \infty)$, such that $x \in U, y \notin U$ and $y \in V, x \notin V$. This shows that the real line R with usual topology is a T_1 -space.

(xv) Show that every two point cofinite topological space is a T_1 -space.

Sol: Let X be a two point cofinite topological space, then we have to show that X is a T_1 -space. For this, let $x, y \in X, x \neq y$. Since X is cofinite topological space and $\{x\}$ and $\{y\}$ are finite, so $U = X \setminus \{y\}$ and $V = X \setminus \{x\}$ are open sets. Since $x \in U, y \notin U$ and $y \in V, x \notin V$, so X is a T_1 -space.

Q.3 Solve / answer the following short questions:

(i) Every two point cocountable topological space is a T_1 -space.

Sol: Let X be a two point cocountable topological space, then we have to show that X is a T_1 -space. For this, let $x, y \in X, x \neq y$. Since X

is cocountable topological space and $\{x\}$ and $\{y\}$ are countable, so $U = X \setminus \{y\}$ and $V = X \setminus \{x\}$ are open sets. Since $x \in U$, $y \notin U$ and $y \in V$, $x \notin V$, so X is a T_1 -space.

(ii) **Show that T_1 property is preserved under closed functions.**

Sol: Let X be a T_1 -space. Let $f: X \rightarrow f(X)$ be a closed function. Since X is a T_1 -space, so for any $x \in X$, $\{x\}$ is a closed set. Since f is closed, so $f(\{x\})$ is closed in $f(X)$.

Since $f(\{x\}) = \{f(x)\}$, so each singleton subset of $f(X)$ is closed.

This shows that $f(X)$ is a T_1 -space. Hence, T_1 property is preserved under closed functions.

(iii) **Define Hausdorff space.**

Sol: A topological space X is said to be a Hausdorff space if for any $x, y \in X$, $x \neq y$, there exist open sets U and V such that

$$x \in U, y \in V \text{ and } U \cap V = \phi$$

(iv) **Show that two point discrete topological space is Hausdorff space.**

Sol: Let X be a two point discrete topological space.
Let $x, y \in X$, $x \neq y$.

Since X is a discrete topological space so each of its subsets is an open set.

Thus, $U = \{x\}$ and $V = \{y\}$, being the subsets of X , are open subsets of X such that

$$x \in U, y \in V \text{ and } U \cap V = \phi$$

This shows that X is a Hausdorff space or T_2 -space.

(v) **Let τ be the topology on the real line $\mathbb{R}$ generated by the open-closed intervals $(a, b]$. Show that $(\mathbb{R}, \tau)$ is Hausdorff.**

Sol: Let $a, b \in \mathbb{R}$ with $a \neq b$, say $a < b$.

Let $G = (a-1, a]$ and $H = (a, b]$, then

$$G, H \in \tau$$

with

$$a \in G, b \in H$$

and

$$G \cap H = (a-1, a] \cap (a, b] = \phi$$

This shows that for any $a, b \in \mathbb{R}$ with $a \neq b$, there exist open sets G and H such that $G \cap H = \phi$, so $(\mathbb{R}, \tau)$ is Hausdorff.

(vi) **Show that every T_2 -space is a T_1 -space.**

Sol: Let X be T_2 -space, then we have to show that X is a T_1 -space. For this let $x, y \in X$, $x \neq y$, then there must exist open sets U and V such that

$$x \in U, y \in V \text{ and } U \cap V = \phi$$

Then obviously,

$$x \in U, y \notin V$$

and

$$y \in V, x \notin U$$

This shows that X is a T_1 -space.

(vii) Define a convergent sequence in a topological space.

Sol: Let (X, τ) be a topological space. A sequence $\{x_n\}$ of points of X is said to converge to a point $x \in X$ if for every open set containing x , there exists a natural n_0 such that

$$x_n \in U \text{ for all } n \geq n_0.$$

In this case, we write

$$x_n \rightarrow x \text{ as } n \rightarrow \infty$$

or simply

$$x_n \rightarrow x$$

We read it as " x_n approaches x ".

We can also write it as $\lim_{n \rightarrow \infty} x_n = x$.

(viii) Define a regular space.

Sol: A topological space X is said to be a regular space if for any $x \in X$ and any closed subset A of X , there exist open sets U and V such that

$$x \in U, A \subseteq V \text{ and } U \cap V = \phi$$

(ix) Give an example to show that a regular space need not be a Hausdorff space.

Sol: Let X be an indiscrete topological space, then the only nonempty closed set X , so for any $x \in X$, there does not exist any closed set A which does not contain x , so X is trivially a regular space. Since for any $x, y \in X, x \neq y$, there is only one open set X itself containing these points, so X is not a Hausdorff space.

(x) Define a T_3 -space.

Sol: A regular T_1 -space is called a T_3 -space.

(xi) Define a Tychonoff space.

Sol: A completely regular T_1 -space is said to be a Tychonoff space or a $T_{3\frac{1}{2}}$ -space.

(xii) Define a normal space.

Sol: A topological space X is said to be a normal space if for any disjoint pair of closed sets A and B , there exist open sets U and V such that

$$A \subseteq U, B \subseteq V \text{ and } U \cap V = \phi$$

(xiii) Define a compact space.

Sol: A topological space (X, τ) is said to be a compact space if every open cover of X has a finite subcover.

Long Questions

Q.4 Show that every subspace of T_0 -space is a T_0 -space.

Sol: Let X be a T_0 -space. Let Y be a subspace of X , then we have to show that Y is a T_0 -space.

For this, let $x, y \in Y, x \neq y$, then $x, y \in X$.

Since X is a T_0 -space, so there exists an open set U in X such that $x \in U$ and $y \notin U$.

Since Y is a subspace of X , so an open set V in Y can be defined as $V = U \cap Y$.

Since $x \in U$ and $x \in Y$, so

$$x \in U \cap Y$$

$$\Rightarrow x \in V$$

But $y \notin U \cap Y = V$, because $y \notin U$. This shows that there is an open subset V of Y such that $x \in V$ and $y \notin V$. Hence, Y is a T_0 -space.

Q.5 If X is a T_0 -space, then show that $a \neq b \Rightarrow \overline{\{a\}} \neq \overline{\{b\}}$ for any $a, b \in X$.

Sol: Let $a, b \in X, a \neq b$, then we have to show that $\overline{\{a\}} \neq \overline{\{b\}}$.

For this, suppose on contrary that $\overline{\{a\}} = \overline{\{b\}}$.

Since $\overline{\{a\}} = \{a\} \cup \{a\}^d$ and $\overline{\{b\}} = \{b\} \cup \{b\}^d$, so each point of $\overline{\{a\}}$ must be in $\overline{\{b\}} = \{b\} \cup \{b\}^d$ and vice versa.

Since X is a T_0 -space, so there is an open set U such that $a \in U$ and $b \notin U$.

Thus, U is an open set which contains a but not any point of $\{b\}$, so a is not a limit point of $\{b\}$.

Since a is neither in $\{b\}$ nor a limit point of $\{b\}$, so $a \notin \overline{\{b\}}$.

This shows that $\overline{\{a\}} \not\subseteq \overline{\{b\}}$, so our assumption that $\overline{\{a\}} = \overline{\{b\}}$ is wrong. Hence $\overline{\{a\}} \neq \overline{\{b\}}$.

Q.6 If for any $a, b \in X, a \neq b \Rightarrow \overline{\{a\}} \neq \overline{\{b\}}$, then show that X is T_0 -space.

Sol: Let, for any $a, b \in X, a \neq b \Rightarrow \overline{\{a\}} \neq \overline{\{b\}}$, then we have to show that X is a T_0 -space. For this, suppose on contrary that X is not a T_0 -space, then for any $a, b \in X, a \neq b$, each open set U containing a also contains b , so for any $x \in \overline{\{a\}}$, we have

$$(\{b\} \cap U) \setminus \{x\} \neq \emptyset$$

This shows that x is a limit point of $\{b\}$, so $x \in \overline{\{b\}}$

$$\Rightarrow \overline{\{a\}} \subseteq \overline{\{b\}} \quad \dots(1)$$

Similarly, each open set V containing b also contains a , so for any $y \in \overline{\{b\}}$,

$$(\{a\} \cap U) \setminus \{y\} \neq \emptyset$$

This shows that y is a limit point of $\{a\}$, so $y \in \overline{\{a\}}$, and therefore,

$$\overline{\{b\}} \subseteq \overline{\{a\}} \quad \dots(2)$$

From (1) and (2), we have $\overline{\{a\}} = \overline{\{b\}}$ which is a contradiction, so X is a T_0 -space.

Q.7 Show that a pseudo metric space is a metric space if and only if it is a T_0 -space.

Sol: Let (X, d) be a metric space, then we have to show that X is T_0 -space. For this, let $x, y \in X, x \neq y$, then $d(x, y) \neq 0$, so let $d(x, y) = r > 0$. Then we can find an open sphere $S_r(x)$ in X centred on x with radius r such that $x \in S_r(x)$ and $y \notin S_r(x)$. Since open spheres are open sets, so X is a T_0 -space.

Conversely, let X be a pseudo metric space. Let X be a T_0 -space, then we have to show that X is metric space. For this it is enough to show that $x \neq y \Rightarrow d(x, y) > 0$. Let $x, y \in X, x \neq y$. Since X is a T_0 -space, so we can find open set U in X such that $x \in U$ and $y \notin U$. By the definition of open set, there exists an open sphere $S_r(x)$ of radius $r > 0$ such that $S_r(x) \subseteq U$. But then $x \in S_r(x)$ and

$$y \notin S_r(x)$$

$$\Rightarrow d(x, y) \geq r$$

$$\Rightarrow d(x, y) > 0 \quad \because r > 0$$

Hence, X is a metric space.

Q.8 Show that any homeomorphic image of a T_0 -space is a T_0 -space.

Sol: Let X be a T_0 -space and Y a topological space. Let $f: X \rightarrow Y$ be a homeomorphism, then we have to show that Y is also a T_0 -space.

For this, let $x, y \in Y, x \neq y$, then $f^{-1}(x), f^{-1}(y) \in X$.

Since $x \neq y$ and f , being a homeomorphism, is one-to-one, so $f^{-1}(x) \neq f^{-1}(y)$.

Since X is a T_0 -space, so there exists an open set V in X such that $f^{-1}(x) \in V$ and $f^{-1}(y) \notin V$.

These give us $x \in f(V)$ and $y \notin f(V)$. Since V is open in X and f , being a homeomorphism, is an open function, so $f(V)$ is open in Y . Since for any $x, y \in Y, x \neq y$, there is an open set $f(V)$ in Y such that $x \in f(V)$ and $y \notin f(V)$, so Y is a T_0 -space.

Q.9 Show that every subspace of T_1 -space is T_1 -space.

Sol: Let X be a T_1 -space. Let Y be a subspace of X , then we have to show that Y is a T_1 -space. For this, let $x, y \in Y, x \neq y$, then $x, y \in X$. Since X is a T_1 -space, so there exist open subsets U_1, V_1 such that $x \in U_1, y \notin U_1$ and $y \in V_1, x \notin V_1$. Since Y is a subspace of X , so open sets U and V in Y are defined as

$$U = U_1 \cap Y \text{ and } V = V_1 \cap Y$$

Now

$$x \in U_1 \cap Y \Rightarrow x \in U$$

And

$$y \notin U_1 \Rightarrow y \notin U_1 \cap Y = V$$

Similarly, $y \in V, y \notin V$.

This shows that Y is a T_1 -space.

Q.10 Show that a topological space is a T_1 -space if and only if its each singleton subset is a closed set.

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Sol: Let X be a T_1 -space, then we have to show that each singleton subset of X is a closed set. For this, let $x \in X$ and consider the arbitrary singleton subset $\{x\}$ of X . To show that $\{x\}$ is closed, it is enough to show that $\{x\}^c$ is open. Let $y \in \{x\}^c$, then $y \notin \{x\}$, so $y \neq x$. Since X is a T_1 -space, so there exist open sets U and V_y such that $x \in U, y \notin U$ and $y \in V_y, x \notin V_y$. Since $\{x\}$ has only x and $x \notin V_y$, so

$$\{x\} \cap V_y = \emptyset$$

$$\Rightarrow V_y \subseteq \{x\}^c$$

$$\Rightarrow y \in V_y \subseteq \{x\}^c$$

$$\Rightarrow \bigcup_{y \in \{x\}^c} \{y\} \subseteq \bigcup_{y \in \{x\}^c} V_y \subseteq \{x\}^c$$

$$\Rightarrow \{x\}^c \subseteq \bigcup_{y \in \{x\}^c} V_y \subseteq \{x\}^c$$

$$\Rightarrow \{x\}^c = \bigcup_{y \in \{x\}^c} V_y$$

Thus, $\{x\}^c$, being the union of open sets, is an open subset of X , so $\{x\}$ is a closed subset of X . This shows that each singleton subset of X is a closed set.

Conversely, let each singleton subset of X be a closed set, then

we have to show that X is a T_1 -space. For this, let $x, y \in X$, $x \neq y$. Since $\{x\}$ and $\{y\}$ are closed sets, so $U = X \setminus \{y\}$ and $V = X \setminus \{x\}$ are open sets. Moreover, $x \in U$, $y \notin U$ and $y \in V$, $x \notin V$. This shows that X is a T_1 -space.

Q.11 Show that a topological space is a T_1 -space if and only if its each finite subset is a closed set.

Sol: Let X be a T_1 -space, then we have to show that each finite subset of X is a closed set. For this, let A be an arbitrary finite subset of X . In order to show that A is closed, let $x \in A$ be an arbitrary point of A . Let $y \in \{x\}^c$, then $y \notin \{x\}$, so $y \neq x$. Since X is a T_1 -space, so there exist open sets U and V_y such that $x \in U$, $y \notin U$ and $y \in V_y$, $x \notin V_y$. Since $\{x\}$ has only x and $x \notin V_y$, so

$$\begin{aligned}\{x\} \cap V_y &= \emptyset \\ \Rightarrow V_y &\subseteq \{x\}^c \\ \Rightarrow y \in V_y &\subseteq \{x\}^c \\ \Rightarrow \bigcup_{y \in \{x\}^c} \{y\} &\subseteq \bigcup_{y \in \{x\}^c} V_y \subseteq \{x\}^c \\ \Rightarrow \{x\}^c &\subseteq \bigcup_{y \in \{x\}^c} V_y \subseteq \{x\}^c \\ \Rightarrow \{x\}^c &= \bigcup_{y \in \{x\}^c} V_y\end{aligned}$$

Thus, $\{x\}^c$, being the union of open sets, is an open subset of X , so $\{x\}$ is a closed subset of X . This shows that each singleton subset of X is a closed set. Since $A = \bigcup_{x \in A} \{x\}$, so A , being the union of any finite number of closed sets, is closed.

Conversely, let each finite subset of X be a closed set, then we have to show that X is a T_1 -space. For this, let $x, y \in X$, $x \neq y$, then $\{x\}$ and $\{y\}$, being the finite subsets of X , are closed sets, so $U = X \setminus \{y\}$ and $V = X \setminus \{x\}$ are open sets. Moreover, $x \in U$, $y \notin U$ and $y \in V$, $x \notin V$. This shows that X is a T_1 -space.

Q.12 Show that every finite T_1 -space is discrete.

Sol: Let X be a finite T_1 -space, then we have to show that X is discrete. For this, let A be an arbitrary subset of X . Since X is a T_1 -space, so each singleton subset of X is closed. For any $x \in A^c$, $\{x\}$ is a closed subset of X .

Since X is finite, so A^c is also finite.

Moreover, $A^c = \bigcup_{x \in A^c} \{x\}$, being the union of finite number of closed sets, is a closed set. Thus, A is an open set. This shows that each subset of X is an open set, so X is a discrete space.

Q.13 Show that the following statements about a topological space X are equivalent.

- (i) X is a T_1 -space.
- (ii) Each singleton subset of X is closed.
- (iii) Each subset A of X is the intersection of its open supersets.

Sol: (i) $\Rightarrow$ (ii): Let X be a T_1 -space, then we have to show that each singleton subset of X is closed. For this, let $x \in X$ and consider the arbitrary singleton subset $\{x\}$ of X . To show that $\{x\}$ is closed, it is enough to show that $\{x\}^c$ is open. Let $y \in \{x\}^c$, then $y \notin \{x\}$, so $y \neq x$. Since X is a T_1 -space, so there exist open sets U and V_y such that $x \in U, y \notin U$ and $y \in V_y, x \notin V_y$. Since $\{x\}$ has only x and $x \notin V_y$, so

$$\begin{aligned}
 \{x\} \cap V_y &= \emptyset \\
 \Rightarrow V_y &\subseteq \{x\}^c \\
 \Rightarrow y \in V_y &\subseteq \{x\}^c \\
 \Rightarrow \bigcup_{y \in \{x\}^c} \{y\} &\subseteq \bigcup_{y \in \{x\}^c} V_y \subseteq \{x\}^c \\
 \Rightarrow \{x\}^c &\subseteq \bigcup_{y \in \{x\}^c} V_y \subseteq \{x\}^c \\
 \Rightarrow \{x\}^c &= \bigcup_{y \in \{x\}^c} V_y
 \end{aligned}$$

Thus, $\{x\}^c$, being the union of open sets, is an open subset of X , so $\{x\}$ is a closed subset of X . This shows that each singleton subset of X is a closed set.

(ii) $\Rightarrow$ (iii): Let each singleton subset of X be closed, then we have to show that each subset A of X is the intersection of its open supersets. For this, let A be an arbitrary subset of X . For any $y \notin A$, $X \setminus \{y\}$ is a superset of A , i.e., $A \subseteq X \setminus \{y\}$ for all $y \notin A$. Thus, the intersection of all such sets is also a superset of A , i.e.

$$A \subseteq \bigcap_{y \notin A} X \setminus \{y\} \quad \dots(1)$$

Let $x \in \bigcap_{y \notin A} X \setminus \{y\}$, then $x \in X \setminus \{y\}$ for all $y \notin A$, so $x \in A$.

$$\bigcap_{y \notin A} X \setminus \{y\} \subseteq A \quad \dots(2)$$

From (1) and (2), we have

$$A = \bigcap_{y \notin A} X \setminus \{y\} \quad \dots(3)$$

Since each singleton subset of X be closed, so each $X \setminus \{y\}$ is open. Thus, from (3) it is clear that A is the intersection of its open supersets. This shows that each subset A of X is the intersection of its open supersets.

(iii) $\Rightarrow$ (i): Let each subset A of X is the intersection of its open supersets, then we have to show that X is a T_1 -space.

For this, let $x, y \in X, x \neq y$, then $\{x\}$ is the intersection of its open supersets and $\{y\}$ is the intersection of its open supersets.

So there is at least one open set U which contains x but not y (for otherwise x and y both will be in the intersection of open supersets of $\{x\}$) and similarly there is an open set V which contains y but not x . Hence x is a T_1 -space.

Q.14 Show that homeomorphic image of a T_1 -space is a T_1 -space.

Sol: Let X be a T_1 -space and Y a topological space. Let $f: X \rightarrow Y$ be a homeomorphism, then we have to show that Y is also a T_1 -space.

For this, let $x, y \in Y, x \neq y$, then $f^{-1}(x), f^{-1}(y) \in X$.

Since $x \neq y$ and f , being a homeomorphism, is one-to-one, so $f^{-1}(x) \neq f^{-1}(y)$.

Since X is a T_1 -space, so there exist open sets U and V in X such that $f^{-1}(x) \in U, f^{-1}(y) \notin U$, and $f^{-1}(y) \in V, f^{-1}(x) \notin V$.

These give us $x \in f(U), y \notin f(U)$ and $y \in f(V), x \notin f(V)$.

Since U and V are open in X and f , being a homeomorphism, is an open function, so $f(U)$ and $f(V)$ are open in Y .

Since for any $x, y \in Y, x \neq y$, there are open sets $f(U)$ and $f(V)$ in Y such that $x \in f(U), y \notin f(U)$ and $y \in f(V), x \notin f(V)$, so Y is a T_1 -space.

Q.15 If x is a limit point of a set A in a T_1 -space X , then show that every open set containing x contains an infinite number of distinct points of A .

Sol: Let X be a T_1 -space and A be a subset of X . Let x be a limit point of A , and let U be an open set containing x . Suppose on contrary that U contains only finite number of distinct points of A and further assume that

$$(U \cap A) \setminus \{x\} = \{x_1, x_2, \dots, x_n\} = F$$

Since X is a T_1 -space, so F , being a finite subset of X , is closed.

Therefore, F^c is an open set containing x and $F^c \cap U$, being the intersection of two open sets, is also open. Since x is in both F^c and U , so $F^c \cap U$ is an open set containing x .

Since F consists of points of A which are contained in U and are different from x , so $F^c \cap U$ will not contain any point of A different from x .

This shows that x is not a limit point of A , which is a contradiction.

Hence, any open set containing the limit point x of A contains an infinite number of distinct points of A in a T_1 -space.

Q.16 Show that a finite set has no limit points in a T_1 -space.

Sol: Let X be a T_1 -space and A be a finite subset of X , then we have to show that A has no limit points. For this, suppose on contrary that x is a limit point of A , then each open set U containing x can contain only finite points of A different from x , so let

$$(U \cap A) \setminus \{x\} = \{x_1, x_2, \dots, x_n\} = F$$

Since X is a T_1 -space, so F , being a finite subset of X , is closed.

Therefore, F^c is an open set containing x and $F^c \cap U$, being the intersection of two open sets, is also open.

Since x is in both F^c and U , so $F^c \cap U$ is an open set containing x .

Since F consists of points of A which are contained in U and are different from x , so $F^c \cap U$ will not contain any point of A different from x .

This shows that x is not a limit point of A , which is a contradiction. Hence, any finite subset A of a T_1 -space has no limit points.

Q.17 Let X be a T_1 -space and let β_x be a local base at $x \in X$. If $y \in X$ is distinct from x , then show that some member of β_x does not contain y .

Sol: Since $x \neq y$ and X is a T_1 -space, so there exists an open set U in X such that

$$x \in U, y \notin U$$

Since β_x is a local base at x and U is an open set containing x , so there is a member B of β_x such that

$$x \in B \subseteq U$$

Since $y \notin U$, so $y \notin B$.

This completes the proof that there is a member B of β_x which does not contain y .

Q.18 Show that every metric space is Hausdorff space.

Sol: Let (X, d) be a metric space, then we have to show that X is a Hausdorff space. For this let $x, y \in X, x \neq y$. Then obviously

$$d(x, y) \neq 0$$

So, let

$$d(x, y) = 2r > 0$$

Since open spheres are open sets in metric spaces, so open spheres $S_r(x)$ and $S_r(y)$ are open sets in X such that

$$x \in S_r(x), y \in S_r(y)$$

To show that

$$S_r(x) \cap S_r(y) = \emptyset$$

Suppose on contrary that

$$S_r(x) \cap S_r(y) \neq \emptyset$$

Let

$$z \in S_r(x) \cap S_r(y)$$

$$\Rightarrow z \in S_r(x) \text{ and } z \in S_r(y)$$

$$\Rightarrow d(x, z) < r \text{ and } d(z, y) < r$$

Using triangle inequality, we have

$$d(x, y) \leq d(x, z) + d(z, y)$$

$$\Rightarrow 2r < r + r \quad \because d(x, y) = 2r, d(x, z) < r, d(z, y) < r$$

$$\Rightarrow 2r < 2r$$

This is not possible, so our assumption that $S_r(x) \cap S_r(y) \neq \emptyset$ is wrong. Consequently

$$S_r(x) \cap S_r(y) = \emptyset$$

Hence X is a Hausdorff space or T_2 -space.

Q.19 Give an example to show that a T_1 -space may not be a T_2 -space.

Sol: Let X be an infinite set with cofinite topology.
Let $x, y \in X, x \neq y$ and consider

$$U = X \setminus \{y\} = \{y\}^c$$

$$V = X \setminus \{x\} = \{x\}^c$$

Then $U^c = \{y\}$ and $V^c = \{x\}$ both are finite, so both U and V are open sets in X . Also

$$x \in \{y\}^c = U, \quad y \notin \{y\}^c = U$$

$$\text{and } y \in \{x\}^c = V, \quad x \notin \{x\}^c = V$$

This shows that X is a T_1 -space.

Next suppose on contrary that X is a T_2 -space, then for any $x, y \in X, x \neq y$, there exist open sets U and V such that

$$x \in U, y \in V$$

and

$$U \cap V = \emptyset$$

$$\Rightarrow (U \cap V)^c = \emptyset^c$$

$$\Rightarrow U^c \cup V^c = X \quad \dots(1)$$

Since U and V are open sets in X and X is a cofinite topological space, so both U^c and V^c are finite. It is also clear from (1) that X , being the union of two finite sets, is finite. This is a contradiction, because x is an infinite set. Thus, our assumption that X is a T_2 -space is wrong. Consequently, X is not a T_2 -space.

Hence an infinite set with cofinite topology is an example of a T_1 -space which is not a T_2 -space.

Q.20 Show that the real line R with usual topology is a Hausdorff space.

Sol: Let $x, y \in R$, $x \neq y$, then $|x - y| \neq 0$, so let $|x - y| = 2r > 0$.

Since the usual topology on R consists of open intervals, so consider the open intervals

$$U =]x - r, x + r[\text{ and } V =]y - r, y + r[$$

Obviously, $x \in]x - r, x + r[= U$

and $y \in]y - r, y + r[= V$

If possible, let

$$z \in]x - r, x + r[\cap]y - r, y + r[$$

then $z \in]x - r, x + r[\text{ and } z \in]y - r, y + r[$

$$|x - z| < r \text{ and } |z - y| < r \quad \dots(1)$$

We know that for any real numbers x , y and z

$$|x - y| = |x - z + z - y| \leq |x - z| + |z - y|$$

$$\Rightarrow 2r \leq |x - z| + |z - y| \quad \because |x - y| = 2r$$

$$\Rightarrow 2r < r + r \quad \because |x - z| < r, |z - y| < r$$

$$\Rightarrow 2r < 2r$$

This is not possible, so

$$]x - r, x + r[\cap]y - r, y + r[= \phi$$

Thus, we have two open sets U and V such that

$$x \in U, y \in V \text{ and } U \cap V = \phi$$

This shows that the real line R with usual topology is a Hausdorff space.

Q.21 Show that every finite T_2 -space is discrete.

Sol: Let X be a finite T_2 -space, then we have to show that X is discrete. For this, let A be an arbitrary subset of X . Since X , being a T_2 -space, is a T_1 -space, so each singleton subset of X is closed.

Thus, for any $x \in A^c$, $\{x\}$ is a closed subset of X .

Since X is finite, so A^c is also finite.

Also

$$A^c = \bigcup_{x \in A^c} \{x\}$$

being the union of finite number of closed sets, is a closed set.

Thus, A is an open set. This shows that each subset of X is an open set, so X is a discrete space.

Q.22 Show that every subspace of T_2 -space is T_2 -space.

Sol: Let X be a T_2 -space and Y be an arbitrary subspace of X , then we have to show that Y is a T_2 -space.

For this let $x, y \in Y, x \neq y$, then obviously, $x, y \in X$, because $Y \subseteq X$.

Since X is a T_2 -space, so for $x, y \in X, x \neq y$, there must exist open sets U_1 and V_1 such that

$$x \in U_1, y \in V_1 \text{ and } U_1 \cap V_1 = \phi$$

Since Y is a subspace of X , so the open sets U and V in Y are defined as

$$U = U_1 \cap Y \text{ and } V = V_1 \cap Y$$

Since $x \in U_1$ and $x \in Y$, so $x \in U_1 \cap Y = U$

Also $y \in V_1$ and $y \in Y$, so $y \in V_1 \cap Y = V$

such that

$$\begin{aligned} U \cap V &= (U_1 \cap Y) \cap (V_1 \cap Y) \\ &= (U_1 \cap V_1) \cap Y \\ &= \phi \cap Y && \because U_1 \cap V_1 = \phi \\ &= \phi \end{aligned}$$

Thus, for any $x, y \in Y, x \neq y$, we can find open sets U and V such that

$$x \in U, y \in V \text{ and } U \cap V = \phi$$

This shows that Y is a T_2 -space.

Q.23 Show that in a Hausdorff space, any point and a disjoint compact subspace can be separated by open sets, in the sense that there exist disjoint open sets one contains the point and the other contains the compact subspace.

Sol: Let X be a Hausdorff space. Let $x \in X$ be any point of X and C be a compact subspace of X disjoint from x , i.e. $x \notin C$.

Let $y \in C$, then obviously $x \neq y$.

Since $C \subseteq X$, so $y \in C \Rightarrow y \in X$.

Since X is a Hausdorff space, so for $x, y \in X, x \neq y$, there exist open sets G_x and H_y such that

$$x \in G_x, y \in H_y \text{ and } G_x \cap H_y = \phi$$

If we allow y to vary over C , then the collection $\{H_y : y \in C\}$ is an open cover of C . Since C is compact, so this open cover must have a finite subcover.

Let $\{H_{y_i} : 1 \leq i \leq n\}$ be the finite subcover of the open cover $\{H_y : y \in C\}$, then

$$C = \bigcup_{i=1}^n H_{y_i} \quad \dots(1)$$

If each G_{x_i} is an open set corresponding to each H_{y_i} , then

$$G_{xi} \cap H_{yi} = \phi \quad \forall i \quad \dots(2)$$

Since $y \in C$, so by (1), $y \in \bigcup_{i=1}^n H_{yi} = H$.

Since $x \in G_{xi}$ for each i , so $x \in \bigcap_{i=1}^n G_{xi} = G$.

H , being the union of open sets, is open and G , being the intersection of finite number of open sets, is open.

Also

$$\begin{aligned} G \cap H &= \left(\bigcap_{i=1}^n G_{xi} \right) \cap \left(\bigcup_{i=1}^n H_{yi} \right) \\ &= (G_{x1} \cap G_{x2} \cap \dots \cap G_{xn}) \cap (H_{y1} \cup H_{y2} \cup \dots \cup H_{yn}) \\ &= (G_{x1} \cap H_{y1}) \cup (G_{x2} \cap H_{y2}) \cup \dots \cup (G_{xn} \cap H_{yn}) \\ &= \bigcup_{i=1}^n (G_{xi} \cap H_{yi}) \\ &= \phi \end{aligned} \quad \because G_{xi} \cap H_{yi} = \phi, \forall i$$

This shows that there exist disjoint open sets G and H such that
 $x \in G, C \subseteq H (\because C = H)$

Q.24 Prove that every compact subspace of Hausdorff space is closed.

Sol: Let X be a Hausdorff space and C be a compact subspace of X , then we have to show that C is closed. For this it is enough to show that C^c is open.

If C^c is empty, then it is trivially open, so let $C^c \neq \phi$ and let $x \in C^c$, then $x \notin C$, so, by previous theorem, there must exist open sets U_x and V such that

$$x \in U_x, C \subseteq V, U_x \cap V = \phi$$

$$\text{Now} \quad C \subseteq V \Rightarrow V^c \subseteq C^c \quad \dots(1)$$

$$\text{And} \quad U_x \subseteq V^c \quad \dots(2)$$

Combining (1) and (2), we have

$$\begin{aligned} U_x &\subseteq V^c \subseteq C^c \\ \Rightarrow x \in U_x &\subseteq V^c \subseteq C^c \quad \because x \in U_x \\ \Rightarrow x \in U_x &\subseteq C^c \\ \Rightarrow \bigcup_{x \in C^c} \{x\} &\subseteq \bigcup_{x \in C^c} U_x \subseteq C^c \\ \Rightarrow C^c &\subseteq \bigcup_{x \in C^c} U_x \subseteq C^c \\ \Rightarrow C^c &= \bigcup_{x \in C^c} U_x \end{aligned}$$

Thus, C^c , being the union of open sets, is open. Consequently, C is closed.

Q.25 Show that one-to-one continuous mapping of a compact space onto a Hausdorff space is homeomorphism.

Sol: Let X be a compact space and Y be a Hausdorff space. Let $f: X \rightarrow Y$ be one-to-one continuous mapping onto Y , then to show that f is homeomorphism, it is enough to show that f is closed as it is given that f is bijective.

Let F be any closed set in X . If F is empty set, then $f(F) = \emptyset$ and in this case f is closed as $\emptyset$ is closed in Y .

If $F \neq \emptyset$, then F , being the closed subset (subspace) of a compact space X , is compact. But then $f(F)$, being the continuous image of a compact space is compact. Now $f(F)$, being the compact subset of a Hausdorff space Y , is closed.

This shows that f is closed. Hence f is a homeomorphism.

Q.26 If the space $C(X, R)$ separates points of X , then show that X is Hausdorff.

Sol: Let the space $C(X, R)$ separate the points of X , then we have to show that X is Hausdorff space. For this let $x, y \in X, x \neq y$.

Since $C(X, R)$ separates the points of X , so for any $x, y \in X, x \neq y$, there exists a function $f \in C(X, R)$ such that $f(x) \neq f(y)$.

Since $f \in C(X, R)$, so f is a real valued function, i.e. $f(x), f(y) \in R$. Since $f(x) \neq f(y)$, so let $f(x) < f(y)$, then by the property of real numbers, there must exist a real number r such that

$$f(x) < r < f(y)$$

Next consider

$$U = \{z: f(z) < r\}$$

$$V = \{z: f(z) > r\}$$

Then obviously, $x \in U$ and $y \in V$.

Also $U \cap V = \emptyset$

By the definitions of U and V

$$U = f^{-1}(-\infty, r)$$

and

$$V = f^{-1}(r, \infty)$$

Since $(-\infty, r)$ and (r, ∞) are open sets in R and f , being the member of $C(X, R)$, is continuous, so $f^{-1}(-\infty, r)$ and $f^{-1}(r, \infty)$ are open in X , i.e. U and V are open in X .

Thus, for any $x, y \in X, x \neq y$, there exist open sets U and V such that

$$x \in U, y \in V, U \cap V = \emptyset$$

This shows that X is a Hausdorff space.

Q.27 Let X be a topological space and Y a Hausdorff space. Let f

and g be continuous functions from X to Y , then show that the set

$$A = \{x \in X : f(x) = g(x)\}$$

is a closed set.

Sol: To show that A is a closed set, it is enough to show that A^c is open in X . If A^c is empty, then it is trivially open, so let $x \in A^c$, then

$$x \notin A$$

$$\Rightarrow f(x) \neq g(x)$$

Since $f(x), g(x) \in Y$ and Y is a Hausdorff space, so there exist open sets U_x and V_x in Y such that

$$f(x) \in U_x, g(x) \in V_x \text{ and } U_x \cap V_x = \phi$$

Now

$$f(x) \in U_x$$

$$\Rightarrow x \in f^{-1}(U_x)$$

And

$$g(x) \in V_x$$

$$\Rightarrow x \in f^{-1}(V_x)$$

Therefore,

$$\begin{aligned} x &\in f^{-1}(U_x) \cap f^{-1}(V_x) \subseteq A^c \\ \Rightarrow \bigcup_{x \in A^c} \{x\} &\subseteq \bigcup_{x \in A^c} \{f^{-1}(U_x) \cap f^{-1}(V_x)\} \subseteq A^c \\ \Rightarrow A^c &\subseteq \bigcup_{x \in A^c} \{f^{-1}(U_x) \cap f^{-1}(V_x)\} \subseteq A^c \\ \Rightarrow A^c &= \bigcup_{x \in A^c} \{f^{-1}(U_x) \cap f^{-1}(V_x)\} \end{aligned} \quad \dots(1)$$

Since f is continuous and both U_x and V_x are open in Y , so both $f^{-1}(U_x)$ and $f^{-1}(V_x)$ are open in X and so $f^{-1}(U_x) \cap f^{-1}(V_x)$ is also open in X .

It is also clear from (1) that, A^c , being the union of open sets, is open. Consequently, A is closed.

Q.28 Let $f: X \rightarrow Y, g: X \rightarrow Y$ be continuous functions from a space X to a Hausdorff space Y and suppose that $f(x) = g(x)$ for all x in a dense subset D of X . Then show that $f(x) = g(x)$ for all x in X .

PU, 2005 (M.Sc. Math)

Sol: If possible, let there be an element $y \in X$ such that

$$f(y) \neq g(y)$$

Since $f(y), g(y) \in Y$, and Y is a Hausdorff space, so there exist open sets U_1 and V_1 in Y such that $f(y) \in U_1, g(y) \in V_1$ and

$$U_1 \cap V_1 = \phi.$$

Now $U = f^{-1}(U_1)$, being the inverse image of an open set U_1 under continuous mapping f , is open in X .

Similarly, $V = g^{-1}(V_1)$, being the inverse image of an open set V_1 under continuous mapping g , is open in X .

Also $y \in f^{-1}(U_1) = U$ and $y \in g^{-1}(V_1) = V$, so

$$y \in U \cap V$$

$$\Rightarrow U \cap V \neq \phi$$

Since both U and V are open sets, so $U \cap V$ is a nonempty open set.

Since D is dense in X , i.e. $\bar{D} = X$, so $y \in X$ is either in D or a limit point of D , therefore, the nonempty set $U \cap V$ containing y must intersect D , i.e.

$$(U \cap V) \cap D \neq \phi$$

Let

$$z \in (U \cap V) \cap D$$

$$\Rightarrow z \in U \cap V \text{ and } z \in D$$

$$\Rightarrow z \in U \text{ and } z \in V$$

$$\Rightarrow z \in f^{-1}(U_1) \text{ and } z \in g^{-1}(V_1)$$

$$\Rightarrow f(z) \in U_1 \text{ and } g(z) \in V_1$$

Since $f(x) = g(x)$ for all $x \in D$, so $f(z) = g(z)$, because $z \in D$.

Let $f(z) = g(z) = h$, then $h \in U_1$ and $h \in V_1$.

So

$$h \in U_1 \cap V_1$$

$$\Rightarrow U_1 \cap V_1 \neq \phi$$

This is a contradiction because $U_1 \cap V_1 = \phi$, so our assumption that $f(y) \neq g(y)$ for some $y \in X$ is wrong. Consequently $f(x) = g(x)$ for all x in X .

Q.29 Show that a space X is Hausdorff space if and only if for any two distinct points x, y in X , there exist closed sets C_1 and C_2 such that

$$x \in C_1, y \notin C_1$$

$$y \in C_2, x \notin C_2$$

and

$$X = C_1 \cup C_2$$

PU, 2001 (M.Sc. Math)

Sol: Let X be a Hausdorff space, then for any $x, y \in X, x \neq y$, there exist open sets U and V such that

$$x \in U, y \notin U$$

$$y \in V, x \notin V$$

and

$$U \cap V = \phi$$

Let $V^c = C_1$, $U^c = C_2$, then both C_1 and C_2 , being the complements of open sets, are closed sets.

Now

$$x \notin V \Rightarrow x \in V^c, \text{ i.e. } x \in C_1$$

$$y \in V \Rightarrow y \notin V^c, \text{ i.e. } y \notin C_1$$

$$y \notin U \Rightarrow y \in U^c, \text{ i.e. } y \in C_2$$

$$x \in U \Rightarrow x \notin U^c, \text{ i.e. } x \notin C_2$$

$$U \cap V = \phi$$

$$\Rightarrow (U \cap V)^c = \phi^c$$

$$\Rightarrow U^c \cup V^c = X \quad \because \phi^c = X$$

$$\Rightarrow C_1 \cup C_2 = X \quad \because U^c = C_1, V^c = C_2$$

Conversely, let for any two distinct points x, y in X , there exist closed sets C_1 and C_2 such that

$$x \in C_1, y \notin C_1$$

$$y \in C_2, x \notin C_2$$

$$X = C_1 \cup C_2$$

and

Now

$$x \notin C_2 \Rightarrow x \in C_2^c$$

$$y \notin C_1 \Rightarrow y \in C_1^c$$

$$C_1 \cup C_2 = X$$

$$\Rightarrow (C_1 \cup C_2)^c = X^c$$

$$\Rightarrow C_1^c \cap C_2^c = \phi \quad \because X^c = \phi$$

Since both C_1 and C_2 are closed sets, so both C_1^c and C_2^c are open sets.

Thus, for any $x, y \in X, x \neq y$, there exist open sets C_1^c and C_2^c such that

$$x \in C_2^c, y \in C_1^c \text{ such that } C_1^c \cap C_2^c = \phi$$

This shows that X is a Hausdorff space.

Q.30 Show that a space X is Hausdorff space if and only if every point a of X is the intersection of its closed neighbourhoods.

PU, 2004 (M.Sc. Math)

Sol: Let X be a Hausdorff and $\{C_\alpha : \alpha \in \Omega\}$ be a collection of all closed neighbourhoods of any $a \in X$, then we have to show that

$$\{a\} = \bigcap_{\alpha \in \Omega} C_\alpha$$

For this let $x \in X, x \neq a$, then since X is a Hausdorff space, so there must exist open sets U and V such that

$$a \in U, x \in V, U \cap V = \phi$$

Now

$$x \in V \Rightarrow x \notin V^c$$

Also $U \cap V = \phi \Rightarrow U \subseteq V^c$
 and $a \in U$ and $U \subseteq V^c$
 gives $a \in U \subseteq V^c$

This shows that V^c is a closed neighbourhood of a , so it must be a member of $\{C_\alpha : \alpha \in \Omega\}$, i.e. $V^c = C_\alpha$ for some $\alpha \in \Omega$.

Since $x \notin V^c$, so $x \notin C_\alpha$ for some $\alpha \in \Omega$, therefore, $x \notin \bigcap_{\alpha \in \Omega} C_\alpha$.

Now an arbitrary point x of X different from a is not in $\bigcap_{\alpha \in \Omega} C_\alpha$.

This shows that $\bigcap_{\alpha \in \Omega} C_\alpha$ contains no point of X different from a , i.e.

$$\bigcap_{\alpha \in \Omega} C_\alpha = \{a\}$$

Conversely, let every point a of X be the intersection of its closed neighbourhoods, then we have to show that X is a hausdorff space.

For this let $x, y \in X$, $x \neq y$, then by the given condition, x must be the intersection of its closed neighbourhoods.

Let $\{C_\alpha : \alpha \in \Omega\}$ be a collection of all closed neighbourhoods of any x , such that

$$\{x\} = \bigcap_{\alpha \in \Omega} C_\alpha$$

Then $x \in C_\alpha$ and $y \notin C_\alpha$ for all $\alpha \in \Omega$.

Since each C_α is a closed neighbourhood of x , so there must exist an open set U such that

$$x \in U \subseteq C_\alpha$$

Since C_α , being the closed neighbourhood of x , is a closed set, so C_α^c is an open set.

Also $y \notin C_\alpha \Rightarrow y \in C_\alpha^c$.

Let $V = C_\alpha^c$, then V is an open set containing y .

Now $C_\alpha \cap C_\alpha^c = \phi$
 $\Rightarrow U \cap C_\alpha^c = \phi \quad \because U \subseteq C_\alpha$
 $\Rightarrow U \cap V = \phi \quad \because C_\alpha^c = V$

Thus, for any $x, y \in X$, $x \neq y$, there exist open sets U and V such that

$$x \in U, y \in V, U \cap V = \phi$$

This shows that X is a Hausdorff space.

Q.31 Give an example to show that in an arbitrary topological space X , a sequence may converge to more than one point.

Sol: Let X be an indiscrete topological space and $\{x_n\}$ be a sequence in X . Let $x, y \in X$ be any points of X such that $x \neq y$.

Since X is the only open set containing x such that $x_n \in X$ for all n . This satisfies the definition of the limit of the sequence, so x is the limit of the sequence $\{x_n\}$, i.e. $x_n \rightarrow x$.

Similarly, X is the only open set containing y such that $x_n \in X$ for all n . This satisfies the definition of the limit of the sequence, so y is the limit of the sequence $\{x_n\}$, i.e. $x_n \rightarrow y$.

This shows that the sequence $\{x_n\}$ converges to both x and y .

Q.32 Show that in a Hausdorff space any sequence can converge to at most one point.

Sol: Let X be a Hausdorff space and $\{x_n\}$ a sequence in X . Suppose on contrary that the sequence $\{x_n\}$ converges to two different points x and y of X .

Since $x, y \in X$, $x \neq y$ and X is a Hausdorff space, so there exist open sets U and V such that

$$x \in U, y \in V, U \cap V = \phi$$

Since $x_n \rightarrow x$ and U is an open set containing x , so there exists a natural number n_1 such that

$$x_n \in U \text{ for all } n \geq n_1 \quad \dots(1)$$

Since $x_n \rightarrow y$ and V is an open set containing y , so there exists a natural number n_2 such that

$$x_n \in V \text{ for all } n \geq n_2 \quad \dots(2)$$

Let $n_0 = \max(n_1, n_2)$, then (1) and (2) take the form

$$x_n \in U \text{ for all } n \geq n_0 \quad \dots(3)$$

and

$$x_n \in V \text{ for all } n \geq n_0 \quad \dots(4)$$

From (3) and (4), we have

$$\begin{aligned} x_n &\in U \cap V \text{ for all } n \geq n_0 \\ &\Rightarrow U \cap V \neq \phi \end{aligned}$$

This is not possible because $U \cap V = \phi$.

Thus, our assumption that the sequence $\{x_n\}$ converges to two points is wrong. Consequently, the sequence $\{x_n\}$ converges to at most one point.

Q.33 Show that a first countable space is a Hausdorff space if and only if every convergent sequence has a unique limit.

University of Gujrat, 2013 (M.Sc. Math)

Sol: Let X be a first countable space.

Let X be a Hausdorff space and $\{x_n\}$ a sequence in X . Suppose on contrary that the sequence $\{x_n\}$ converges to two different points x and y of X .

Since $x, y \in X$, $x \neq y$ and X is a Hausdorff space, so there exist open sets U and V such that $x \in U$, $y \in V$, $U \cap V = \emptyset$.

Since $x_n \rightarrow x$ and U is an open set containing x , so there exists a natural number n_1 such that

$$x_n \in U \text{ for all } n \geq n_1 \quad \dots(1)$$

Since $x_n \rightarrow y$ and V is an open set containing y , so there exists a natural number n_2 such that

$$x_n \in V \text{ for all } n \geq n_2 \quad \dots(2)$$

Let $n_0 = \max(n_1, n_2)$; then (1) and (2) take the form

$$x_n \in U \text{ for all } n \geq n_0 \quad \dots(3)$$

and

$$x_n \in V \text{ for all } n \geq n_0 \quad \dots(4)$$

From (3) and (4), we have

$$\begin{aligned} x_n &\in U \cap V \text{ for all } n \geq n_0 \\ &\Rightarrow U \cap V \neq \emptyset \end{aligned}$$

This is not possible because $U \cap V = \emptyset$.

Thus, our assumption that the sequence $\{x_n\}$ converges to two points is wrong. Consequently, the sequence $\{x_n\}$ converges to at most one point. This shows that the convergent sequence has a unique limit.

Conversely, let X be first countable space in which every convergent sequence has a unique limit, then we have to show that X is a Hausdorff space.

Suppose on contrary that X is not a Hausdorff space, then for a pair a, b of distinct points in X , every open set which contains a has a nonempty intersection with every open set which contains b . If $\{U_n\}$ and $\{V_n\}$ are countable local bases at a and b respectively, then

$$U_n \cap V_n \neq \emptyset \text{ for all } n$$

If $a_n \in U_n \cap V_n$, $n = 1, 2, 3, \dots$

Then the sequence $\{a_n\}$ converges to both a and b . This is a contradiction, so X is a Hausdorff space.

Q.34 Show that every subspace of a regular space is a regular space.

Sol: Let X be a regular space. Let Y be a subspace of X , then we have to show that Y is a regular space. For this, let $x \in Y$ and A be a closed subset of Y not containing x , i.e. $x \notin A$.

Since Y is a subspace of X , so a closed set B in X is defined as

$$A = Y \cap B$$

Since $x \notin A$, so $x \notin Y \cap B \Rightarrow x \notin B$, because $x \in Y$.

Q.35
Sol:

Since $x \in Y$ and $Y \subseteq X$, so $x \in X$.

This shows that B is a closed subset of X not containing x . Since X is a regular space, so there must exist open sets U_1 and V_1 in X such that

$$x \in U_1, B \subseteq V_1 \text{ and } U_1 \cap V_1 = \emptyset$$

Since Y is a subspace of X , so open sets U and V in Y are defined as

$$U = Y \cap U_1 \text{ and } V = Y \cap V_1$$

Since $x \in Y$ and $x \in U_1$, so $x \in Y \cap U_1$, i.e. $x \in U$.

Since $A = Y \cap B \Rightarrow A \subseteq B$ and $B \subseteq V_1$, so $A \subseteq V_1$.

Since $A \subseteq Y$ and $A \subseteq V_1$, so $A \subseteq Y \cap V_1$, i.e. $A \subseteq V$.

We conclude that for a point $x \in Y$ and a closed subset A of Y , there exist open sets U and V in Y such that

$$x \in U, A \subseteq V \text{ and } U \cap V = \emptyset$$

This shows that Y is a regular space.

Q.35 Show that every T_3 -space is a Hausdorff space.

Sol: Let X be a T_3 -space, then we have to show that X is a Hausdorff space.

For this let $x, y \in X, x \neq y$.

Since X , being a T_3 -space, is a T_1 -space. Since the singleton subsets of T_1 -space are closed, so $A = \{y\}$ is a closed subset of X .

Since X , being a T_3 -space, is a regular space, so for $x \in X$ and the closed set A not containing x , there exist open sets U and V such that

$$x \in U, A \subseteq V \text{ and } U \cap V = \emptyset$$

$$\Rightarrow x \in U, y \in V \text{ and } U \cap V = \emptyset \quad \because A = \{y\}$$

Thus, for $x, y \in X, x \neq y$, there exist open sets U and V such that

$$x \in U, y \in V \text{ and } U \cap V = \emptyset$$

This shows that x is a Hausdorff space.

Q.36 Show that the following statements about a topological space are equivalent.

(a) X is regular.

(b) For any open set U in X and $x \in U$, there is an open set V containing x such that $x \in \bar{V} \subseteq U$.

(c) Each element of X has a neighbourhood base consisting of closed sets.

Sol: (a) $\Rightarrow$ (b) Let x be a regular space. Let $x \in X$ and U be any open set containing x , then U^c is a closed set not containing x .

Since X is a regular space, so for $x \in X$ and the closed set U^c

not containing x , there exist open sets V and V_1 such that

$$x \in V, U^c \subseteq V_1 \text{ and } V \cap V_1 = \emptyset$$

Now

$$U^c \subseteq V_1$$

$$\Rightarrow V_1^c \subseteq U \quad \dots(1)$$

and

$$V \cap V_1 = \emptyset$$

$$\Rightarrow V \subseteq V_1^c \quad \dots(2)$$

Since V_1 is an open set, so (2) shows that V_1^c is a closed superset of V . Since the smallest closed superset of V is $\bar{V}$, so

$$V \subseteq \bar{V} \subseteq V_1^c$$

$$\Rightarrow x \in V \subseteq \bar{V} \subseteq V_1^c \quad \because x \in V$$

$$\Rightarrow x \in \bar{V} \subseteq V_1^c \quad \dots(3)$$

Combining (1) and (3), we have

$$x \in \bar{V} \subseteq V_1^c \subseteq U$$

$$\Rightarrow x \in \bar{V} \subseteq U$$

(b) $\Rightarrow$ (c) For any open set U in X and $x \in U$, there is an open set V containing x such that $x \in \bar{V} \subseteq U$, then we have to show that each element of X has a neighbourhood base consisting of closed sets. For this let y be any element of X , then for every open set U containing y , there is an open set V containing y such that

$$y \in \bar{V} \subseteq U$$

Thus, for each $y \in X$, the sets of the form $\bar{V}$ form a closed neighbourhood base consisting of closed subsets of X .

(c) $\Rightarrow$ (a) Let each element of X have a neighbourhood base consisting of closed sets, then we have to show that X is a regular space.

For this let $x \in X$ and A be any closed set not containing x , then A^c is an open set containing x . Since each point of X has a neighbourhood base, so let $\beta = \{B_\alpha : \alpha \in \Omega\}$ be a neighbourhood base of x , where each B_α is a closed set.

Since β is a neighbourhood base of x and A^c is an open set containing x , so by the definition of neighbourhood base, there is some $B_\alpha \in \beta$ such that

$$x \in B_\alpha \subseteq A^c$$

$$\Rightarrow B_\alpha \subseteq A^c$$

$$\Rightarrow A \subseteq B_\alpha^c \quad \dots(1)$$

If U is an open set containing x , then by the definition of base, U is

the union of members of β , i.e.

$$U = \bigcup_{\gamma \in \Omega_\gamma} B_\gamma, \text{ where } \Omega_\gamma \subseteq \Omega$$

Now

$$B_\gamma \cap B_\alpha^c = \phi$$

$$\Rightarrow \bigcup_{\gamma \in \Omega_\gamma} B_\gamma \cap B_\alpha^c = \phi$$

$$\Rightarrow U \cap B_\alpha^c = \phi$$

Thus, for $x \in X$ and a closed set A not containing x , there exist open sets U and B_α^c such that

$$x \in U, A \subseteq B_\alpha^c \text{ and } U \cap B_\alpha^c = \phi$$

This shows that X is a regular space.

Q.37 Prove that every subspace of a T_3 -space is a T_3 -space.

Sol: Let X be a T_3 -space. Let Y be a subspace of X . To show that Y is a T_3 -space, we have to show that Y is a regular T_1 -space.

For this, let $x \in Y$ and A be a closed subset of Y not containing x , i.e. $x \notin A$.

Since Y is a subspace of X , so a closed set B in X is defined as

$$A = Y \cap B$$

Since $x \notin A$, so $x \notin Y \cap B \Rightarrow x \notin B$, because $x \in Y$.

Since $x \in Y$ and $Y \subseteq X$, so $x \in X$.

This shows that B is a closed subset of X not containing x . Since X is a regular space, so there must exist open sets U_1 and V_1 in X such that

$$x \in U_1, B \subseteq V_1 \text{ and } U_1 \cap V_1 = \phi$$

Since Y is a subspace of X , so open sets U and V in Y are defined as

$$U = Y \cap U_1 \text{ and } V = Y \cap V_1$$

Since $x \in Y$ and $x \in U_1$, so $x \in Y \cap U_1$, i.e. $x \in U$.

Since $A = Y \cap B \Rightarrow A \subseteq B$ and $B \subseteq V_1$, so $A \subseteq V_1$.

Since $A \subseteq Y$ and $A \subseteq V_1$, so $A \subseteq Y \cap V_1$, i.e. $A \subseteq V$.

We conclude that for a point $x \in Y$ and a closed subset A of Y , there exist open sets U and V in Y such that

$$x \in U, A \subseteq V \text{ and } U \cap V = \phi$$

This shows that Y is a regular space.

Next suppose that $x, y \in Y, x \neq y$, then $x, y \in X$. Since X is a T_1 -space, so there exist open subsets U_1, V_1 such that $x \in U_1, y \notin U_1$ and $y \in V_1, x \notin V_1$. Since Y is a subspace of X , so open sets U and V in Y are defined as

$$U = U_1 \cap Y \text{ and } V = V_1 \cap Y$$

Now

$$x \in U_1 \cap Y \Rightarrow x \in U$$

And $y \notin U_1 \Rightarrow y \notin U_1 \cap Y = V$

Similarly, $y \in V, y \notin V$.

This shows that Y is a T_1 -space.

Thus, Y being a regular T_1 -space, is a T_3 -space.

Q.38 Define a completely regular space. Prove that every subspace of a completely regular space is completely regular.

PU, 2013; 2002 (M.Sc. Math)

Sol: Completely Regular Space:

A topological space X is said to be a completely regular space if for any $x \in X$ and a closed set A not containing x , there exists a continuous function $f: X \rightarrow [0,1]$ such that

$$f(x) = 0 \text{ and } f(A) = 1$$

Let X be a completely regular space and Y a subspace of X , then we have to show that Y is a completely regular space. For this let $x \in Y$ and A be a closed subset of Y not containing x .

Since A is closed in Y and Y is a subspace of X , so a closed set A_1 in X is defined as

$$A = Y \cap A_1$$

Since $x \notin A$, so $x \notin Y \cap A_1 \Rightarrow x \notin A_1$, because $x \in Y$.

Since $Y \subseteq X$ and $x \in Y$, so $x \in X$.

Since X is a completely regular space, so for $x \in X$ and a closed set A_1 in X not containing x , there exists a continuous function $f: X \rightarrow [0,1]$ such that

$$f(x) = 0 \text{ and } f(A_1) = 1$$

Next consider the function $g: Y \rightarrow [0,1]$ defined as

$$g(x) = f(x), \quad \forall x \in Y$$

Then g , being the restriction of a continuous function f to Y , is also continuous

Now $g(x) = f(x) = 0$

And $g(A) = f(A) = f(Y \cap A_1)$
 $= f(Y) \cap f(A_1)$
 $= [0,1] \cap \{1\} = 1$

Thus, for any $x \in Y$ and a closed set $A \subseteq Y$ not containing x , there exists a continuous function $g: Y \rightarrow [0,1]$ such that

$$g(x) = 0 \text{ and } g(A) = 1$$

This shows that Y is also a completely regular space.

Hence every subspace of a completely regular space is a completely regular space.

Q.39 Show that a completely regular space is regular.

PU, 2013 (M.Sc. Math)

Sol: Let X be a completely regular space, then we have to show that X is a regular space. For this let $x \in X$ and A be a closed set not containing x .

Since X is a completely regular space, so for $x \in X$ and a closed set A in X not containing x , there exists a continuous function $f: X \rightarrow [0,1]$ such that

$$f(x) = 0 \text{ and } f(A) = 1$$

Since $[0,1]$ is a subspace of R and $\left]-\frac{1}{2}, \frac{1}{2}\right[$ is an open set in R , so

$$\left[0, \frac{1}{2}\right[= [0,1] \cap \left]-\frac{1}{2}, \frac{1}{2}\right[$$

is an open set in $[0,1]$.

Similarly,

$$\left]\frac{1}{2}, 1\right] = [0,1] \cap \left]\frac{1}{2}, 1\right]$$

is an open set in $[0,1]$.

Thus, $U = \left[0, \frac{1}{2}\right[$ and $V = \left]\frac{1}{2}, 1\right]$ are open sets in $[0,1]$.

Since $f: X \rightarrow [0,1]$ is continuous, so $f^{-1}(U)$ and $f^{-1}(V)$ are open in X .

Now

$$0 \in \left[0, \frac{1}{2}\right[$$

$$\Rightarrow f(x) \in \left[0, \frac{1}{2}\right[\quad \because f(x) = 0$$

$$\Rightarrow x \in f^{-1}\left(\left[0, \frac{1}{2}\right[\right)$$

$$\Rightarrow x \in f^{-1}(U) \quad \because U = \left[0, \frac{1}{2}\right[$$

Similarly

$$1 \in \left]\frac{1}{2}, 1\right]$$

$$\Rightarrow f(A) \in \left]\frac{1}{2}, 1\right] \quad \because f(A) = 1$$

$$\Rightarrow A \subseteq f^{-1}\left(\left]\frac{1}{2}, 1\right]\right)$$

$$\Rightarrow A \subseteq f^{-1}(V) \quad \because V = \left] \frac{1}{2}, 1 \right]$$

$$\begin{aligned} \text{Also } f^{-1}(U) \cap f^{-1}(V) &= f^{-1}(U \cap V) \\ &= f^{-1}(U \cap V) \\ &= f^{-1}\left(\left[0, \frac{1}{2}\right] \cap \left] \frac{1}{2}, 1 \right]\right) \\ &= f^{-1}(\emptyset) \\ &= \emptyset \end{aligned}$$

Thus, for any $x \in X$ and a closed set A not containing x , there exist open sets $f^{-1}(U)$ and $f^{-1}(V)$ in X such that

$$x \in f^{-1}(U), A \subseteq f^{-1}(V) \text{ and } f^{-1}(U) \cap f^{-1}(V) = \emptyset$$

This shows that X is a regular space.

Q.40 Show that every subspace of Tychonoff space is Tychonoff.
 University of Sargodha, 2010 (M.Sc. Math)

Sol: Let X be a Tychonoff space and Y a subspace of X . To show that Y is a Tychonoff space, first we show that Y is a completely regular space. For this let $x \in Y$ and A be a closed subset of Y not containing x .

Since A is closed in Y and Y is a subspace of X , so a closed set A_1 in X is defined as

$$A = Y \cap A_1$$

Since $x \notin A$, so $x \notin Y \cap A_1 \Rightarrow x \notin A_1$, because $x \in Y$.

Since $Y \subseteq X$ and $x \in Y$, so $x \in X$.

Since X , being a Tychonoff space, is a completely regular space, so for $x \in X$ and a closed set A_1 in X not containing x , there exists a continuous function $f: X \rightarrow [0, 1]$ such that

$$f(x) = 0 \text{ and } f(A_1) = 1$$

Next consider the function $g: Y \rightarrow [0, 1]$ defined as

$$g(x) = f(x), \quad \forall x \in Y$$

Then g , being the restriction of a continuous function f to Y , is also continuous

$$\text{Now } g(x) = f(x) = 0$$

$$\begin{aligned} \text{And } g(A) &= f(A) = f(Y \cap A_1) \\ &= f(Y) \cap f(A_1) \\ &= [0, 1] \cap \{1\} \\ &= 1 \end{aligned}$$

Thus, for any $x \in Y$ and a closed set $A \subseteq Y$ not containing x , there exists a continuous function $g: Y \rightarrow [0, 1]$ such that

$$g(x) = 0 \text{ and } g(A) = 1$$

This shows that Y is also a completely regular space.

Next we show that Y is a T_1 -space. For this, let $x, y \in Y, x \neq y$, then $x, y \in X$. Since X , being a Tychonoff space, is a T_1 -space, so there exist open subsets U_1, V_1 of X such that $x \in U_1, y \notin U_1$ and $y \in V_1, x \notin V_1$. Since Y is a subspace of X , so open sets U and V in Y are defined as

$$U = U_1 \cap Y \text{ and } V = V_1 \cap Y$$

Now

$$x \in U_1 \cap Y \Rightarrow x \in U$$

And

$$y \notin U_1 \Rightarrow y \notin U_1 \cap Y = V$$

Similarly, $y \in V, x \notin V$.

This shows that Y is a T_1 -space.

Thus, Y , being a completely regular T_1 -space, is a Tychonoff space.

Q.41 Show that every completely regular T_1 -space (Tychonoff space) is Hausdorff.

PU, 2012; 2006; 2000 (M.Sc. Math)

Sol: Let X be a Tychonoff space, then we have to show that X is a Hausdorff space. For this let $x, y \in X, x \neq y$.

Since X , being a Tychonoff space, is a T_1 -space. Since the singleton subsets of T_1 -space are closed, so $A = \{y\}$ is a closed subset of X .

Since X , being a Tychonoff space, is a completely regular space. By Theorem 3-5.3, X is regular, so for $x \in X$ and the closed set A not containing x , there exist open sets U and V such that

$$x \in U, A \subseteq V \text{ and } U \cap V = \phi$$

$$\Rightarrow x \in U, y \in V \text{ and } U \cap V = \phi \quad \because A = \{y\}$$

Thus, for $x, y \in X, x \neq y$, there exist open sets U and V such that

$$x \in U, y \in V \text{ and } U \cap V = \phi$$

This shows that x is a Hausdorff space.

Q.42 Give an example to show that a normal space may not be a Hausdorff space.

Sol: Let $X = \{a, b, c\}$ and $\tau = \{\phi, \{a\}, \{b, c\}, X\}$.

Since $\{a\}$ is open, so $\{a\}^c = \{b, c\}$ is closed.

Similarly, since $\{b, c\}$ is open, so $\{b, c\}^c = \{a\}$ is closed.

Also $\{a\}$ and $\{b, c\}$ are the only disjoint nonempty closed subsets of X , and there exist open sets $\{a\}$ and $\{b, c\}$ such that

$$\{a\} \subseteq \{a\}, \{b, c\} \subseteq \{b, c\} \text{ and } \{a\} \cap \{b, c\} = \phi$$

This shows that X is a normal space.

Since $b \neq c$ and we cannot find open sets U and V such that

$$b \in U, c \in V \text{ and } U \cap V = \emptyset$$

This shows that X is not a Hausdorff space.

Q.43 Show that every closed subspace of a normal space is normal.

PU, 2005 (M.Sc. Math)

Sol: Let X be a normal space and Y a closed subspace of X , then we have to show that Y is also a normal space. For this let A and B be any disjoint closed subsets of Y . Since Y is a subspace of X , so the closed sets A_1 and B_1 in X are defined as

$$A = Y \cap A_1, \quad B = Y \cap B_1$$

Since Y and A_1 are closed in X , so A , being the intersection of two closed sets, is closed in X .

Since Y and B_1 are closed in X , so B , being the intersection of two closed sets, is closed in X .

Thus, A and B are disjoint closed sets in X .

Since X is normal space, so there exist open sets U_1 and V_1 in X such that

$$A \subseteq U_1, B \subseteq V_1 \text{ and } U_1 \cap V_1 = \emptyset$$

The open sets U and V in Y are defined as

$$U = Y \cap U_1, \quad V = Y \cap V_1$$

Since $A \subseteq U_1$ and $A \subseteq Y$, so $A \subseteq Y \cap U_1$, i.e. $A \subseteq U$.

Since $B \subseteq V_1$ and $B \subseteq Y$, so $B \subseteq Y \cap V_1$, i.e. $B \subseteq V$.

Now $U \cap V = (Y \cap U_1) \cap (Y \cap V_1)$

$$= Y \cap (U_1 \cap V_1)$$

$$= Y \cap \emptyset \quad \because U_1 \cap V_1 = \emptyset$$

$$= \emptyset$$

Thus, for any disjoint closed sets A and B in Y , there exist open sets U and V in Y such that

$$A \subseteq U, B \subseteq V \text{ and } U \cap V = \emptyset$$

This shows that Y is a normal space.

Q.44 Show that a closed continuous image of a normal space is normal.

PU, 2007; 2006 (M.Sc. Math)

Sol: Let X be a normal space and $Y = f(X)$ be its closed continuous image.

Then we have to show that Y is also normal. For this let A and B be disjoint closed subsets of Y . Then by the continuity of f , $f^{-1}(A)$ and $f^{-1}(B)$ are closed subsets of X .

Also

$$\begin{aligned}
 f^{-1}(A) \cap f^{-1}(B) &= f^{-1}(A \cap B) \\
 &= f^{-1}(\phi) = \phi \quad \because A \cap B = \phi
 \end{aligned}$$

This shows that $f^{-1}(A)$ and $f^{-1}(B)$ are disjoint closed sets in X .

Since X is normal space, so there must exist open sets U_1 and V_1 in X such that

$$f^{-1}(A) \subseteq U_1, f^{-1}(B) \subseteq V_1 \text{ and } U_1 \cap V_1 = \phi$$

Since U_1 is open in X , so U_1^c is closed in X .

Since f is closed, so $f(U_1^c)$ is closed in Y , i.e. $f(Y \setminus U_1)$ is closed in Y . This shows that $f(Y) \setminus f(U_1)$ is closed in Y , so $f(U_1)$ is open in Y .

Similarly, $f(V_1)$ is open in Y .

Let $U = f(U_1)$ and $V = f(V_1)$

Now

$$\begin{array}{l|l}
 f^{-1}(A) \subseteq U_1 & f^{-1}(B) \subseteq V_1 \\
 \Rightarrow A \subseteq f(U_1) & \Rightarrow B \subseteq f(V_1) \\
 \Rightarrow A \subseteq U & \Rightarrow B \subseteq V
 \end{array}$$

Also

$$\begin{aligned}
 U \cap V &= f(U_1) \cap f(V_1) \\
 &= f(U_1 \cap V_1) \\
 &= f(\phi) \because U_1 \cap V_1 = \phi \\
 &= \phi
 \end{aligned}$$

Thus, for disjoint closed sets A and B in Y , there exist open sets U and V in Y such that

$$A \subseteq U, B \subseteq V \text{ and } U \cap V = \phi$$

This shows that $Y = f(X)$ is normal.

Hence, the closed continuous image of a normal space is normal.

Q.45 Show that a topological space is normal if and only if for any closed set A and an open set U containing A , there is at least one open set V containing A such that $A \subseteq V \subseteq \bar{V} \subseteq U$.

PU, 2008 (M.Sc. Math)

Sol: Let X be a normal space and A be a close subset of X . Let U be an open subset of X containing A . Then U^c is a closed set such that

$$A \cap U^c = \phi \quad \because A \subseteq U$$

Thus, A and U^c are disjoint closed subsets of X . Since X is normal, so there must exist disjoint open sets V and V_1 such that

$$A \subseteq V, U^c \subseteq V_1$$

Since $V \cap V_1 = \phi$, so $V \subseteq V_1^c$.

But V_1 is open, so V_1^c is the closed superset of V . Since the smallest closed superset of V is $\bar{V}$, so

$$\bar{V} \subseteq V_1^c \quad \dots(1)$$

Also

$$V \subseteq \bar{V} \quad \dots(2)$$

and

$$A \subseteq V \quad \dots(3)$$

From $U^c \subseteq V_1$, we have

$$V_1^c \subseteq U \quad \dots(4)$$

Combining (1), (2), (3) and (4), we have

$$\begin{aligned} A \subseteq V \subseteq \bar{V} \subseteq V_1^c \subseteq U \\ \Rightarrow A \subseteq V \subseteq \bar{V} \subseteq U \end{aligned}$$

Conversely, let for any closed set A and an open set U containing A , there is at least one open set V containing A such that $A \subseteq V \subseteq \bar{V} \subseteq U$, then we have to show that X is normal space. For this let A_1, B_1 be two disjoint closed sets.

Since $A_1 \cap B_1 = \phi$, so $A_1 \subseteq B_1^c$.

Since B_1 is closed, so B_1^c is an open set containing A_1 . Therefore, by the given condition, there exists at least one open set V such that

$$\begin{aligned} A_1 \subseteq V \subseteq \bar{V} \subseteq B_1^c \\ \Rightarrow A_1 \subseteq V, \bar{V} \subseteq B_1^c \\ \Rightarrow A_1 \subseteq V, B_1 \subseteq (\bar{V})^c \end{aligned}$$

Since $\bar{V}$ is closed, so $(\bar{V})^c$ is open.

Also

$$\begin{aligned} \bar{V} \cap (\bar{V})^c &= \phi \\ \Rightarrow V \cap (\bar{V})^c &= \phi \quad \because V \subseteq \bar{V} \end{aligned}$$

Thus, for any disjoint closed sets A_1, B_1 in X , there exist disjoint open sets V and $(\bar{V})^c$ such that

$$A_1 \subseteq V, B_1 \subseteq (\bar{V})^c$$

This shows that X is a normal space.

Q.46 Show that every closed subspace of a T_4 -space is a T_4 -space.

PU, 2005 (M.Sc. Math)

Sol: Let X be T_4 -space and Y a closed subspace of X , then we have to

show that Y is a T_4 -space. We first show that Y is a normal space. For this let A and B be any disjoint closed subsets of Y . Since Y is a subspace of X , so the closed sets A_1 and B_1 in X are defined as

$$A = Y \cap A_1, \quad B = Y \cap B_1$$

Since Y and A_1 are closed in X , so A , being the intersection of two closed sets, is closed in X .

Since Y and B_1 are closed in X , so B , being the intersection of two closed sets, is closed in X .

Thus, A and B are disjoint closed sets in X .

Since X is normal space, so there exist open sets U_1 and V_1 in X such that $A \subseteq U_1, B \subseteq V_1$ and $U_1 \cap V_1 = \phi$

The open sets U and V in Y are defined as

$$U = Y \cap U_1, \quad V = Y \cap V_1$$

Since $A \subseteq U_1$ and $A \subseteq Y$, so $A \subseteq Y \cap U_1$, i.e. $A \subseteq U$.

Since $B \subseteq V_1$ and $B \subseteq Y$, so $B \subseteq Y \cap V_1$, i.e. $B \subseteq V$.

Now

$$\begin{aligned} U \cap V &= (Y \cap U_1) \cap (Y \cap V_1) \\ &= Y \cap (U_1 \cap V_1) \\ &= Y \cap \phi \quad \because U_1 \cap V_1 = \phi \\ &= \phi \end{aligned}$$

Thus, for any disjoint closed sets A and B in Y , there exist open sets U and V in Y such that

$$A \subseteq U, B \subseteq V \text{ and } U \cap V = \phi$$

This shows that Y is a normal space.

Next we show that Y is a T_1 -space. For this let $x, y \in Y, x \neq y$, then $x, y \in X$. Since X , being a normal space, is a T_1 -space, so there exist open subsets U_1, V_1 of X such that $x \in U_1, y \notin U_1$ and $y \in V_1, x \notin V_1$. Since Y is a subspace of X , so open sets U and V in Y are defined as

$$U = U_1 \cap Y \text{ and } V = V_1 \cap Y$$

Now $x \in U_1 \cap Y \Rightarrow x \in U$

And $y \notin U_1 \Rightarrow y \notin U_1 \cap Y = V$

Similarly, $y \in V, x \notin V$. This shows that Y is a T_1 -space.

Thus, Y being the normal T_1 -space, is a T_4 -space.

- Q.47** Consider $X = \{a, b\}$ to be a cofinite topological space. Determine whether X is a normal space. Test the normality condition on the pair $\{a\}$ and $\{b\}$ of closed subsets of X .

PU, 2004 (M.Sc. Math)

Sol: $X = \{a, b\}$

If τ_c is a cofinite topology on X , then by the definition of cofinite topology, we have

$$\phi \in \tau_c$$

$$X \in \tau_c$$

$$\{a\} \in \tau_c$$

$$\{b\} \in \tau_c$$

$$\because X^c = \phi = \text{finite}$$

$$\because \{a\}^c = \{b\} = \text{finite}$$

$$\because \{b\}^c = \{a\} = \text{finite}$$

Thus, $\tau_c = \{\phi, \{a\}, \{b\}, X\}$.

Now for disjoint pair of closed sets $\{a\}$ and $\{b\}$, there exist open sets $\{a\}$ and $\{b\}$ such that

$$\{a\} \subseteq \{a\}, \{b\} \subseteq \{b\} \quad \text{and} \quad \{a\} \cap \{b\} = \phi$$

This shows that (X, τ_c) is a normal space.

Chapter 4

CONNECTEDNESS

EXERCISE 4

Short Questions

Q.2 Solve / answer the following short questions:

(i) Define separated sets.

Sol: Two subsets A and B of a topological space X are said to be separated iff $\overline{A} \cap B = \phi$ and $A \cap \overline{B} = \phi$.

(ii) Show that intervals $A = (-1, 0)$ and $B = (0, 1)$ on the real line R are separated.

Sol: Since all real numbers between -1 and 0 including -1 and 1 are limit points of $A = (-1, 0)$, so derived set of A is

$$A^d = [-1, 0]$$

Now

$$\overline{A} = A \cup A^d = (-1, 0) \cup [-1, 0] = [-1, 0]$$

Similarly,

$$\overline{B} = B \cup B^d = (0, 1) \cup [0, 1] = [0, 1]$$

Next consider

$$\overline{A} \cap B = [-1, 0] \cap (0, 1) = \phi$$

$$A \cap \overline{B} = (-1, 0) \cap [0, 1] = \phi$$

This shows that A and B are separated sets.

(iii) Show that intervals $A = (0, \frac{1}{2})$ and $B = [\frac{1}{2}, 1)$ on the real line R are not separated.

Sol: Since all real numbers between 0 and $\frac{1}{2}$ including 0 and $\frac{1}{2}$ are limit points of $A = (0, \frac{1}{2})$, so derived set of A is

$$A^d = [0, \frac{1}{2}]$$

Now

$$\overline{A} = A \cup A^d = (0, \frac{1}{2}) \cup [0, \frac{1}{2}] = [0, \frac{1}{2}]$$

Next consider

$$\overline{A} \cap B = [0, \frac{1}{2}] \cap [\frac{1}{2}, 1) = \{\frac{1}{2}\} \neq \emptyset$$

This shows that A and B are not separated sets.

(iv) **Define disconnected space.**

Sol: A topological space X is said to be disconnected space if X can be represented as the union of two nonempty disjoint open sets.

(v) **Show that two point discrete space is disconnected.**

Sol: Let X be a two point discrete space. If A is any proper subset of X , then both A and A^c are nonempty open subsets of X such that

$$A \cap A^c = \emptyset \text{ and } A \cup A^c = X$$

This shows that $\{A, A^c\}$ is a disconnection of X , so X is a disconnected space.

(vi) **Define connected space.**

Sol: A topological space which cannot be written as the union of two nonempty disjoint open sets is said to be a connected space.

(vii) **Define connected subspace.**

Sol: A subspace Y of a topological space is said to be connected subspace if Y is connected as a topological space in its own right.

(viii) **Give an example to show that a subset of a connected space may not be connected.**

Sol: The set of real numbers R cannot be represented as the union of two disjoint nonempty sets, so R is a connected space. Next suppose that

$$A = Q \cap]-\infty, \sqrt{2}[\text{ and } B = Q \cap]\sqrt{2}, \infty[$$

Since $]-\infty, \sqrt{2}[$ and $]\sqrt{2}, \infty[$ are open subsets of R , so A and B are open subsets of Q . Also

$$A \cap B = \emptyset$$

and

$$A \cup B = Q$$

This shows that Q is a disconnected subspace of R .

(ix) **Show that every indiscrete space is connected.**

Sol: Let X be an indiscrete space, then X is the only nonempty open set, so we cannot find the disconnection of X . Hence X is connected.

(x) **Define the component of a topological space X .**

PU, 2011 (M.Sc. Math)

Sol: A connected subspace of a topological space X is said to be the component of X if it is not properly contained in any connected subspace of X .

(xi) **Show that the singleton subsets of a two-point discrete space are its components.**

Sol: Let X be a two-point discrete space, then the only possible connected subsets of X are its singleton subsets. As no singleton

subset is properly contained in any other singleton subset, so these singleton subsets are components of X .

Hence the singleton subsets of a two-point discrete space are its components.

(xii) **Define totally disconnected space.**

Sol: A topological space X is said to be totally disconnected space if for any two points x and y of X , there is a disconnection $\{A, B\}$ of X such that

$$x \in A \text{ and } y \in B$$

(xiii) **Every discrete space is totally disconnected.**

Sol: Let X be a discrete space. Let $x, y \in X$, $x \neq y$, then

$$A = \{x\} \text{ and } A^c = X \setminus \{x\}$$

are open subsets of X such that $x \in A$, $y \in A^c$.

Since $\{A, A^c\}$ is a disconnection of X , so X is totally disconnected.

Long Questions

Q.3 Show that a topological space is connected if and only if it cannot be represented as the union of two disjoint nonempty closed sets.

Sol: Let X be a connected space. Suppose on contrary that X can be represented as the union of two disjoint nonempty closed sets, i.e. there exist nonempty closed sets A and B such that

$$A \cap B = \phi \text{ and } A \cup B = X$$

Now

$$A \cap B = \phi$$

$$\Rightarrow (A \cap B)^c = \phi^c$$

$$\Rightarrow A^c \cup B^c = X$$

Similarly

$$A \cup B = X$$

$$\Rightarrow (A \cup B)^c = X^c$$

$$\Rightarrow A^c \cap B^c = \phi$$

...(1)

...(2)

Since $A \neq \phi$, $B \neq \phi$ and $A \cap B = \phi$, so $A \neq X$, $B \neq X$.

This shows that $A^c \neq \phi$, $B^c \neq \phi$.

Since both A and B are closed sets, so A^c and B^c are open sets.

Since A^c and B^c are nonempty disjoint open sets whose union is X , so X is disconnected space. This is a contradiction, so X cannot be written as the union of two disjoint nonempty closed sets.

Conversely, let X cannot be written as the union of two disjoint nonempty closed sets, then we have to show that X is connected. For this suppose on contrary that X is disconnected, then there must exist disjoint nonempty open sets A and B such that

Now

$$A \cup B = X$$

$$A \cap B = \phi$$

$$\Rightarrow (A \cap B)^c = \phi^c$$

$$\Rightarrow A^c \cup B^c = X$$

...(3)

Similarly

$$A \cup B = X$$

$$\Rightarrow (A \cup B)^c = X^c$$

$$\Rightarrow A^c \cap B^c = \phi$$

...(4)

Since $A \neq \phi$, $B \neq \phi$ and $A \cap B = \phi$, so $A \neq X$, $B \neq X$.

This shows that $A^c \neq \phi$, $B^c \neq \phi$.

Since both A and B are open sets, so A^c and B^c are closed sets. It is clear from (3) and (4) that X can be written as the union of two nonempty disjoint closed sets which is a contradiction, so X is a connected space.

Q.4 Show that an infinite set with cofinite topology is a connected space.

PU, 2012; 2008 (M.Sc. Math)

Sol: Let X be an infinite set. Let τ be cofinite topology on X , then we have to show that (X, τ) is a connected space. For this suppose on contrary that (X, τ) is a disconnected space, then we must have nonempty open sets A and B such that

$$A \cap B = \phi \quad \dots(1)$$

and

$$A \cup B = X \quad \dots(2)$$

From (1), we have

$$(A \cap B)^c = \phi^c$$

$$\Rightarrow A^c \cup B^c = X \quad \dots(3)$$

Since A and B are open sets and X is a cofinite topological space, so both A^c and B^c are finite sets. Thus, from (3) it is clear that X , being the union of two finite sets, is a finite set. This is a contradiction, because X is an infinite set.

Hence X is a connected space.

Q.5 Show that a cofinite topological space (X, τ_c) is disconnected if X is finite.

PU, 2008 (M.Sc. Math)

Sol: Let A be a nonempty open subset of X . Since X is finite, so A must be finite. Since (X, τ_c) is a cofinite topological space and a finite subset of a cofinite topological space is closed, so A , being finite subset of a cofinite topological space, is closed, i.e. A^c is open.

Hence, we have open sets A and A^c such that

$$A \cap A^c = \phi$$

and

$$A \cup A^c = X$$

This shows that $\{A, A^c\}$ is a disconnection of X .

Hence a cofinite topological space (X, τ_c) is disconnected if X is finite.

Q.6 Show that a subspace of the real line R is connected if and only if it is an interval. In particular, R is connected.

University of Gujrat 2013; PU, 2007; 2006 (M.Sc. Math)

Sol: Let X be a connected subspace of R , then we have to show that X is an interval. For this suppose on contrary that X is not an interval, then there exist real numbers x, y, z , $x < y < z$ such that $x, z \in X$ and $y \notin X$.

The sets

$$A = X \cap]-\infty, y[\text{ and } B = X \cap]y, \infty[$$

are nonempty open subsets of X such that

$$\begin{aligned} A \cup B &= (X \cap]-\infty, y[) \cup (X \cap]y, \infty[) \\ &= X \cap (]-\infty, y[\cup]y, \infty[) \\ &= X \cap (R - \{y\}) \\ &= X \qquad \qquad \qquad \because y \notin X \\ A \cap B &= (X \cap]-\infty, y[) \cap (X \cap]y, \infty[) \\ \Rightarrow A \cap B &= X \cap (]-\infty, y[\cap]y, \infty[) \\ &= X \cap \emptyset \\ &= \emptyset \end{aligned}$$

Thus, $\{A, B\}$ is a disconnection of X , which is a contradiction, so our assumption that X is not an interval is wrong. Consequently X is an interval.

Conversely, let X be an interval, then we have to show that X is connected.

For this suppose on contrary that X disconnected, then there must exist closed sets

$$A \neq \emptyset, B \neq \emptyset, A \cap B = \emptyset, A \cup B = X$$

Next suppose that $x \in A$ and $z \in B$, then obviously $x \neq z$.

We assume that $x < z$.

Now $x \in A$ and $z \in B \Rightarrow x, z \in A \cup B$, so $x, z \in X$, because $A \cup B = X$.

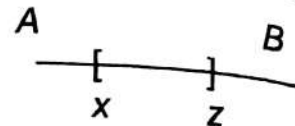
Since X is an interval, so

$$\begin{aligned} x, z \in X &\Rightarrow [x, z] \subseteq X \\ &\Rightarrow [x, z] \subseteq A \cup B \end{aligned}$$

This shows that each point of $[x, z]$ is either in A or in B .

Next we define

$$y = \sup(A \cap [x, z])$$



Then

$$y = \sup(A)$$

Since A is closed, so its sup must be in it, i.e. $y \in A$.

But then $y \neq z$.

Therefore, $x \leq y < z$.

Again by the definition of y , $y + \varepsilon$ is in B for every $\varepsilon > 0$ such that $y + \varepsilon \leq z$

This shows that y is the greatest lower bound of B . Since B is closed, so $y \in B$.

Now $y \in A$ and $y \in B$ shows that

$$y \in A \cap B$$

$$\Rightarrow A \cap B \neq \phi$$

This is a contradiction because $A \cap B = \phi$.

This shows that our assumption that X is disconnected is wrong. Hence X is a connected.

Since R is the subspace of R and R is an interval, so R is connected.

Q.7 Show that the continuous image of a connected space is connected.

University of Sargodha, 2010; PU, 2012; 2009; 2006; 2005 (M.Sc. Math)

Sol: Let X be a connected space. Let $f: X \rightarrow f(X)$ be a continuous function.

To show that $f(X)$ connected, suppose on contrary that $f(X)$ is disconnected. Let $\{A, B\}$ be a disconnection of $f(X)$, that is both A and B are nonempty open subsets of $f(X)$ such that

$$A \cap B = \phi \text{ and } A \cup B = f(X)$$

Since A and B are open sets in $f(X)$ and f is continuous, so $f^{-1}(A)$ and $f^{-1}(B)$ are open sets in X .

Now

$$\begin{aligned} f^{-1}(A) \cup f^{-1}(B) &= f^{-1}(A \cup B) \\ &= f^{-1}(f(X)) \quad \because A \cup B = f(X) \\ &= X \end{aligned}$$

And

$$\begin{aligned} f^{-1}(A) \cap f^{-1}(B) &= f^{-1}(A \cap B) \\ &= f^{-1}(\phi) \quad \because A \cap B = \phi \\ &= \phi \end{aligned}$$

Since $A \neq \phi$, so $f^{-1}(A) \neq \phi$.

Similarly, $B \neq \phi \Rightarrow f^{-1}(B) \neq \phi$.

This shows that $\{f^{-1}(A), f^{-1}(B)\}$ is a disconnection of X which is a contradiction, so $f(X)$ is connected.

Q.8 Show that the range of a continuous real function defined on a connected space is an interval.

Sol: Let f be a continuous real function defined on a connected space X , i.e. X is a connected space and $f: X \rightarrow R$ be a continuous function. Then we have to show that $f(X)$ is an interval. For this first we show that $f(X)$ is connected.

To show that $f(X)$ connected, suppose on contrary that $f(X)$ is disconnected. Let $\{A, B\}$ be a disconnection of $f(X)$, that is both A and B are nonempty open subsets of $f(X)$ such that

$$A \cap B = \phi \text{ and } A \cup B = f(X)$$

Since A and B are open sets in $f(X)$ and f is continuous, so $f^{-1}(A)$ and $f^{-1}(B)$ are open sets in X .

$$\begin{aligned} \text{Now } f^{-1}(A) \cup f^{-1}(B) &= f^{-1}(A \cup B) \\ &= f^{-1}(f(X)) \quad \because A \cup B = f(X) \\ &= X \end{aligned}$$

$$\begin{aligned} \text{and } f^{-1}(A) \cap f^{-1}(B) &= f^{-1}(A \cap B) \\ &= f^{-1}(\phi) \quad \because A \cap B = \phi \\ &= \phi \end{aligned}$$

Since $A \neq \phi$, so $f^{-1}(A) \neq \phi$.

Similarly, $B \neq \phi \Rightarrow f^{-1}(B) \neq \phi$.

This shows that $\{f^{-1}(A), f^{-1}(B)\}$ is a disconnection of X which is a contradiction, so $f(X)$ is connected.

Now $f(X)$, being the connected subspace of R is an interval.

Q.9 Show that a topological space X is disconnected if and only if X contains a nonempty clopen set.

PU, 2013; 2012 (M.Sc. Math)

Sol: Let X be a disconnected space, then there exists a disconnection $\{A, B\}$ of X , i.e. A and B are nonempty open sets such that

$$A \cap B = \phi \quad \dots(1)$$

$$\text{and } A \cup B = X \quad \dots(2)$$

But (1) and (2) are only possible if $B = A^c$.

Since B is open, i.e. A^c is open, so A is closed.

Now A , being open and closed set, is a clopen set.

Conversely, let A be a nonempty clopen subset of X , then A is

open as well as closed set.

Since A is closed set, so A^c is open, therefore, $\{A, A^c\}$ is a disconnection of X . This shows that X is disconnected.

Q.10 Show that a topological space X is connected if and only if X does not contain a nonempty clopen set.

Sol: Let X be a connected space, then we have to show that X does not contain a nonempty clopen set. For this suppose on contrary that A is a nonempty clopen subset of X , then A is open as well as closed set.

Since A is closed set, so A^c is open, therefore, $\{A, A^c\}$ is a disconnection of X . This shows that X is disconnected which is a contradiction, so X does not contain a nonempty clopen set.

Conversely, let X does not contain a nonempty clopen set, then we have to show that X is connected. For this suppose on contrary that X is disconnected, then there exists a disconnection $\{A, B\}$ of X , i.e. A and B are nonempty open sets such that

$$A \cap B = \emptyset \quad \dots(1)$$

$$\text{and} \quad A \cup B = X \quad \dots(2)$$

But (1) and (2) are only possible if $B = A^c$.

Since B is open, i.e. A^c is open, so A is closed.

Now A , being open and closed set, is a clopen set which is a contradiction, so X is a connected space.

Q.11 If $\{A, B\}$ is a disconnection of X and C is a connected subspace of X , then show that C is contained either in A or in B .

Sol: Suppose on contrary that C is contained neither in A nor in B , then

$$C \cap A \neq \emptyset$$

$$\text{and} \quad C \cap B \neq \emptyset$$

Since A and B are open subsets of X and C is a subspace of X , so $C \cap A$ and $C \cap B$ are nonempty open subsets of C .

Also

$$\begin{aligned} (C \cap A) \cup (C \cap B) &= C \cap (A \cup B) \\ &= C \cap X \quad \because A \cup B = X \\ &= C \quad \because C \subseteq X \end{aligned}$$

$$\begin{aligned} \text{and} \quad (C \cap A) \cap (C \cap B) &= C \cap (A \cap B) \\ &= C \cap \emptyset \quad \because A \cap B = \emptyset \\ &= \emptyset \end{aligned}$$

This shows that $\{C \cap A, C \cap B\}$ is a disconnection of C , which is a contradiction, so our assumption that C is contained neither in A nor in B is wrong.

Consequently C is contained either in A or in B .

Q.12 State and prove Intermediate Value Theorem.

Intermediate Value Theorem:

Let $f: R \rightarrow R$ be a continuous function. Let $x, z \in f(R)$ with $x < z$. Then for every $y \in R$ with $x < y < z$, there is an $r \in R$ such that $f(r) = y$.

Proof: Since R is connected and f is continuous, so $f(R)$, being the continuous image of a connected space, is connected.

Suppose on contrary that for $x, z \in f(R)$ and for some $y \in R$ with $x < y < z$, there is no $r \in R$ such that $f(r) = y$, i.e. for all $r \in R$, $f(r) \neq y$, so $y \notin f(R)$.

Since $(-\infty, y)$ and (y, ∞) are open subsets of R and $f(R)$ is also a subset of R , so the open sets in $f(R)$ are defined as

$$A = f(R) \cap (-\infty, y)$$

$$B = f(R) \cap (y, \infty)$$

Now

$$\begin{aligned} A \cap B &= (f(R) \cap (-\infty, y)) \cap (f(R) \cap (y, \infty)) \\ &= f(R) \cap ((-\infty, y) \cap (y, \infty)) \\ &= f(R) \cap \emptyset \\ &= \emptyset \end{aligned}$$

and

$$\begin{aligned} A \cup B &= (f(R) \cap (-\infty, y)) \cup (f(R) \cap (y, \infty)) \\ &= f(R) \cap ((-\infty, y) \cup (y, \infty)) \\ &= f(R) \cap (R \setminus \{y\}) \\ &= f(R) \end{aligned}$$

$$\because y \notin f(R)$$

This shows that $\{A, B\}$ is a disconnection of $f(R)$, which is a contradiction.

Hence, for every $y \in R$ with $x < y < z$, there is an $r \in R$ such that $f(r) = y$.

Q.13 Let $X = \bigcup_{\alpha \in I} X_\alpha$, where each X_α is connected and $\bigcap_{\alpha \in I} X_\alpha \neq \emptyset$, then X is connected.

Sol: Suppose on contrary that X is disconnected and let $\{A, B\}$ be a disconnection of X . Since X_α is connected, so $X_\alpha \subseteq A$ or $X_\alpha \subseteq B$.

Since $\bigcap_{\alpha \in I} X_\alpha \neq \emptyset$, so each X_α is contained in A or in B , i.e.

$\bigcup_{\alpha \in I} X_\alpha$ is contained in A or in B .

If $\bigcup_{\alpha \in I} X_\alpha \subseteq A$, then $X \subseteq A$, so $B = \emptyset$.

Similarly, if $\bigcup_{\alpha \in I} X_\alpha \subseteq B$, then $X \subseteq B$, so $A = \emptyset$.

Both cases contradict the fact that $\{A, B\}$ is a disconnection of X . This shows that our assumption that X is disconnected is wrong, so X is connected.

Q.14 Show that a topological space X is connected if and only if for every pair of points in X , there is some connected subspace of X which contains both.

PU, 2001 (M.Sc. Math)

Sol: Let X be a connected space, then for every pair of points of X , X itself is a connected subspace of X which contains both of them. Conversely, let for every pair of points in X there be some connected subspace of X which contains both, then to show that X is connected, let a be a fixed point of X and for any $x \in X$, $C_{a,x}$ a connected subspace of X containing x and a . Since $C_{a,x}$ is a subspace of X , so

$$\begin{aligned} C_{a,x} &\subseteq X \\ \Rightarrow \bigcup_{x \in X} C_{a,x} &\subseteq X \end{aligned} \quad \dots(1)$$

Now

$$\begin{aligned} x &\in C_{a,x} \\ \Rightarrow \bigcup_{x \in X} \{x\} &\subseteq \bigcup_{x \in X} C_{a,x} \\ \Rightarrow X &\subseteq \bigcup_{x \in X} C_{a,x} \end{aligned} \quad \dots(2)$$

Combining (1) and (2), we have

$$X = \bigcup_{x \in X} C_{a,x} \quad \dots(3)$$

Since

$$a \in \bigcap_{x \in X} C_{a,x}$$

So

$$\bigcap_{x \in X} C_{a,x} \neq \emptyset$$

Hence, X , being the union of connected subspaces with nonempty intersection, is connected.

Q.15 Let C be a connected subset of X and for some subset A of X , $C \subset A \subseteq \bar{C}$. Show that A is connected. In particular $\bar{C}$ is connected.

PU, 2002 (M.Sc. Math)

Sol: Suppose on contrary that A is disconnected and $\{U, V\}$ is the disconnection of A , then C , being the connected subset of a disconnected set A , is contained either in U or in V , i.e.

$$C \subseteq U \text{ or } C \subseteq V$$

If C is contained in U , then $C \cap V = \emptyset$

$$\begin{aligned} \Rightarrow C &\subseteq V^c \\ \Rightarrow \bar{C} &\subseteq \overline{V^c} \end{aligned} \quad \dots(1)$$

Since V is open, so V^c is closed, therefore,

$$\overline{V^c} = V^c$$

Making this substitution in (1), we have

$$\bar{C} \subseteq V^c \quad \dots(2)$$

It is given that

$$C \subset A \subseteq \bar{C} \quad \dots(3)$$

Combining (2) and (3), we have

$$C \subset A \subseteq \bar{C} \subseteq V^c$$

$$\Rightarrow A \subseteq V^c$$

$$\Rightarrow A \cap V = \phi \quad \dots(4)$$

Since $\{U, V\}$ is the disconnection of A , so

$$A = U \cup V$$

$$\Rightarrow V \subseteq A \quad \dots(5)$$

(4) and (5) are only possible if $V = \phi$ which is a contradiction, so our assumption that A is disconnected is wrong. Consequently A is connected.

Same arguments show that $\bar{C}$ is connected.

Q.16 If $A_1, A_2, \dots, A_n, \dots$ is a sequence of connected subspaces of a topological space each of which intersects its successor, then show that $\bigcup_{n=1}^{\infty} A_n$ is connected.

Sol: Let X be a topological space and $\{A_n : n \in N\}$ a collection of connected subspaces of X such that

$$A_n \cap A_{n+1} \neq \phi, \quad n \in N$$

Then we have to show that $\bigcup_{n=1}^{\infty} A_n$ is connected. We shall prove it by induction.

For $n = 1, 2$, i.e., for two subspaces, it is true, because the union of any two connected subspaces whose intersection is nonempty is connected.

Thus,

$$\bigcup_{n=1}^2 A_n = A_1 \cup A_2$$

is connected, because $A_1 \cap A_2 \neq \phi$ and A_1, A_2 are connected.

Next suppose that it is true for k space, i.e.

$$G = \bigcup_{n=1}^k A_n$$

is connected and $A_k \cap A_{k+1} \neq \phi$.

But then

$$G \cap A_{k+1} \neq \phi \quad \because A_n \cap A_{n+1} \neq \phi$$

Therefore, $G \cap A_{k+1}$, being the union of connected spaces, is connected.

Thus,

$$\left(\bigcup_{n=1}^k A_n \right) \cup A_{k+1} = \bigcup_{n=1}^{k+1} A_n$$

is connected. Hence $\bigcup_{n=1}^{\infty} A_n$ is connected.

Q.17 Show that a topological space X is disconnected $\Leftrightarrow$ there exists a continuous mapping of X onto the two-point discrete space $\{0,1\}$.

Sol: Let X be a disconnected space. Let $\{A,B\}$ be the disconnection of X . Next define a function $f: X \rightarrow \{0,1\}$ such that

$$f(x) = \begin{cases} 0, & \forall x \in A \\ 1, & \forall x \in B \end{cases}$$

Then obviously,

$$f(A) = \{0\}$$

$$\Rightarrow A = f^{-1}\{0\} \quad \dots(1)$$

and

$$f(B) = \{1\}$$

$$\Rightarrow B = f^{-1}\{1\} \quad \dots(2)$$

Since $\{0,1\}$ is discrete space, so $\{0\}$ and $\{1\}$ are open subsets of $\{0,1\}$.

Now (1) and (2) show that $f^{-1}\{0\}$ and $f^{-1}\{1\}$ are open sets in X , because A and B are open sets in X .

Since the inverse images of open sets $\{0\}$ and $\{1\}$ under f are open, so f is a continuous function.

Also

$$\begin{aligned} f(X) &= f(A \cup B) & \because X &= A \cup B \\ &= f(f^{-1}\{0\} \cup f^{-1}\{1\}) \\ &= f(f^{-1}(\{0\} \cup \{1\})) \\ &= f(f^{-1}\{0,1\}) \\ &= \{0,1\} \end{aligned}$$

Thus, $f: X \rightarrow \{0,1\}$ is onto.

Hence $f: X \rightarrow \{0,1\}$ is a continuous mapping of X onto the two-point discrete space $\{0,1\}$.

Conversely, let there exist a continuous mapping of X onto the two-point discrete space $\{0,1\}$, then we have to show that X is disconnected.

Let $f: X \rightarrow \{0,1\}$ be a continuous mapping of X onto $\{0,1\}$.

Since $\{0,1\}$ is a discrete space, so $\{0\}$ and $\{1\}$ are open subsets of $\{0,1\}$.

But then

$$f^{-1}\{0\} \text{ and } f^{-1}\{1\}$$

being the inverse images of open sets under a continuous function, are open.

Also

$$f^{-1}\{0\} \neq \emptyset \quad \because \{0\} \neq \emptyset$$

$$f^{-1}\{1\} \neq \emptyset \quad \because \{1\} \neq \emptyset$$

and

$$\begin{aligned} f^{-1}\{0\} \cap f^{-1}\{1\} &= f^{-1}(\{0\} \cap \{1\}) \\ &= f^{-1}(\emptyset) \\ &= \emptyset \end{aligned}$$

Similarly

$$\begin{aligned} f^{-1}\{0\} \cup f^{-1}\{1\} &= f^{-1}(\{0\} \cup \{1\}) \\ &= f^{-1}\{0,1\} \\ &= X \end{aligned}$$

Thus, $\{f^{-1}\{0\}, f^{-1}\{1\}\}$ is a disconnection of X .

Hence X is a disconnected space.

Q.18 Show that a topological space X is connected $\Leftrightarrow$ there does not exist a continuous mapping of X onto the two-point discrete space $\{0,1\}$.

PU, 2011; 2007 (M.Sc. Math)

Sol: Let X be a connected space and D a two-point discrete space, then we have to show that there does not exist a continuous function from X onto D . For this we suppose on contrary that $f: X \rightarrow D$ is a continuous onto function. But then $f(X)$, being the continuous image of a connected space, is connected. Since f is onto, so $f(X) = D$. Therefore, D is connected, which is a contradiction to the fact that the two-point discrete space is disconnected.

Hence there does not exist a continuous function from X onto the two-point discrete space.

Conversely, let there do not exist a continuous function from X onto the two-point discrete space, then we have to show that X is connected.

For this suppose on contrary that X is a disconnected space. Let $\{A, B\}$ be the disconnection of X . Next define a function $f: X \rightarrow \{0,1\}$ such that

$$f(x) = \begin{cases} 0, & \forall x \in A \\ 1, & \forall x \in B \end{cases}$$

Then obviously,

$$f(A) = \{0\}$$

$$\Rightarrow A = f^{-1}\{0\} \quad \dots(1)$$

and

$$\begin{aligned} f(B) &= \{1\} \\ \Rightarrow B &= f^{-1}\{1\} \end{aligned} \quad \dots(2)$$

Since $\{0,1\}$ is discrete space, so $\{0\}$ and $\{1\}$ are open subsets of $\{0,1\}$.

Now (1) and (2) show that $f^{-1}\{0\}$ and $f^{-1}\{1\}$ are open sets in X , because A and B are open sets in X .

Since the inverse images of open sets $\{0\}$ and $\{1\}$ under f are open, so f is a continuous function.

Also

$$\begin{aligned} f(X) &= f(A \cup B) & \because X &= A \cup B \\ &= f(f^{-1}\{0\} \cup f^{-1}\{1\}) \\ &= f(f^{-1}(\{0\} \cup \{1\})) \\ &= f(f^{-1}\{0,1\}) \\ &= \{0,1\} \end{aligned}$$

Thus, $f: X \rightarrow \{0,1\}$ is onto.

Hence $f: X \rightarrow \{0,1\}$ is a continuous mapping of X onto the two-point discrete space $\{0,1\}$. This contradicts the given condition, so X is connected.

Q.19 Show that a topological space X is connected if and only if every continuous mapping f from X into the two-point discrete space $\{0,1\}$ is constant.

Sol: Let X be a connected space and $f: X \rightarrow \{0,1\}$ a continuous mapping, where $\{0,1\}$ is a two-point discrete space, then we have to show that f is a constant function.

For this let $y \in \{0,1\}$, then $\{y\}$, being the subset of a discrete space, is a clopen set.

Since f is continuous mapping, so $f^{-1}\{y\}$, being the inverse image of a clopen set is clopen.

Since $\{y\} \neq \emptyset$, so $f^{-1}\{y\} \neq \emptyset$.

This shows that $f^{-1}\{y\}$ is a nonempty clopen subset of X . Since X is connected, so it does not contain a proper clopen subset, therefore, $f^{-1}\{y\}$ is not proper subset of X , i.e.

$$\begin{aligned} f^{-1}\{y\} &= X \\ \Rightarrow \{y\} &= f(X) \end{aligned}$$

This shows that the range of f is a single element. Hence f is a constant mapping.

Conversely, let every continuous mapping f from X into two-point discrete space $\{0,1\}$ is constant, then we have to show that X is connected.

For this suppose on contrary that X is disconnected and $\{A,B\}$ is a disconnection of X .

Then the function $f: X \rightarrow \{0,1\}$ defined by

$$f(x) = \begin{cases} 0, & \forall x \in A \\ 1, & \forall x \in B \end{cases}$$

is continuous, because $A = f^{-1}\{0\}$ is open in X when $\{0\}$ is open in $\{0,1\}$.

Similarly, $B = f^{-1}\{1\}$ is open in X when $\{1\}$ is open in $\{0,1\}$.

Also

$$\begin{aligned} f(X) &= f(A \cup B) & \because X &= A \cup B \\ &= f(f^{-1}\{0\} \cup f^{-1}\{1\}) \\ &= f(f^{-1}(\{0\} \cup \{1\})) \\ &= f(f^{-1}\{0,1\}) \\ &= \{0,1\} \end{aligned}$$

This shows that f is not a constant function which is a contradiction, so X is connected.

Q.20 Let X be a topological space and $\sim$ be a relation in X defined as $x \sim y \Leftrightarrow x$ and y belong to a connected subspace A of X . Then show that $\sim$ is an equivalence relation.

Sol: (i) Reflexive: Since the singleton subsets of a topological space are connected, therefore, for $x \in X$, $\{x\}$ is connected subspace of X such that $x \in \{x\}$. This shows that $x \sim x$.

Hence $\sim$ is reflexive.

(ii) Symmetric: Let $x \sim y \Leftrightarrow x, y \in A$, where A is connected subspace of X .

$$\Leftrightarrow y, x \in A$$

$$\Leftrightarrow y \sim x$$

This shows that $x \sim y \Leftrightarrow y \sim x$

Hence $\sim$ is symmetric.

(iii) Transitive: Let $x \sim y$ and $y \sim z$, then

$x \sim y \Rightarrow x, y \in C_1$, where C_1 is a connected subspace of X .

and $y \sim z \Rightarrow y, z \in C_2$, where C_2 is a connected subspace of X .

Since $y \in C_1$ and $y \in C_2$, so

$$y \in C_1 \cap C_2$$

$$\Rightarrow C_1 \cap C_2 \neq \emptyset$$

Let $C = C_1 \cup C_2$, then obviously, C is also a connected subspace of X .

Now $x \in C_1 \Rightarrow x \in C_1 \cup C_2 \Rightarrow x \in C$

And $z \in C_2 \Rightarrow z \in C_1 \cup C_2 \Rightarrow z \in C$

Since $x, z \in C$, where C is connected, so $x \sim z$.

This shows that $x \sim y, y \sim z \Rightarrow x \sim z$.

Hence $\sim$ is transitive.

Thus, $\sim$, being reflexive, symmetric and transitive, is an equivalence relation.

Q.21 Let X be a topological space, then show that for each $x \in X$, there is exactly one component of X containing x .

PU, 2011 (M.Sc. Math)

Sol: Let X be a topological space. Let x be any arbitrary point of X .

Let $\{C_\alpha : \alpha \in I\}$ be a collection of all connected subspaces of X containing x .

Let

$$C = \bigcup_{\alpha \in I} C_\alpha$$

Since each C_α contains x , so $x \in \bigcap_{\alpha \in I} C_\alpha$.

Thus,

$$\bigcap_{\alpha \in I} C_\alpha \neq \emptyset.$$

Then C is connected subspace of X containing x .

Next we show that C is a component of X . For this suppose on contrary that C is not a component of X , then there must exist a connected subspace C^* of X such that

$$C \subset C^*$$

Since $x \in C$, so $x \in C^*$.

This shows that C^* is one of C_α 's.

Since each $C_\alpha \subseteq C$, so $C^* \subseteq C$, which is a contradiction to the fact $C \subset C^*$, so our assumption that C is not a component of X is wrong. Consequently C is a component of X .

Next we have to show that x is contained in exactly one component.

For this let C_1 and C_2 be two distinct components of X . Suppose on contrary that $x \in C_1$ and $x \in C_2$, then $x \in C_1 \cap C_2$.

Then, $C_1 \cup C_2$, being the overlapping connected subspaces, is connected subspace of X such that

$$C_1 \subset C_1 \cup C_2 \text{ and } C_2 \subset C_1 \cup C_2$$

This shows that C_1 and C_2 are not components of X . Therefore, our assumption that x is contained in two components is wrong. Consequently x is contained in exactly one component.

Q.22 Let X be a topological space, then show that each connected subset of X is contained in a component of X .

PU, 2011 (M.Sc. Math)

Sol: Let X be a topological space and A a connected subspace of X . Let $\{C_\alpha : \alpha \in I\}$ be a collection of all connected subspaces of X containing A .

Let

$$C = \bigcup_{\alpha \in I} C_\alpha$$

Since each C_α contains A , so $A \subseteq \bigcap_{\alpha \in I} C_\alpha$.

Thus,

$$\bigcap_{\alpha \in I} C_\alpha \neq \emptyset.$$

Then C is connected subspace of X containing A .

Next we show that C is a component of X . For this suppose on contrary that C is not a component of X , then there must exist a connected subspace C^* of X such that

$$C \subset C^*$$

Since $A \subseteq C$, so $A \subseteq C^*$.

This shows that C^* is one of C_α 's.

Since each $C_\alpha \subseteq C$, so $C^* \subseteq C$, which is a contradiction to the fact $C \subset C^*$, so our assumption that C is not a component of X is wrong. Consequently C is a component of X .

Next we have to show that A is contained in exactly one component.

For this let C_1 and C_2 be two distinct components of X . Suppose on contrary that $A \subseteq C_1$ and $A \subseteq C_2$, then $A \subseteq C_1 \cap C_2 \Rightarrow C_1 \cap C_2 \neq \emptyset$.

Then, $C_1 \cup C_2$, being the overlapping connected subspaces, is connected subspace of X such that

$$C_1 \subset C_1 \cup C_2 \text{ and } C_2 \subset C_1 \cup C_2$$

This shows that C_1 and C_2 are not components of X . Therefore, our assumption that A is contained in two components is wrong. Consequently A is contained in exactly one component.

Q.23 Let X be a topological space, then show that a connected clopen subspace of X is a component of X .

Sol: Let X be a topological space and C a connected clopen subspace of X , then we have to show that C is a component of X . For this suppose on contrary that C is not a component of X , then there exists a connected subspace C^* of X such that

$$C \subset C^*$$

Since C is open in X , so $A = C \cap C^*$ is open in C^* .

Since C is closed in X , so C^c is open in X , so $B = C^c \cap C^*$ is

open in C^* .

Since C is a proper subset of C^* , so

$$A = C \cap C^* \neq \emptyset \text{ and } B = C^c \cap C^* \neq \emptyset$$

Now

$$\begin{aligned} A \cap B &= (C \cap C^*) \cap (C^c \cap C^*) \\ &= (C \cap C^c) \cap C^* \\ &= \emptyset \cap C^* \quad \because C \cap C^c = \emptyset \\ \Rightarrow A \cap B &= \emptyset \end{aligned}$$

Similarly

$$\begin{aligned} A \cup B &= (C \cap C^*) \cup (C^c \cap C^*) \\ &= (C \cup C^c) \cap C^* \\ &= X \cap C^* \quad \because C \cup C^c = X \\ &= C^* \end{aligned}$$

This shows that $\{A, B\}$ is a disconnection of C^* , which is a contradiction. Hence C is a component.

Q.24 Let X be a topological space, then every component is closed in X .

PU, 2011 (M.Sc. Math)

Sol: Let C be a component of X . If possible, suppose that C is not closed in X . Since $\overline{C}$ is closed and connected and contains C , which is a contradiction to the fact that C is a component. Hence $C = \overline{C}$, so C is closed.

Q.25 Give an example of a totally disconnected space which is not discrete.

Sol: Consider the set of rational numbers, Q . Let $p, q \in Q$, then by the property of real numbers, there is an irrational number r such that $p < r < q$, then

$$p \in A = Q \cap]-\infty, r[\text{ and } q \in B = Q \cap]r, \infty[$$

Also

$$\begin{aligned} A \cap B &= (Q \cap]-\infty, r[) \cap (Q \cap]r, \infty[) \\ &= Q \cap (]-\infty, r[\cap]r, \infty[) \\ &= Q \cap \emptyset = \emptyset \quad \because]-\infty, r[\cap]r, \infty[= \emptyset \end{aligned}$$

and

$$\begin{aligned} A \cup B &= (Q \cap]-\infty, r[) \cup (Q \cap]r, \infty[) \\ &= Q \cap (]-\infty, r[\cup]r, \infty[) \\ &= Q \cap (R \setminus \{r\}) \quad \because]-\infty, r[\cup]r, \infty[= R \setminus \{r\} \\ &= Q \quad \because r \notin Q \end{aligned}$$

This shows that $\{A, B\}$ is a disconnection of Q .

Thus, for any $p, q \in Q$, there exist a disconnection $\{A, B\}$ of Q such that $p \in A$ and $q \in B$.

This shows that Q is a totally disconnected space.

But Q is not discrete, because the singleton subsets of Q are not open.

Q.26 Prove that every totally disconnected space is Hausdorff space.

Sol: Let X be a totally disconnected space, then we have to show that X is a Hausdorff space. For this let $x, y \in X, x \neq y$.

Since X is a totally disconnected space, so for $x, y \in X, x \neq y$, there is a disconnection $\{A, B\}$ of X such that $x \in A, y \in B$.

Since $\{A, B\}$ is a disconnection of X , so both A and B are open sets with

$$A \cap B = \phi \text{ and } A \cup B = X$$

Since for $x, y \in X, x \neq y$, we have open sets A and B such that

$$x \in A, y \in B \text{ and } A \cap B = \phi$$

This shows that X is a Hausdorff space.

Q.27 Show that the components of a totally disconnected space are its singleton subsets.

Sol: Let X be a totally disconnected space, then we have to show that components of X are its singleton subsets. For this it is enough to show that every two-point subset of X is disconnected.

For this let $C = \{x, y\}$ be any two-point subset of X , then obviously $x, y \in X, x \neq y$. Therefore, there is a disconnection $\{A_1, B_1\}$ of X such that

$$x \in A_1 \text{ and } y \in B_1$$

But then

$$A = C \cap A_1 \text{ and } B = C \cap B_1$$

are open subsets of C .

Now

$$\begin{aligned} A \cap B &= (C \cap A_1) \cap (C \cap B_1) \\ &= C \cap (A_1 \cap B_1) \\ &= C \cap \phi \quad \because A_1 \cap B_1 = \phi \\ &= \phi \end{aligned}$$

Also

$$\begin{aligned} A \cup B &= (C \cap A_1) \cup (C \cap B_1) \\ &= C \cap (A_1 \cup B_1) \\ &= C \cap X \quad \because A_1 \cup B_1 = X \\ &= C \quad \because C \subseteq X \end{aligned}$$

Since $x \in C \cap A_1 = A$ and $y \in C \cap B_1 = B$, so $A \neq \emptyset$ and $B \neq \emptyset$.

This shows that $\{A, B\}$ is a disconnection of C . Hence every two-point subset of X is disconnected. Consequently the components of a totally disconnected space are its singleton subsets.

Q.28 If a Hausdorff space X has an open base whose sets are also closed then show that X is totally disconnected.

Sol: Let X be a Hausdorff space.

Let $\beta = \{B_\alpha : \alpha \in I\}$ be an open base whose sets are also closed, then we have to show that X is totally disconnected.

For this let $x, y \in X, x \neq y$.

Since X is a Hausdorff space, so there exist open sets U and V such that

$$x \in U, y \in V \quad \text{and} \quad U \cap V = \emptyset$$

Since $x \in U$ and U is an open set, so by the definition of an open base, there exists $B_\alpha \in \beta$ such that

$$x \in B_\alpha \subseteq U$$

$$\Rightarrow x \in B_\alpha$$

Since $y \in V$ and $U \cap V = \emptyset$, so $y \notin U$ and so $y \notin B_\alpha$

This shows that $y \in B_\alpha^c$. Since each B_α^c is also open, so if we do so for every pair of distinct points of X , then

$$\left\{ \bigcup_{\alpha \in I} B_\alpha, \bigcup_{\alpha \in I} B_\alpha^c \right\}$$

will be a disconnection of X such that

$$x \in \bigcup_{\alpha \in I} B_\alpha \quad \text{and} \quad y \in \bigcup_{\alpha \in I} B_\alpha^c$$

Hence X is a totally disconnected space.

Chapter 5

COMPACTNESS

EXERCISE 5

Short Questions

Q.2 Solve / answer the following short questions:

(i) Define a cover.

Sol: Let X be a topological space. A collection $\{A_\alpha : \alpha \in I\}$ of subsets of X is said to be a cover of X if

$$X = \bigcup_{\alpha \in I} A_\alpha$$

(ii) Define subcover.

Sol: Let X be a topological space. A collection $\{A_\alpha : \alpha \in I\}$ of subsets of X is said to be a cover of X if

$$X = \bigcup_{\alpha \in I} A_\alpha$$

A sub-collection of $\{A_\alpha : \alpha \in I\}$ is said to be a subcover of X if it itself is a cover of X .

(iii) Define open cover.

Sol: Let X be a topological space. A collection $\{U_\alpha : \alpha \in I\}$ of open subsets of X is said to be an open cover of X if

$$X = \bigcup_{\alpha \in I} U_\alpha$$

(iv) Define an open subcover.

Sol: Let X be a topological space. A collection $\{U_\alpha : \alpha \in I\}$ of open subsets of X is said to be an open cover of X if

$$X = \bigcup_{\alpha \in I} U_\alpha$$

The sub-collection of $\{U_\alpha : \alpha \in I\}$ is said to be an open subcover of X if it itself is an open cover of X .

(v) Show that the singleton subsets of a discrete topological space X form an open cover of X .

Sol: Since the singleton subsets of a discrete topological space X are open sets such that their union is X . This shows that the singleton subsets of the discrete topological space X form an open cover of X .

(vi) **Define a compact space.**

Sol: A compact space is a topological space in which every open cover has a finite subcover.

(vii) **Define a compact space.**

Sol: A compact subspace of a topological space is a subspace which is compact as a topological space in its own right.

(viii) **Show that every finite topological space is compact.**

Sol: If X is a finite topological space, i.e. X is a finite set, then every open cover of X is finite, so no open cover of X can have infinite subcover, therefore X is a compact topological space.

(ix) **Let (X, τ) be a topological space where τ consists of finite number of elements, then show that X is a compact space.**

Sol: In this case every open cover of X is finite, so no open cover of X can have infinite subcover, therefore X is a compact topological space.

(x) **The real line R is not compact.**

Sol: It is clear that the collection of open intervals

$$C = \{]n, n+2[: n \in \mathbb{N} \}$$

is an open cover of R .

Since C contains no finite sub-collection that covers R , so R is not compact.

(xi) **Define finite intersection property.**

Sol: A collection A of subsets of a nonempty set X is said to have the finite intersection property if every finite sub-collection of A has nonempty intersection.

In other words, the collection

$$A = \{ A_\alpha : \alpha \in I \}$$

of subsets of the topological space X is said to have finite intersection property if every finite sub-collection of A has nonempty intersection, i.e. for any finite subset I_1 of I ,

$$\bigcap_{\beta \in I_1} A_\beta \neq \emptyset$$

Long Questions

Q.3 An infinite set with cofinite topology is a compact space.

PU, 2012 (M.Sc. Math)

Sol: Let X be an infinite set with cofinite topology τ , then we have to show that (X, τ) is a compact topological space. For this let

$$C = \{ U_\alpha : \alpha \in I \}$$

be an open cover of X , then

$$X = \bigcup_{\alpha \in I} U_\alpha \quad \dots(1)$$

Take a $U_\beta \in C$, then U_β is an open set. Since τ is a cofinite

topology on X , so U_β^c must be finite.

Let

$$U_\beta^c = \{x_1, x_2, \dots, x_n\}$$

Since

$$x_i \in X, \quad i = 1, 2, \dots, n$$

From (1), we have

$$x_i \in \bigcup_{\alpha \in I} U_\alpha, \quad i = 1, 2, \dots, n$$

Therefore, there must exist $U_{\alpha_i} \in C$ such that

$$x_i \in U_{\alpha_i}, \quad i = 1, 2, \dots, n$$

Then obviously

$$C^* = \{U_\beta, U_{\alpha_1}, U_{\alpha_2}, \dots, U_{\alpha_n}\}$$

is the finite sub-collection of C such that

$$X = U_\beta \cup U_{\alpha_1} \cup U_{\alpha_2} \cup \dots \cup U_{\alpha_n}$$

This shows that C^* is a finite subcover of C .

Since an open cover C of X has a finite subcover C^* , so X is a compact space.

Q.4 Prove that every closed subspace of a compact space is compact.

Sol: **PU, 2013 (BS Math); PU, 2013; 2006; University of Sargodha, 2010 (M.Sc. Math)**

Let X be a compact space. Let Y be a closed subspace of X , then we have to show that Y is compact. For this let

$$C = \{U_\alpha : \alpha \in I\}$$

be an open cover of Y . i.e. U_α are open subsets of Y such that

$$Y = \bigcup_{\alpha \in I} U_\alpha \quad \dots(1)$$

Since Y is a subspace of X , so the open sets V_α in X are defined as

$$U_\alpha = Y \cap V_\alpha, \quad \alpha \in I$$

Using this in (1), we have

$$\begin{aligned} Y &= \bigcup_{\alpha \in I} (Y \cap V_\alpha) \\ \Rightarrow Y &= Y \cap \left(\bigcup_{\alpha \in I} V_\alpha \right) \\ \Rightarrow Y &\subseteq \bigcup_{\alpha \in I} V_\alpha \\ \Rightarrow Y^c \cup Y &\subseteq Y^c \cup \left(\bigcup_{\alpha \in I} V_\alpha \right) \\ \Rightarrow X &\subseteq Y^c \cup \left(\bigcup_{\alpha \in I} V_\alpha \right) \quad \because Y^c \cup Y = X \end{aligned}$$

But

$$Y^c \cup \left(\bigcup_{\alpha \in I} V_\alpha \right) \subseteq X$$

so

$$X = Y^c \cup \left(\bigcup_{\alpha \in I} V_\alpha \right) \quad \dots(2)$$

Since Y is closed in X , so Y^c is open in X . Thus from (2), we note

that

$$\{Y^c, V_\alpha : \alpha \in I\}$$

is an open cover of X .

Since X is compact, so this open cover must have finite subcover.

Let

$$\{Y^c, V_{\alpha_i} : i = 1, 2, \dots, n\}$$

be a finite subcover of X , i.e.

$$X = Y^c \cup (\bigcup_{i=1}^n V_{\alpha_i})$$

$$\Rightarrow Y \cap X = Y \cap (Y^c \cup (\bigcup_{i=1}^n V_{\alpha_i}))$$

$$\Rightarrow Y = (Y \cap Y^c) \cup (\bigcup_{i=1}^n (Y \cap V_{\alpha_i})) \quad \because Y \cap X = Y$$

$$\Rightarrow Y = \phi \cup (\bigcup_{i=1}^n (Y \cap V_{\alpha_i})) \quad \because Y \cap Y^c = \phi$$

$$\Rightarrow Y = \bigcup_{i=1}^n U_{\alpha_i}$$

This shows that $\{U_{\alpha_i} : i = 1, 2, \dots, n\}$ is a finite subcover of Y , so Y is compact.

Q.5 Show that the continuous image of compact space is compact.

PU, 2004; 2001 (M.Sc. Math)

Sol: Let X be a compact space. Let f be continuous function defined on X , then we have to show that $Y = f(X)$ is compact.

For this let

$$C = \{U_\alpha : \alpha \in I\}$$

be an open cover for Y , i.e. for all $\alpha \in I$, U_α are open subsets of Y such that

$$Y = \bigcup_{\alpha \in I} U_\alpha \quad \dots(1)$$

Since f is continuous and U_α are open subsets of Y , so

$$V_\alpha = f^{-1}(U_\alpha)$$

are open subsets of X .

Now

$$V_\alpha = f^{-1}(U_\alpha)$$

$$\Rightarrow \bigcup_{\alpha \in I} V_\alpha = \bigcup_{\alpha \in I} f^{-1}(U_\alpha)$$

$$= f^{-1}(\bigcup_{\alpha \in I} U_\alpha)$$

$$= f^{-1}(Y)$$

$$= X$$

$$\because \bigcup_{\alpha \in I} U_\alpha = Y$$

$$\because f^{-1}(Y) = X$$

$$\Rightarrow \bigcup_{\alpha \in I} V_\alpha = X \quad \dots(2)$$

This shows that $\{V_\alpha : \alpha \in I\}$ is an open cover of X . Since X is compact, so this open cover must have a finite subcover. Let that finite subcover be

i.e.

$$\{V_{\alpha_i} : i = 1, 2, \dots, n\}$$

$$\begin{aligned} X &= \bigcup_{i=1}^n V_{\alpha_i} & \dots(3) \\ \Rightarrow X &= \bigcup_{i=1}^n f^{-1}(U_{\alpha_i}) & \because V_{\alpha_i} = f^{-1}(U_{\alpha_i}) \\ \Rightarrow X &= f^{-1}\left(\bigcup_{i=1}^n U_{\alpha_i}\right) \\ \Rightarrow f(X) &= \bigcup_{i=1}^n U_{\alpha_i} \\ \Rightarrow Y &= \bigcup_{i=1}^n U_{\alpha_i} & \because f(X) = Y \end{aligned}$$

This shows that $\{U_{\alpha_i} : i = 1, 2, \dots, n\}$ is a finite subcover of C .

Thus, an open cover of Y has a finite subcover of Y .

Hence Y is a compact space.

Q.6 Show that the homeomorphic image of compact space is compact.

PU, 2004 (M.Sc. Math)

Sol: Let X be a compact space. Let f be homeomorphism defined on X , then we have to show that $Y = f(X)$ is compact.

For this let

$$C = \{U_{\alpha} : \alpha \in I\}$$

be an open cover for Y , i.e. for all $\alpha \in I$, U_{α} are open subsets of Y such that

$$Y = \bigcup_{\alpha \in I} U_{\alpha} \quad \dots(1)$$

Since f , being a homeomorphism, is a continuous function and U_{α} are open subsets of Y , so

$$V_{\alpha} = f^{-1}(U_{\alpha})$$

are open subsets of X .

Now

$$\begin{aligned} V_{\alpha} &= f^{-1}(U_{\alpha}) \\ \Rightarrow \bigcup_{\alpha \in I} V_{\alpha} &= \bigcup_{\alpha \in I} f^{-1}(U_{\alpha}) \\ &= f^{-1}\left(\bigcup_{\alpha \in I} U_{\alpha}\right) \\ &= f^{-1}(Y) & \because \bigcup_{\alpha \in I} U_{\alpha} = Y \\ &= X & \because f^{-1}(Y) = X \\ \Rightarrow \bigcup_{\alpha \in I} V_{\alpha} &= X & \dots(2) \end{aligned}$$

This shows that $\{V_{\alpha} : \alpha \in I\}$ is an open cover of X . Since X is compact, so this open cover must have a finite subcover. Let that finite subcover be

$$\{V_{\alpha_i} : i = 1, 2, \dots, n\}$$

i.e.

$$\begin{aligned}
 X &= \bigcup_{i=1}^n V_{\alpha_i} \\
 \Rightarrow X &= \bigcup_{i=1}^n f^{-1}(U_{\alpha_i}) & \because V_{\alpha_i} = f^{-1}(U_{\alpha_i}) \\
 \Rightarrow X &= f^{-1}\left(\bigcup_{i=1}^n U_{\alpha_i}\right) \\
 \Rightarrow f(X) &= \bigcup_{i=1}^n U_{\alpha_i} \\
 \Rightarrow Y &= \bigcup_{i=1}^n U_{\alpha_i} & \because f(X) = Y
 \end{aligned}
 \tag{3}$$

This shows that $\{U_{\alpha_i} : i = 1, 2, \dots, n\}$ is a finite subcover of C .

Thus, an open cover of Y has a finite subcover of Y .

Hence Y is a compact space.

Thus, the homeomorphic image of a compact space is compact.

Q.7 Show that any continuous bijective function from a compact space X to a Hausdorff space Y is a homeomorphism.

Sol: Let X be a compact space and Y a Hausdorff space.

Let $f: X \rightarrow Y$ be a continuous bijective function, then we have to show that f is homeomorphism.

For this it is enough to show that f is closed.

For this let C be a closed subset of X , then C , being the closed subset of the compact space X , is a compact set. Then $f(C)$, being the continuous image of a compact set, is a compact subset of Y .

Since Y is a Hausdorff space, so $f(C)$, being the compact subset of Hausdorff space, is closed in Y .

This shows that f is closed.

Hence, f being the closed, continuous bijective function, is a homeomorphism.

Q.8 Show that a topological space X is compact if and only if every collection $A = \{A_\alpha : \alpha \in I\}$ of closed sets in X which satisfies the finite intersection property itself has nonempty intersection.

Sol: Let X be a compact topological space.

Let $A = \{A_\alpha : \alpha \in I\}$ be any collection of closed sets which satisfy the finite intersection property. Suppose on contrary that

$$\begin{aligned}
 \bigcap_{\alpha \in I} A_\alpha &= \phi & \dots(1) \\
 \Rightarrow \left(\bigcap_{\alpha \in I} A_\alpha\right)^c &= \phi^c \\
 \Rightarrow \bigcup_{\alpha \in I} A_\alpha^c &= X & \because \phi^c = X
 \end{aligned}$$

This shows that $\{A_\alpha^c : \alpha \in I\}$ is an open cover of X , because each A_α^c , being the complement of a closed set, is an open set.

Since X is compact, so its open cover $\{A_\alpha^c : \alpha \in I\}$ must have finite subcover $\{A_{\alpha_i}^c : i = 1, 2, \dots, n\}$, i.e.

$$\begin{aligned}
 \bigcup_{i=1}^n A_{\alpha i}^c &= X \\
 \Rightarrow \left(\bigcup_{i=1}^n A_{\alpha i}^c \right)^c &= X^c \\
 \Rightarrow \bigcap_{i=1}^n A_{\alpha i} &= \phi \quad \because X^c = \phi
 \end{aligned}$$

This contradicts the finite intersection property of the collection A .
Hence

$$\bigcap_{\alpha \in I} A_{\alpha} \neq \phi$$

Conversely, let every collection of closed sets in X which satisfies the finite intersection property itself has nonempty intersection, then we have to show that X is a compact space.
For this let

$$\{U_{\alpha} : \alpha \in I\}$$

be any open cover of X , i.e.

$$\bigcup_{\alpha \in I} U_{\alpha} = X \quad \dots(2)$$

Suppose on contrary that the open cover $\{U_{\alpha} : \alpha \in I\}$ has no finite subcover, then for every finite subset I_1 of I ,

$$\bigcup_{\beta \in I_1} U_{\beta} \neq X$$

This shows that for every finite subset I_1 of I ,

$$\begin{aligned}
 \bigcap_{\beta \in I_1} U_{\beta}^c &\neq X^c \\
 \Rightarrow \bigcap_{\beta \in I_1} U_{\beta}^c &\neq \phi \quad \dots(3)
 \end{aligned}$$

This shows that the collection $\{U_{\alpha}^c : \alpha \in I\}$ of closed sets satisfies the finite intersection property, therefore,

$$\begin{aligned}
 \bigcap_{\alpha \in I} U_{\alpha}^c &\neq \phi \\
 \Rightarrow \left(\bigcap_{\alpha \in I} U_{\alpha}^c \right)^c &\neq \phi^c \\
 \Rightarrow \bigcup_{\alpha \in I} U_{\alpha} &\neq X \quad \because \phi^c = X
 \end{aligned}$$

This shows that $\{U_{\alpha} : \alpha \in I\}$ is not an open cover of X . This is a contradiction, so our assumption that this open cover has no finite subcover is wrong. Hence the open cover $\{U_{\alpha} : \alpha \in I\}$ has a finite subcover. Consequently X is compact.

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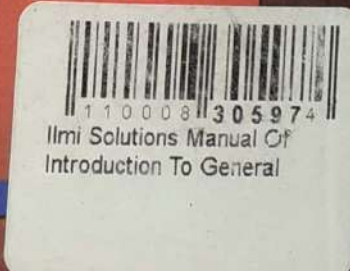
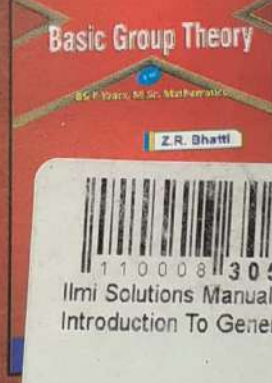
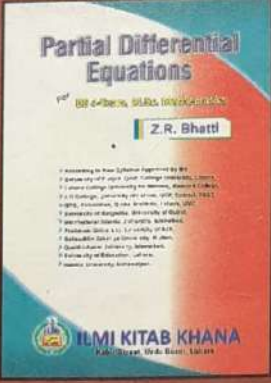
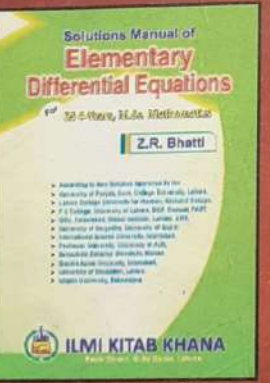
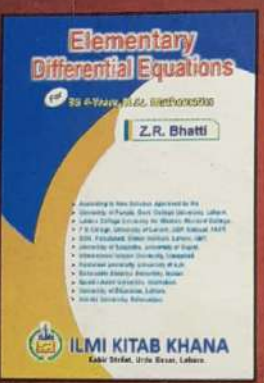
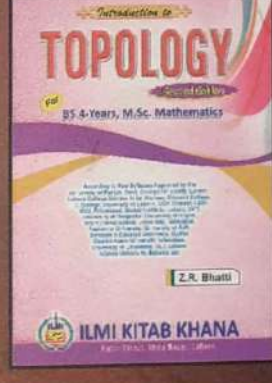
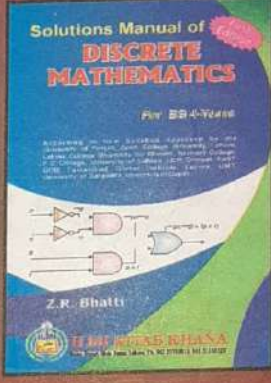
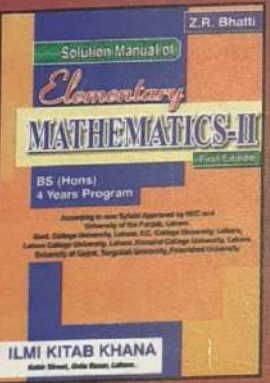
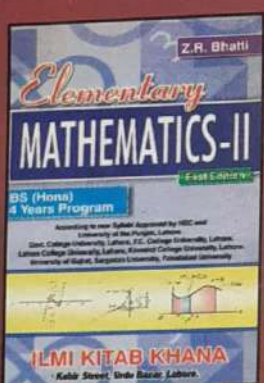
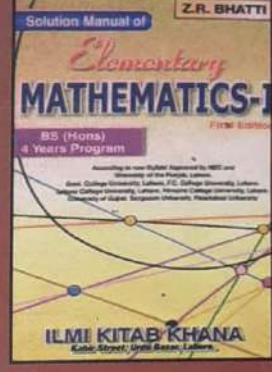
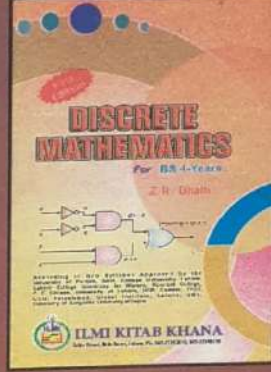
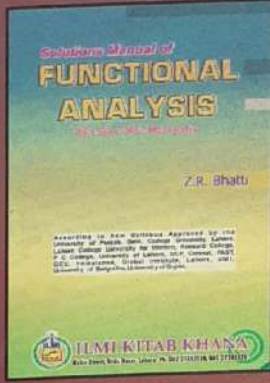
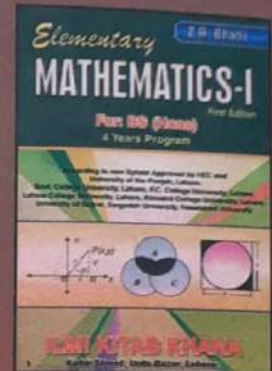
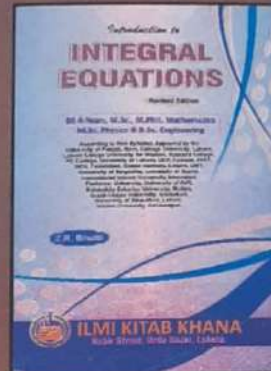
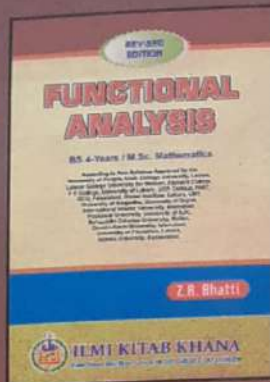
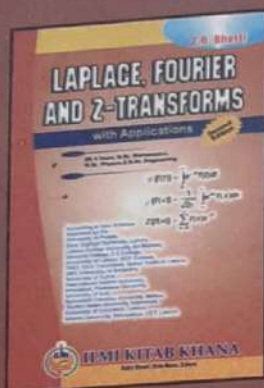
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